

# MATH 521

Real Analysis

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## Final Practice Book

Write-On Workbook

University of Wisconsin–Madison

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*Current build: Homework 1 through Homework 12*

Name: \_\_\_\_\_

Date: \_\_\_\_\_

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## How to Use This Workbook

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This workbook is for active proof practice. For each problem, work through the diagnosis, recipe card, self-explanation, skeleton, and full proof attempt before reading the reference solution.

- Pass 1**     Diagnose the topic, task, relevant definitions, and likely proof method.
- Pass 2**     Convert the problem into hypotheses, conclusion, tools, and a proof plan.
- Pass 3**     Answer targeted questions that expose the proof's logic.
- Pass 4**     Reconstruct the proof from a skeleton without reading the full solution.
- Pass 5**     Write a complete proof from scratch in the blank writing space.

Only after Pass 5 should you compare your work to the official or reference solution.

# Homework 1

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## Problem 1

---

### Problem Statement

### Homework 1, Problem 1

Using induction, prove that for all  $n \in \mathbb{N}$  we have

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n.$$

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What is the statement  $P(n)$  that should be proved by induction?

---

2. What is the base case, and what are both sides when  $n = 1$ ?

---

3. What is the next term added when passing from  $n$  to  $n + 1$ ?

---

4. How does  $4n^2 - n + (8n + 3)$  simplify?

---

5. Why is that expression equal to  $4(n + 1)^2 - (n + 1)$ ?

---

Define  $P(n)$  precisely:

---

Base case  $n = 1$ : verify left-hand side equals right-hand side:

---

Inductive hypothesis: assume  $P(n)$  for some  $n \in \mathbb{N}$ :

---

Add the next term  $8(n+1) - 5 =$  to both sides:

---

Simplify the right-hand side and match the formula for  $P(n+1)$ :

---

Invoke the principle of mathematical induction:

---

Use this page for the complete induction proof.

### Student-Derived Solution (No official solution available)

For  $n \in \mathbb{N}$ , let  $P(n)$  denote the statement

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n.$$

We argue by induction.

Base case. For  $n = 1$ , the left-hand side is 3, while the right-hand side is

$$4 \cdot 1^2 - 1 = 3.$$

Hence  $P(1)$  holds.

Inductive step. Assume  $P(n)$  holds for some  $n \in \mathbb{N}$ , i.e.

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n.$$

We must show that  $P(n + 1)$  holds. The next term in the sum is

$$8(n + 1) - 5 = 8n + 3.$$

Adding this term to both sides of the inductive hypothesis gives

$$3 + 11 + \cdots + (8n - 5) + (8n + 3) = (4n^2 - n) + (8n + 3).$$

The right-hand side simplifies to

$$4n^2 + 7n + 3.$$

On the other hand,

$$4(n + 1)^2 - (n + 1) = 4(n^2 + 2n + 1) - n - 1 = 4n^2 + 7n + 3.$$

Therefore

$$3 + 11 + \cdots + (8(n + 1) - 5) = 4(n + 1)^2 - (n + 1),$$

so  $P(n + 1)$  holds. By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

---

### Problem Statement

### Homework 1, Problem 2

Compute the following sum:

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \cdots + \frac{1}{(3n-2)(3n+1)}.$$

Prove that your answer is true for all  $n \in \mathbb{N}$  using induction.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What closed form does the telescoping computation suggest?

---

2. What is the partial fraction decomposition of  $\frac{1}{(3k-2)(3k+1)}$ ?

---

3. What is the induction statement  $P(n)$ ?

---

4. In the inductive step, what is the next term?

---

5. What algebra turns  $\frac{n}{3n+1} + \frac{1}{(3n+1)(3n+4)}$  into  $\frac{n+1}{3n+4}$ ?

---

First compute the suspected formula using partial fractions/telescoping:

---

State  $P(n)$ :

---

Base case  $n = 1$ :

---

Inductive hypothesis: assume the sum through  $n$  equals  $\frac{n}{3n+1}$ :

---

Add the term  $\frac{1}{(3n+1)(3n+4)}$  and combine fractions:

---

Conclude  $P(n+1)$  and invoke induction:

---

Use this page for the telescoping computation and induction proof.

### Student-Derived Solution (No official solution available)

For  $n \in \mathbb{N}$ , let  $P(n)$  denote the statement

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \cdots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}.$$

We prove  $P(n)$  for all  $n \in \mathbb{N}$  by induction.

Base case.  $P(1)$  says

$$\frac{1}{1 \cdot 4} = \frac{1}{3 \cdot 1 + 1},$$

which is true.

Inductive step. Assume  $P(n)$  is true for some  $n \in \mathbb{N}$ , that is,

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \cdots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}.$$

Adding the next term  $\frac{1}{(3n+1)(3n+4)}$  to both sides gives

$$\frac{1}{1 \cdot 4} + \cdots + \frac{1}{(3n-2)(3n+1)} + \frac{1}{(3n+1)(3n+4)} = \frac{n}{3n+1} + \frac{1}{(3n+1)(3n+4)}.$$

The right-hand side is

$$\frac{1}{3n+1} \left( n + \frac{1}{3n+4} \right) = \frac{1}{3n+1} \cdot \frac{n(3n+4) + 1}{3n+4}.$$

Since

$$n(3n+4) + 1 = 3n^2 + 4n + 1 = (3n+1)(n+1),$$

we obtain

$$\frac{n}{3n+1} + \frac{1}{(3n+1)(3n+4)} = \frac{n+1}{3n+4} = \frac{n+1}{3(n+1)+1}.$$

Thus  $P(n+1)$  holds. By induction,

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \cdots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$

for all  $n \in \mathbb{N}$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

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#### Problem Statement

#### Homework 1, Problem 3

Let  $f : A \rightarrow B$  be a function.

- (a) Prove that  $f$  is injective if and only if  $f^{-1}(f(X)) = X$  for all  $X \subseteq A$ .
- (b) Prove that  $f$  is surjective if and only if  $f(f^{-1}(Y)) = Y$  for all  $Y \subseteq B$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why is  $X \subseteq f^{-1}(f(X))$  always true?

---

2. Where is injectivity used to prove  $f^{-1}(f(X)) \subseteq X$ ?

---

3. To prove injectivity from the set identity, why choose  $X = \{a_1\}$ ?

---

4. Why is  $f(f^{-1}(Y)) \subseteq Y$  always true?

---

5. Where is surjectivity used to prove  $Y \subseteq f(f^{-1}(Y))$ ?

---

**Part (a), injective  $\Rightarrow$  identity.**

**Fix  $X \subseteq A$ . Prove  $X \subseteq f^{-1}(f(X))$ :**

---

**Let  $a \in f^{-1}(f(X))$ . Choose  $x \in X$  with  $f(a) = f(x)$ , then use injectivity:**

---

**Part (a), identity  $\Rightarrow$  injective.**

**Assume  $f(a_1) = f(a_2)$ . Let  $X = \{a_1\}$  and show  $a_2 \in f^{-1}(f(X)) = X$ :**

---

**Part (b).**

**For surjective  $\Rightarrow$  identity, prove both inclusions for  $Y \subseteq B$ :**

---

**For identity  $\Rightarrow$  surjective, apply the identity to  $Y = \{b\}$  for arbitrary  $b \in B$ :**

---

Use this page for part (a).

Use this page for part (b).

### Student-Derived Solution (No official solution available)

(a) Suppose first that  $f$  is injective. Let  $X \subseteq A$ . If  $x \in X$ , then  $f(x) \in f(X)$ , so  $x \in f^{-1}(f(X))$ . Hence  $X \subseteq f^{-1}(f(X))$ .

Conversely, let  $a \in f^{-1}(f(X))$ . Then  $f(a) \in f(X)$ , so there exists  $x \in X$  such that  $f(a) = f(x)$ . Since  $f$  is injective,  $a = x \in X$ . Thus  $f^{-1}(f(X)) \subseteq X$ , so  $f^{-1}(f(X)) = X$ .

Now suppose  $f^{-1}(f(X)) = X$  for all  $X \subseteq A$ . Let  $a_1, a_2 \in A$  and assume  $f(a_1) = f(a_2)$ . Set  $X = \{a_1\}$ . Then  $f(a_2) = f(a_1) \in f(X)$ , so

$$a_2 \in f^{-1}(f(X)) = X = \{a_1\}.$$

Hence  $a_2 = a_1$ , and  $f$  is injective.

(b) Suppose first that  $f$  is surjective. Let  $Y \subseteq B$ . For any function,  $f(f^{-1}(Y)) \subseteq Y$ : if  $b \in f(f^{-1}(Y))$ , then  $b = f(a)$  for some  $a \in f^{-1}(Y)$ , hence  $f(a) \in Y$ , so  $b \in Y$ .

For the reverse inclusion, let  $y \in Y$ . Since  $f$  is surjective, there exists  $a \in A$  such that  $f(a) = y$ . Then  $a \in f^{-1}(Y)$ , so  $y = f(a) \in f(f^{-1}(Y))$ . Thus  $Y \subseteq f(f^{-1}(Y))$ , and  $f(f^{-1}(Y)) = Y$ .

Conversely, suppose  $f(f^{-1}(Y)) = Y$  for all  $Y \subseteq B$ . Fix  $b \in B$ . Let  $Y = \{b\}$ . Then

$$f(f^{-1}(\{b\})) = \{b\}.$$

Thus  $f^{-1}(\{b\}) \neq \emptyset$ , so there exists  $a \in A$  with  $f(a) = b$ . Since  $b \in B$  was arbitrary,  $f$  is surjective.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

---

### Problem Statement

### Homework 1, Problem 4

Let  $f : A \rightarrow B$  be a function. Prove that the following two statements are equivalent:

- (a) The function  $f$  is surjective.
- (b) For any set  $C$  and for any functions  $g : B \rightarrow C$  and  $h : B \rightarrow C$  such that  $g \circ f = h \circ f$ , we have  $g = h$ .



Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why does surjectivity let you test equality of  $g, h : B \rightarrow C$  after composing with  $f$ ?

---

2. In the forward direction, after fixing  $b \in B$ , what element of  $A$  do you need?

---

3. To prove the reverse direction, what does it mean for  $f$  not to be surjective?

---

4. Why choose  $C = \{0, 1\}$  in the counterexample construction?

---

5. How can  $g \circ f = h \circ f$  hold while  $g \neq h$ ?

---

**Surjective  $\Rightarrow$  cancellation.**

**Assume  $g \circ f = h \circ f$ . Fix  $b \in B$ , choose  $a \in A$  with  $f(a) = b$ :**

---

**Compute  $g(b) = g(f(a)) = (g \circ f)(a) = \cdots = h(b)$ :**

---

**Cancellation  $\Rightarrow$  surjective.**

**Assume  $f$  is not surjective; choose  $b_0 \in B \setminus f(A)$ :**

---

**Set  $C = \{0, 1\}$ , define  $g$  constantly 0, and define  $h$  to differ only at  $b_0$ :**

---

**Show  $g \circ f = h \circ f$  because  $f(a) \neq b_0$  for all  $a$ :**

---

Contradict the cancellation property and conclude  $f$  is surjective:

---

Use this page for the forward direction.

Use this page for the reverse direction.

### Student-Derived Solution (No official solution available)

$\Rightarrow$  Assume that  $f$  is surjective. Let  $C$  be any set, and let  $g, h : B \rightarrow C$  satisfy  $g \circ f = h \circ f$ . We show  $g = h$ . Fix  $b \in B$ . Since  $f$  is surjective, there exists  $a \in A$  such that  $f(a) = b$ . Then

$$g(b) = g(f(a)) = (g \circ f)(a) = (h \circ f)(a) = h(f(a)) = h(b).$$

Since  $b \in B$  was arbitrary,  $g = h$ .

$\Leftarrow$  Assume the cancellation property in (b). Suppose, for contradiction, that  $f$  is not surjective. Then there exists  $b_0 \in B \setminus f(A)$ . Let  $C = \{0, 1\}$ , and define  $g, h : B \rightarrow C$  by

$$g(b) = 0 \quad \text{for all } b \in B,$$

and

$$h(b) = \begin{cases} 1, & b = b_0, \\ 0, & b \neq b_0. \end{cases}$$

Then  $g \neq h$ , since  $g(b_0) = 0$  but  $h(b_0) = 1$ . However, for every  $a \in A$ , we have  $f(a) \neq b_0$ , so

$$(g \circ f)(a) = g(f(a)) = 0 = h(f(a)) = (h \circ f)(a).$$

Thus  $g \circ f = h \circ f$ , and by the assumed property this forces  $g = h$ , a contradiction. Therefore  $f$  is surjective.



What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 5

---

### Problem Statement

### Homework 1, Problem 5

Let  $f : B \rightarrow C$  be a function. Prove that the following two statements are equivalent:

- (a) The function  $f$  is injective.
- (b) For any set  $A$  and for any functions  $g : A \rightarrow B$  and  $h : A \rightarrow B$  such that  $f \circ g = f \circ h$ , we have  $g = h$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why is this a left-cancellation property for injective maps?

---

2. In the forward direction, what equality do we get after evaluating at  $a \in A$ ?

---

3. Where is injectivity used to conclude  $g(a) = h(a)$ ?

---

4. In the reverse direction, why does choosing  $A = \{1\}$  suffice?

---

5. How do the test maps  $g(1) = b_1$ ,  $h(1) = b_2$  force  $b_1 = b_2$ ?

---

**Injective  $\Rightarrow$  cancellation.**

**Assume  $f \circ g = f \circ h$ . Fix  $a \in A$  and write the equality after evaluating at  $a$ :**

---

**Use injectivity of  $f$  to conclude  $g(a) = h(a)$ :**

---

**Since  $a$  was arbitrary, conclude  $g = h$ :**

---

**Cancellation  $\Rightarrow$  injective.**

**Fix  $b_1, b_2 \in B$  with  $f(b_1) = f(b_2)$ :**

---

**Define  $g, h : \{1\} \rightarrow B$  by  $g(1) = b_1$ ,  $h(1) = b_2$ :**

---

Show  $f \circ g = f \circ h$ , apply the property, and conclude  $b_1 = b_2$ :

---

Use this page for both directions.



### Official Solution (Professor)

$(a) \Rightarrow (b)$ : Suppose that  $f$  is injective. We want to show that property (b) is true. Let  $A$  be a set and  $g, h : A \rightarrow B$  be functions such that  $f \circ g = f \circ h$ . This means that for any  $a \in A$ , we have

$$f(g(a)) = f(h(a)).$$

In other words,  $g(a)$  and  $h(a)$  are two points in  $B$  which  $f$  maps to the same point in  $C$ . As  $f$  is injective, we must have

$$g(a) = h(a).$$

This holds for any  $a \in A$ , and so the functions  $g$  and  $h$  are equal.

$(a) \Leftarrow (b)$ : Suppose that  $f$  satisfies property (b). We want to show that  $f$  is injective. Fix  $b_1, b_2 \in B$  such that  $f(b_1) = f(b_2)$ . Consider the functions  $g, h : \{1\} \rightarrow B$  given by

$$g(1) = b_1 \quad \text{and} \quad h(1) = b_2.$$

Then we have  $f \circ g = f \circ h$ , since

$$f(g(1)) = f(b_1) = f(b_2) = f(h(1)).$$

Therefore, by property (b) we must have  $g = h$ , and so

$$b_1 = g(1) = h(1) = b_2.$$

Thus  $b_1 = b_2$ , as desired.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 6

---

### Problem Statement

### Homework 1, Problem 6

Let  $f : A \rightarrow B$  be a function. Suppose that there exists a function  $g : B \rightarrow A$  such that  $(g \circ f)(a) = a$  for all  $a \in A$  and  $(f \circ g)(b) = b$  for all  $b \in B$ . Prove that  $f$  is bijective.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What do the equations  $g \circ f = \text{id}_A$  and  $f \circ g = \text{id}_B$  mean pointwise?

---

2. To prove injectivity, what should we assume about  $f(a_1)$  and  $f(a_2)$ ?

---

3. Why does applying  $g$  to both sides prove  $a_1 = a_2$ ?

---

4. To prove surjectivity, how should an arbitrary  $b \in B$  be written as  $f(a)$ ?

---

5. Why do injective plus surjective imply bijective?

---

**Injectivity.**

**Assume  $f(a_1) = f(a_2)$ . Apply  $g$  and use  $g(f(a)) = a$ :**

---

**Conclude  $a_1 = a_2$ , so  $f$  is injective:**

---

**Surjectivity.**

**Fix  $b \in B$ . Set  $a =$**

---

**Use  $f(g(b)) = b$  to show  $b \in f(A)$ :**

---

**Conclude every  $b \in B$  is hit, and finish:**

---

Use this page for the complete proof.



### Official Solution (Professor)

First, we will show that  $f$  is injective. Suppose that  $f(a_1) = f(a_2)$  for some  $a_1, a_2 \in A$ . Then

$$a_1 = g(f(a_1)) = g(f(a_2)) = a_2.$$

Next, we will show that  $f$  is surjective. Fix  $b \in B$ . Set  $a = g(b)$ . Then  $a \in A$  and

$$f(a) = f(g(b)) = b.$$

Together, we conclude that  $f$  is bijective.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Homework 2

---

## Problem 1

---

### Problem Statement

### Homework 2, Problem 1

Prove that if  $A$  and  $B$  are both countable sets and  $A \cap B = \emptyset$ , then  $A \cup B$  is countable. (Hint: We proved that  $\mathbb{Z}$  is countable in class. Try using a similar idea here.)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does it mean for  $A$  and  $B$  to be countable in terms of bijections with  $\mathbb{N}$ ?

---

2. Why does disjointness make the odd/even coding well-defined?

---

3. Which natural numbers should encode elements of  $A$ , and which should encode elements of  $B$ ?

---

4. How is the inverse map  $\phi : \mathbb{N} \rightarrow A \cup B$  defined on odd and even integers?

---

5. Why do the two composition identities prove bijectivity?

---

Choose bijections  $g : A \rightarrow \mathbb{N}$  and  $h : B \rightarrow \mathbb{N}$ :

---

Define  $f : A \cup B \rightarrow \mathbb{N}$  by odd values on  $A$  and even values on  $B$ :

---

Define  $\phi : \mathbb{N} \rightarrow A \cup B$  separately for odd and even  $n$ :

---

Verify  $f \circ \phi = \text{id}_{\mathbb{N}}$ :

---

Verify  $\phi \circ f = \text{id}_{A \cup B}$ :

---



Apply HW1#6 and conclude  $A \cup B$  is countable:

---

Use this page for the complete proof.

### Official Solution (Professor)

As  $A$  and  $B$  are countable, there exist functions  $g : A \rightarrow \mathbb{N}$  and  $h : B \rightarrow \mathbb{N}$  that are bijective.

Define the function  $f : A \cup B \rightarrow \mathbb{N}$  by

$$f(x) = \begin{cases} 2g(x) - 1, & x \in A, \\ 2h(x), & x \in B. \end{cases}$$

As  $A \cap B = \emptyset$ , these two cases are mutually exclusive and so  $f$  is indeed a function. We want to show that  $f$  is bijective. To this end, consider the function  $\phi : \mathbb{N} \rightarrow A \cup B$  given by

$$\phi(n) = \begin{cases} g^{-1}\left(\frac{n+1}{2}\right), & n \text{ odd}, \\ h^{-1}\left(\frac{n}{2}\right), & n \text{ even}. \end{cases}$$

For  $n \in \mathbb{N}$  odd, note that  $\phi(n) = g^{-1}\left(\frac{n+1}{2}\right) \in A$ , and so

$$(f \circ \phi)(n) = f\left(g^{-1}\left(\frac{n+1}{2}\right)\right) = 2g\left(g^{-1}\left(\frac{n+1}{2}\right)\right) - 1 = n.$$

Similarly, for  $n \in \mathbb{N}$  even,  $\phi(n) = h^{-1}\left(\frac{n}{2}\right) \in B$ , and

$$(f \circ \phi)(n) = f\left(h^{-1}\left(\frac{n}{2}\right)\right) = 2h\left(h^{-1}\left(\frac{n}{2}\right)\right) = n.$$

Together,  $(f \circ \phi)(n) = n$  for all  $n \in \mathbb{N}$ .

On the other hand, for  $x \in A$ ,  $f(x) = 2g(x) - 1$  is odd, and so

$$(\phi \circ f)(x) = \phi(2g(x) - 1) = g^{-1}\left(\frac{2g(x) - 1 + 1}{2}\right) = g^{-1}(g(x)) = x.$$

Likewise, for  $x \in B$ ,  $f(x) = 2h(x)$  is even, and so

$$(\phi \circ f)(x) = \phi(2h(x)) = h^{-1}\left(\frac{2h(x)}{2}\right) = h^{-1}(h(x)) = x.$$

Together,  $(\phi \circ f)(x) = x$  for all  $x \in A \cup B$ . Combining the previous two paragraphs,  $f$  is bijective by HW1#6.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

---

### Problem Statement

### Homework 2, Problem 2

Let  $A_n$  be a countable set for each  $n \in \mathbb{N}$ , and suppose that  $A_m \cap A_n = \emptyset$  whenever  $m \neq n$ . Prove that the union  $A = A_1 \cup A_2 \cup A_3 \cup \cdots$  is countable. (Hint: We proved that  $\mathbb{N} \times \mathbb{N}$  is countable in class. Try using a similar idea here.)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. For each  $A_n$ , what enumeration or bijection should be chosen?

---

2. Why does an element of the union have a unique first coordinate  $n$ ?

---

3. What map  $h : \mathbb{N} \times \mathbb{N} \rightarrow A$  naturally lists all elements?

---

4. How does disjointness prove injectivity of  $h$ ?

---

5. How does countability of  $\mathbb{N} \times \mathbb{N}$  finish the proof?

---



For each  $n$ , choose a bijection  $f_n : \mathbb{N} \rightarrow A_n$ :

---

Define  $h : \mathbb{N} \times \mathbb{N} \rightarrow A$  by  $h(n, k) =$

---

Prove  $h$  is surjective by taking  $x \in A$  and locating  $x \in A_n$ :

---

Prove  $h$  is injective using pairwise disjointness and injectivity of  $f_n$ :

---

Compose with a bijection  $\mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$  and conclude  $A$  is countable:

---

Use this page for the complete proof.

### Student-Derived Solution (No official solution available)

For each  $n \in \mathbb{N}$ , since  $A_n$  is countable, choose a bijection

$$f_n : \mathbb{N} \rightarrow A_n.$$

Define  $h : \mathbb{N} \times \mathbb{N} \rightarrow A$  by

$$h(n, k) = f_n(k).$$

We show that  $h$  is bijective.

Surjectivity. Let  $x \in A$ . Since  $A = \bigcup_{n \in \mathbb{N}} A_n$ , there exists  $n \in \mathbb{N}$  such that  $x \in A_n$ . Since  $f_n$  is surjective, there exists  $k \in \mathbb{N}$  such that  $f_n(k) = x$ . Hence  $h(n, k) = x$ .

Injectivity. Suppose  $h(n, k) = h(m, \ell)$ . Then

$$f_n(k) = f_m(\ell).$$

The left side lies in  $A_n$  and the right side lies in  $A_m$ . If  $n \neq m$ , this contradicts the assumption that  $A_n \cap A_m = \emptyset$ . Hence  $n = m$ . With  $n = m$ , we have  $f_n(k) = f_n(\ell)$ , and since  $f_n$  is injective,  $k = \ell$ . Therefore  $(n, k) = (m, \ell)$ , so  $h$  is injective.

Thus  $h$  is a bijection  $\mathbb{N} \times \mathbb{N} \rightarrow A$ . Since  $\mathbb{N} \times \mathbb{N}$  is countable, there exists a bijection  $\phi : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ . Then  $h \circ \phi : \mathbb{N} \rightarrow A$  is a bijection, so  $A$  is countable.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

---

#### Problem Statement

#### Homework 2, Problem 3

Recall that given two sets  $A$  and  $B$ , we write  $A \sim B$  if and only if there exists a bijective function  $f : A \rightarrow B$ . Prove that  $\sim$  is an equivalence relation.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What are the three properties of an equivalence relation?

---

2. Which bijection proves  $A \sim A$ ?

---

3. Why does a bijection  $A \rightarrow B$  give a bijection  $B \rightarrow A$ ?

---

4. Why is the composition of bijections bijective?

---

5. How do these three facts match reflexivity, symmetry, and transitivity?

---



**Reflexivity:** define the identity map  $f : A \rightarrow A$  and show it is bijective:

---

**Symmetry:** start with a bijection  $g : A \rightarrow B$  and use  $g^{-1} : B \rightarrow A$ :

---

**Transitivity:** compose bijections  $h_1 : A \rightarrow B$  and  $h_2 : B \rightarrow C$ :

---

**State that  $\sim$  is reflexive, symmetric, and transitive:**

---

Use this page for the complete proof.

### Official Solution (Professor)

Reflexivity: Let  $A$  be a set. We want to show that  $A \sim A$ . Consider the function  $f : A \rightarrow A$  given by  $f(a) = a$ . Notice that  $(f \circ f)(a) = a$  for all  $a \in A$ , and so  $f$  is bijective by HW1#6. Therefore  $A \sim A$ .

Symmetry: Suppose that  $A, B$  are sets such that  $A \sim B$ . Then there exists a bijective function  $g : A \rightarrow B$ . Consider the function  $g^{-1} : B \rightarrow A$ . By Lecture 3 we know that  $(g^{-1} \circ g)(a) = a$  for all  $a \in A$  and  $(g \circ g^{-1})(b) = b$  for all  $b \in B$ , and so  $g^{-1}$  is bijective by HW1#6. Therefore  $B \sim A$ .

Transitivity: Suppose that  $A, B, C$  are sets such that  $A \sim B$  and  $B \sim C$ . Then there exist bijective functions  $h_1 : A \rightarrow B$  and  $h_2 : B \rightarrow C$ . By Lecture 3, the composition  $h_2 \circ h_1 : A \rightarrow C$  is also bijective, and so  $A \sim C$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

---

### Problem Statement

### Homework 2, Problem 4

For  $a, b \in \mathbb{Z}$ , define  $a \sim b$  if and only if  $b - a$  is divisible by 2.

- (a) Prove that  $\sim$  is an equivalence relation on  $\mathbb{Z}$ .
- (b) Find the equivalence classes.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does it mean for an integer to be divisible by 2?

---

2. Why is  $a - a = 0$  enough for reflexivity?

---

3. Why does  $b - a = 2k$  imply  $a - b$  is divisible by 2?

---

4. How does adding  $b - a$  and  $c - b$  prove transitivity?

---

5. Why are the equivalence classes exactly the even and odd integers?

---



**Part (a).**

**Reflexive:** show  $a - a$  is divisible by 2:

---

**Symmetric:** assume  $b - a = 2k$ , then write  $a - b =$

---

**Transitive:** assume  $b - a = 2k$  and  $c - b = 2\ell$ , then compute  $c - a$ :

---

**Part (b).**

**Show**  $[0] = \{2k : k \in \mathbb{Z}\}$ :

---

**Show**  $[1] = \{2k + 1 : k \in \mathbb{Z}\}$ :

---

Explain why these are the only two classes:

---

Use this page for part (a).

Use this page for part (b).

### Student-Derived Solution (No official solution available)

(a) We show that  $\sim$  is reflexive, symmetric, and transitive on  $\mathbb{Z}$ .

Reflexive. Let  $a \in \mathbb{Z}$ . Then  $a - a = 0 = 2 \cdot 0$ , so  $a - a$  is divisible by 2. Hence  $a \sim a$ .

Symmetric. Suppose  $a \sim b$ . Then  $b - a = 2k$  for some  $k \in \mathbb{Z}$ . Hence

$$a - b = -(b - a) = -2k = 2(-k),$$

so  $a - b$  is divisible by 2. Thus  $b \sim a$ .

Transitive. Suppose  $a \sim b$  and  $b \sim c$ . Then  $b - a = 2k$  and  $c - b = 2\ell$  for some  $k, \ell \in \mathbb{Z}$ .

Adding,

$$c - a = (c - b) + (b - a) = 2\ell + 2k = 2(k + \ell),$$

so  $c - a$  is divisible by 2. Hence  $a \sim c$ .

Therefore  $\sim$  is an equivalence relation.

(b) The equivalence classes are the parity classes. First,

$$[0] = \{n \in \mathbb{Z} : n - 0 \text{ is divisible by } 2\} = \{2k : k \in \mathbb{Z}\} = 2\mathbb{Z}.$$

Similarly,

$$[1] = \{n \in \mathbb{Z} : n - 1 \text{ is divisible by } 2\} = \{2k + 1 : k \in \mathbb{Z}\} = 2\mathbb{Z} + 1.$$

Every integer is either even or odd, so these are the only equivalence classes.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 5

---

### Problem Statement

### Homework 2, Problem 5

Let  $(F, +, \cdot)$  be a field.

- (a) Prove that for any  $x \in F$ , the function  $f : F \rightarrow F$ ,  $f(y) = x + y$  is bijective.
- (b) Prove that for any  $x \in F \setminus \{0\}$ , the function  $g : F \rightarrow F$ ,  $g(y) = x \cdot y$  is bijective.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What operation undoes adding a fixed  $x$ ?

---

2. Which field axiom gives the additive inverse  $-x$ ?

---

3. What operation undoes multiplying by a fixed nonzero  $x$ ?

---

4. Which field axiom gives  $x^{-1}$ ?

---

5. Why does exhibiting a two-sided inverse function prove bijectivity?

---

**Part (a).**

Define  $h : F \rightarrow F$  by  $h(z) =$

---

Compute  $(f \circ h)(z)$  using associativity, inverse, and identity axioms:

---

Compute  $(h \circ f)(y)$ :

---

Conclude  $f$  is bijective by HW1#6:

---

**Part (b).**

Define  $h : F \rightarrow F$  by  $h(z) =$

---

**Compute  $(g \circ h)(z)$  and  $(h \circ g)(y)$ :**

---

**Conclude  $g$  is bijective:**

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Fix  $x \in F$ . By (A5), there exists an additive inverse  $-x \in F$  for  $x$ , so that

$$x + (-x) = 0 = -x + x.$$

Consider the function  $h : F \rightarrow F$  given by  $h(z) = -x + z$ . Then, for any  $z \in F$ ,

$$(f \circ h)(z) = f(-x + z) = x + (-x + z) = (x + (-x)) + z = 0 + z = z.$$

Similarly, for  $y \in F$ ,

$$(h \circ f)(y) = h(x + y) = -x + (x + y) = (-x + x) + y = 0 + y = y.$$

Therefore  $f$  is bijective by HW1#6.

(b) Fix  $x \in F \setminus \{0\}$ . As  $x \neq 0$ , there exists a multiplicative inverse  $x^{-1} \in F$  for  $x$ , so that

$$x \cdot x^{-1} = 1 = x^{-1} \cdot x.$$

Consider the function  $h : F \rightarrow F$  given by  $h(z) = x^{-1} \cdot z$ . Then, for any  $z \in F$ ,

$$(g \circ h)(z) = g(x^{-1} \cdot z) = x \cdot (x^{-1} \cdot z) = (x \cdot x^{-1}) \cdot z = 1 \cdot z = z.$$

Similarly, for  $y \in F$ ,

$$(h \circ g)(y) = h(x \cdot y) = x^{-1} \cdot (x \cdot y) = (x^{-1} \cdot x) \cdot y = 1 \cdot y = y.$$

Therefore  $g$  is bijective by HW1#6.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---



## Problem 6

### Problem Statement

### Homework 2, Problem 6

Let  $(F, +, \cdot)$  be a field. Suppose that  $F = \{0, 1, a, b\}$  is a set with exactly four elements, where 0 and 1 denote the identities for  $+$  and  $\cdot$  respectively, and  $a, b$  denote the remaining two elements of  $F$ . Fill in the addition and multiplication tables below, and justify your answer.

| $+$ | 0 | 1 | $a$ | $b$ |
|-----|---|---|-----|-----|
| 0   |   |   |     |     |
| 1   |   |   |     |     |
| $a$ |   |   |     |     |
| $b$ |   |   |     |     |

| $\cdot$ | 0 | 1 | $a$ | $b$ |
|---------|---|---|-----|-----|
| 0       |   |   |     |     |
| 1       |   |   |     |     |
| $a$     |   |   |     |     |
| $b$     |   |   |     |     |

(Hint: Using the previous problem, show that each row and column in the addition table contains every element of  $F$  exactly once, like a Sudoku puzzle. Show that the same is true for the nonzero rows and columns of the multiplication table. Note that for each entry, there is only one correct answer.)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. How does Problem 5 imply each addition row/column is a permutation of  $F$ ?

---

2. How does Problem 5 imply each nonzero multiplication row/column is a permutation of  $F \setminus \{0\}$ ?

---

3. Which entries are forced immediately by additive and multiplicative identities?

---

4. Why must  $a \cdot b = 1$  in the multiplication table?

---

5. How does distributivity rule out the wrong addition-table entries?

---

Fill the 0 and 1 rows/columns in the multiplication table:

---

Use the nonzero row/column permutation rule to force  $a \cdot b = 1$ ,  $a^2 = b$ , and  $b^2 = a$ :

---

Fill the 0 row/column in the addition table:

---

Use distributivity to rule out  $1 + a = 0$ , hence force  $1 + a = b$ :

---

Similarly force  $1 + b = a$  and  $a + b = 1$ :

---

**Write the completed tables:**

---

Use this page for the multiplication table.

Use this page for the addition table and justification.



### Official Solution (Professor)

There are many ways to do this problem. Here's one option that works.

Step 1: Start with the multiplication table. We can fill in the 0 and 1 rows and columns using what we know from lecture:

| $\cdot$ | 0 | 1   | $a$ | $b$ |
|---------|---|-----|-----|-----|
| 0       | 0 | 0   | 0   | 0   |
| 1       | 0 | 1   | $a$ | $b$ |
| $a$     | 0 | $a$ |     |     |
| $b$     | 0 | $b$ |     |     |

Step 2: The only possibility for the  $a \cdot b$  entry is 1, since the other three elements already appear in this row and column. This in turn leaves only one option for the remaining entries:

| $\cdot$ | 0 | 1   | $a$ | $b$ |
|---------|---|-----|-----|-----|
| 0       | 0 | 0   | 0   | 0   |
| 1       | 0 | 1   | $a$ | $b$ |
| $a$     | 0 | $a$ | $b$ | 1   |
| $b$     | 0 | $b$ | 1   | $a$ |

Step 3: Now turn to the addition table. We can fill in the 0 row and column using (A4):

| $+$ | 0   | 1 | $a$ | $b$ |
|-----|-----|---|-----|-----|
| 0   | 0   | 1 | $a$ | $b$ |
| 1   | 1   |   |     |     |
| $a$ | $a$ |   |     |     |
| $b$ | $b$ |   |     |     |

Step 4: Consider the entry for  $1 + a$ . There are two elements that we can choose from, 0 and  $b$ , since the other two elements already appear in this row and column. Suppose  $1 + a = 0$ . This forces

$$0 = 0 \cdot b = (1 + a) \cdot b = 1 \cdot b + a \cdot b = b + 1,$$

by distributivity and the multiplication table. But this would mean that 0 appears twice in a row/column, which is a contradiction. Therefore  $1 + a = b$ . Similarly, if  $1 + b = 0$ , then

$$0 = 0 \cdot b = (1 + b) \cdot a = 1 \cdot a + b \cdot a = a + 1,$$

which contradicts the current entry for  $a + 1$ . So  $1 + b = a$ .

Step 5: Consider the entry for  $a + b$ . There are two elements that we can choose from, 0 and 1, since the other two elements already appear in this row and column. Suppose  $a + b = 0$ . This forces

$$0 = 0 \cdot b = (a + b) \cdot b = a \cdot b + b \cdot b = 1 + a,$$

by distributivity and the multiplication table. This contradicts the current entry for  $1 + a$ , so  $a + b = 1$ . This leaves the completed addition table:

| $+$ | $0$ | $1$ | $a$ | $b$ |
|-----|-----|-----|-----|-----|
| $0$ | $0$ | $1$ | $a$ | $b$ |
| $1$ | $1$ | $0$ | $b$ | $a$ |
| $a$ | $a$ | $b$ | $0$ | $1$ |
| $b$ | $b$ | $a$ | $1$ | $0$ |

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Homework 3

---

## Problem 1

---

### Problem Statement

### Homework 3, Problem 1

Let  $(F, +, \cdot)$  be a field.

- (a) Prove that the multiplicative identity is unique: If  $b \in F$  satisfies  $b \cdot a = a \cdot b = a$  for all  $a \in F$ , then  $b = 1$ .
- (b) Prove that multiplicative inverses are unique: If  $a \in F \setminus \{0\}$  and  $b \in F$  satisfy  $b \cdot a = a \cdot b = 1$ , then  $b = a^{-1}$ .
- (c) Prove that if  $a \cdot b = a \cdot c$  and  $a \neq 0$ , then  $b = c$ .
- (d) Prove that if  $a, b \in F$  then  $-a + (-b) = -(a + b)$ .
- (e) Prove that if  $a, b \in F$  and  $a, b \neq 0$ , then  $a^{-1} \cdot b^{-1} = (ab)^{-1}$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. In part (a), what two ways can you compute  $b \cdot 1$ ?

---

2. In part (b), why is multiplying by  $a^{-1}$  legitimate?

---

3. In part (c), what field axioms justify cancellation?

---

4. In part (d), how do you prove an element is the additive inverse of  $a + b$ ?

---

5. In part (e), what product must be checked to show  $a^{-1}b^{-1}$  is  $(ab)^{-1}$ ?

---



Parts (a)–(c).

For (a), use the usual identity with  $a = b$  and the alternative identity with  $a = 1$ :

---

For (b), start from  $b \cdot a = 1$  and multiply on the right by  $a^{-1}$ :

---

For (c), multiply  $a \cdot b = a \cdot c$  on the left by  $a^{-1}$ :

---

Parts (d)–(e).

For (d), compute  $((-a) + (-b)) + (a + b)$  and rearrange using commutativity/associativity:

---

For (e), compute  $(a^{-1}b^{-1})(ab)$  and use commutativity/associativity:

---

Invoke uniqueness from earlier parts where needed:

---

Use this page for parts (a)–(c).

Use this page for parts (d)–(e).

### Official Solution (Professor)

(a) We know that  $1, b \in F$  satisfy:

- $a \cdot 1 = 1 \cdot a = a$  for all  $a \in F$ ;
- $a \cdot b = b \cdot a = a$  for all  $a \in F$ .

Taking  $a = b$  in the first bullet point, we see that

$$b \cdot 1 = b.$$

Taking  $a = 1$  in the second bullet point, we obtain

$$b \cdot 1 = 1.$$

Comparing the previous two equalities, we conclude  $b = 1$ .

(b) As  $a \neq 0$ , by (M5) there exists  $a^{-1} \in F$  so that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

Multiplying  $b \cdot a = 1$  by  $a^{-1}$  on the right, we obtain

$$(b \cdot a) \cdot a^{-1} = 1 \cdot a^{-1} \implies b \cdot (a \cdot a^{-1}) = a^{-1} \implies b \cdot 1 = a^{-1} \implies b = a^{-1}.$$

(c) As  $a \neq 0$ , by (M5) there exists  $a^{-1} \in F$  so that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

Multiplying  $a \cdot b = a \cdot c$  by  $a^{-1}$  on the left, we obtain

$$a^{-1} \cdot (a \cdot b) = a^{-1} \cdot (a \cdot c) \implies (a^{-1} \cdot a) \cdot b = (a^{-1} \cdot a) \cdot c \implies 1 \cdot b = 1 \cdot c \implies b = c.$$

(d) Fix  $a, b \in F$ . Then

$$\begin{aligned} ((-a) + (-b)) + (a + b) &= ((-a) + (-b)) + (b + a) \\ &= -a + ((-b) + b) + a = -a + 0 + a = -a + a = 0. \end{aligned}$$

In other words,  $(-a) + (-b)$  is the additive inverse of  $a + b$ . As additive inverses are unique, we conclude

$$-a + (-b) = -(a + b).$$

(e) Fix  $a, b \neq 0$ . Then

$$(a^{-1} \cdot b^{-1}) \cdot (ab) = (a^{-1} \cdot b^{-1}) \cdot (ba) = a^{-1} \cdot (b^{-1}b) \cdot a = a^{-1} \cdot 1 \cdot a = a^{-1} \cdot a = 1.$$

In other words,  $a^{-1} \cdot b^{-1}$  is the multiplicative inverse of  $ab$ . As multiplicative inverses are unique by part (b), we conclude

$$a^{-1} \cdot b^{-1} = (ab)^{-1}.$$

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

---

### Problem Statement

### Homework 3, Problem 2

Given two points  $(x_1, y_1)$  and  $(x_2, y_2)$  in  $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$ , we write

$$(x_1, y_1) < (x_2, y_2) \iff x_1 < x_2 \text{ or } (x_1 = x_2 \text{ and } y_1 < y_2).$$

- (a) Prove that  $<$  is an order relation on  $\mathbb{R}^2$ .
- (b) Prove that  $(\mathbb{R}^2, <)$  does not satisfy the least upper bound property. (Hint: Consider the set of all points on the  $y$ -axis.)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What are trichotomy and transitivity for an order relation?

---

2. Why does the first coordinate decide most comparisons in lexicographic order?

---

3. What set on the  $y$ -axis should be used in part (b)?

---

4. Why is that set bounded above by  $(1, 0)$ ?

---

5. Why must any upper bound have positive first coordinate, and why can it be lowered?

---

Part (a).

For trichotomy, split into  $x < x'$ ,  $x > x'$ , and  $x = x'$ :

---

For transitivity, split into the cases from the definition of  $<$ :

---

Conclude  $<$  is an order relation:

---

Part (b).

Let  $Y = \{(0, y) : y \in \mathbb{R}\}$ . Show  $Y$  is nonempty and bounded above:

---

Assume  $(a, b)$  is a least upper bound. Prove  $a > 0$ :

---

Show  $(a/2, b)$  is also an upper bound and is strictly smaller:

---

State the contradiction to leastness:

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) By definition of an order relation, there are two properties to check.

Trichotomy: Fix  $(x, y), (x', y') \in \mathbb{R}^2$ . We prove that in every possible case, exactly one of the statements  $<$ ,  $=$ , or  $>$  is true.

- Case:  $x < x'$ . Then  $(x, y) < (x', y')$ .
- Case:  $x > x'$ . Then  $(x, y) > (x', y')$ .
- Case:  $x = x'$ . If  $y < y'$ , then  $(x, y) < (x', y')$ . If  $y > y'$ , then  $(x, y) > (x', y')$ . If  $y = y'$ , then  $(x, y) = (x', y')$ .

Transitivity: Suppose  $(x_1, y_1) < (x_2, y_2)$  and  $(x_2, y_2) < (x_3, y_3)$ . By definition of  $<$ , there are four possible cases.

- If  $x_1 < x_2$  and  $x_2 < x_3$ , then  $x_1 < x_3$ .
- If  $x_1 < x_2$ ,  $x_2 = x_3$ , and  $y_2 < y_3$ , then  $x_1 < x_3$ .
- If  $x_1 = x_2$ ,  $y_1 < y_2$ , and  $x_2 < x_3$ , then  $x_1 < x_3$ .
- If  $x_1 = x_2$ ,  $y_1 < y_2$ ,  $x_2 = x_3$ , and  $y_2 < y_3$ , then  $x_1 = x_3$  and  $y_1 < y_3$ .

So, in all cases,  $(x_1, y_1) < (x_3, y_3)$ .

(b) Suppose toward a contradiction that  $(\mathbb{R}^2, <)$  satisfies the least upper bound property. Set

$$Y = \{(0, y) : y \in \mathbb{R}\}.$$

Then  $Y$  is nonempty, since  $(0, 0) \in Y$ , and  $Y$  is bounded above by  $(1, 0)$ . Therefore there exists a least upper bound  $(a, b) \in \mathbb{R}^2$  for  $Y$ .

Claim:  $a > 0$ . If  $a < 0$ , then  $(0, 0) \in Y$  would satisfy  $(0, 0) > (a, b)$ , contradicting that  $(a, b)$  is an upper bound for  $Y$ . If  $a = 0$ , then  $(0, b + 1) \in Y$  and  $(0, b + 1) > (a, b)$ , again contradicting that  $(a, b)$  is an upper bound. Thus  $a > 0$ .

As  $a > 0$ ,  $(a/2, b)$  is also an upper bound for  $Y$ , since

$$0 < a/2 \implies (0, y) < (a/2, b) \text{ for all } y \in \mathbb{R}.$$

But  $a/2 < a$ , so  $(a/2, b) < (a, b)$ . This contradicts that  $(a, b)$  was the least upper bound.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---



### Problem 3

#### Problem Statement

#### Homework 3, Problem 3

Recall that the set of complex numbers  $\mathbb{C}$  consists of all numbers of the form  $a + ib$  for  $a, b \in \mathbb{R}$ . We can add and multiply complex numbers as follows:

$$(a + ib) + (c + id) = (a + c) + i(b + d)$$

and

$$(a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc).$$

With these operations, it turns out  $(\mathbb{C}, +, \cdot)$  is a field (but you do not need to prove this).

- (a) The definition of a field says that for any  $a + ib \neq 0 + i0$ , there exists  $c + id \in \mathbb{C}$  so that  $(a + ib) \cdot (c + id) = 1 + i0$ . What are  $c, d \in \mathbb{R}$  in terms of  $a, b \in \mathbb{R}$ ? Prove your answer.
- (b) Prove that there is no order relation  $<$  on  $\mathbb{C}$  that makes  $(\mathbb{C}, +, \cdot, <)$  an ordered field. (Hint: Can  $i = 0 + i1$  be positive? Negative? Zero?)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What equations result from  $(a + ib)(c + id) = 1 + i0$ ?

---

2. Why is  $a^2 + b^2 \neq 0$  when  $a + ib \neq 0 + i0$ ?

---

3. What are the formulas for  $c$  and  $d$ ?

---

4. In an ordered field, why is  $x^2 > 0$  for every nonzero  $x$ ?

---

5. Why does  $i^2 = -1$  contradict ordered-field behavior?

---

Part (a).

Set  $c = \frac{a}{a^2+b^2}$ ,  $d = -\frac{b}{a^2+b^2}$ :

---

Verify  $ac - bd = 1$ :

---

Verify  $ad + bc = 0$ :

---

Part (b).

Assume  $\mathbb{C}$  is an ordered field. Since  $i \neq 0$ , use the ordered-field fact that  $i^2 > 0$ :

---

Use  $i^2 = -1$  and  $1 > 0$  to get a contradiction:

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Set

$$c = \frac{a}{a^2 + b^2}, \quad d = -\frac{b}{a^2 + b^2}.$$

Then

$$ac - bd = \frac{a^2}{a^2 + b^2} + \frac{b^2}{a^2 + b^2} = 1,$$

and

$$ad + bc = -\frac{ab}{a^2 + b^2} + \frac{ab}{a^2 + b^2} = 0.$$

Therefore  $(a + ib) \cdot (c + id) = 1$ .

(b) Suppose toward a contradiction that  $(\mathbb{C}, +, \cdot, <)$  is an ordered field. As  $i \neq 0$ , a proposition from lecture gives

$$0 < i^2 = -1.$$

On the other hand, we also proved in lecture that

$$1 > 0, \quad \text{and so} \quad -1 < 0.$$

Altogether we have both  $-1 > 0$  and  $-1 < 0$ , which is a contradiction.



What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

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### Problem Statement

### Homework 3, Problem 4

Let  $A \subseteq \mathbb{R}$  be a nonempty and bounded set, let  $c \in \mathbb{R}$  be a number, and define

$$B = \{c \cdot a : a \in A\}.$$

- (a) Prove that if  $c > 0$ , then  $\sup B = c \cdot \sup A$ .
- (b) Prove that if  $c < 0$ , then  $\sup B = c \cdot \inf A$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What two facts prove that a candidate is a supremum?

---

2. Why does multiplying by  $c > 0$  preserve inequalities?

---

3. Why does multiplying by  $c < 0$  reverse inequalities?

---

4. In part (a), how does an upper bound  $L$  for  $B$  produce an upper bound  $L/c$  for  $A$ ?

---

5. In part (b), how does an upper bound  $L$  for  $B$  produce a lower bound  $L/c$  for  $A$ ?

---

**Part (a),  $c > 0$ .**

**Show  $c \sup A$  is an upper bound for  $B$ :**

---

**Let  $L$  be any upper bound for  $B$ . From  $ca \leq L$ , prove  $a \leq L/c$ :**

---

**Use  $\sup A \leq L/c$  to conclude  $c \sup A \leq L$ :**

---

**Part (b),  $c < 0$ .**

**Show  $c \inf A$  is an upper bound for  $B$ :**

---

**Let  $L$  be any upper bound for  $B$ . From  $ca \leq L$ , prove  $a \geq L/c$ :**

---

Use  $L/c \leq \inf A$ , then multiply by  $c < 0$ :

---

Use this page for part (a).



Use this page for part (b).

### Official Solution (Professor)

(a) As  $A$  is nonempty and bounded,  $\sup A \in \mathbb{R}$  exists.

Claim:  $c \sup A$  is an upper bound for  $B$ . As  $\sup A$  is an upper bound for  $A$ , we know that

$$a \leq \sup A \quad \text{for all } a \in A,$$

and so

$$ca \leq c \sup A \quad \text{for all } a \in A.$$

Therefore  $c \sup A$  is an upper bound for  $B$ .

Claim: If  $L$  is an upper bound for  $B$ , then  $L \geq c \sup A$ . As  $L$  is an upper bound for  $B$ , we have

$$ca \leq L \quad \text{for all } a \in A,$$

and so

$$a \leq \frac{L}{c} \quad \text{for all } a \in A.$$

In other words,  $\frac{L}{c}$  is an upper bound for  $A$ . As  $\sup A$  is the least upper bound,  $\sup A \leq \frac{L}{c}$ , and therefore  $c \sup A \leq L$ . Collecting the previous two claims,  $\sup B = c \sup A$ .

(b) As  $A$  is nonempty and bounded,  $\inf A \in \mathbb{R}$  exists.

Claim:  $c \inf A$  is an upper bound for  $B$ . As  $\inf A$  is a lower bound for  $A$ , we know that

$$a \geq \inf A \quad \text{for all } a \in A,$$

and so

$$ca \leq c \inf A \quad \text{for all } a \in A.$$

Therefore  $c \inf A$  is an upper bound for  $B$ .

Claim: If  $L$  is an upper bound for  $B$ , then  $L \geq c \inf A$ . As  $L$  is an upper bound for  $B$ , we have

$$ca \leq L \quad \text{for all } a \in A,$$

and so

$$a \geq \frac{L}{c} \quad \text{for all } a \in A.$$

Thus  $\frac{L}{c}$  is a lower bound for  $A$ . As  $\inf A$  is the greatest lower bound,  $\inf A \geq \frac{L}{c}$ , and therefore  $c \inf A \leq L$ . Collecting the previous two claims,  $\sup B = c \inf A$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 5

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### Problem Statement

### Homework 3, Problem 5

Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded sets, and define

$$S = \{a + b : a \in A \text{ and } b \in B\}.$$

Prove that  $\sup S = \sup A + \sup B$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why do  $\sup A$  and  $\sup B$  exist?

---

2. Why is  $\sup A + \sup B$  an upper bound for  $S$ ?

---

3. How does the proof by contradiction use  $L < \sup A + \sup B$ ?

---

4. Why can you find  $a_0 \in A$  with  $a_0 > \sup A - \varepsilon$ ?

---

5. How do the choices of  $a_0, b_0$  contradict  $L$  being an upper bound?

---

Let  $\alpha = \sup A$  and  $\beta = \sup B$ :

---

Show  $\alpha + \beta$  is an upper bound for  $S$ :

---

Let  $L$  be an upper bound and suppose  $L < \alpha + \beta$ . Set  $\varepsilon =$

---

Use non-leastness of  $\alpha - \varepsilon$  and  $\beta - \varepsilon$  to choose  $a_0, b_0$ :

---

Show  $a_0 + b_0 > L$ , contradiction:

---

Conclude  $\sup S = \alpha + \beta$ :

---



Use this page for the complete proof.

### Official Solution (Professor)

As  $A, B \subseteq \mathbb{R}$  are nonempty and bounded,  $\sup A$  and  $\sup B$  exist.

Claim:  $\sup A + \sup B$  is an upper bound for  $S$ . As  $\sup A$  is an upper bound for  $A$ , we know

$$a \leq \sup A \quad \text{for all } a \in A.$$

Similarly,  $\sup B$  is an upper bound for  $B$ , and so

$$b \leq \sup B \quad \text{for all } b \in B.$$

Adding these two inequalities, we conclude that

$$a + b \leq \sup A + \sup B \quad \text{for all } a \in A, b \in B.$$

Therefore  $\sup A + \sup B$  is an upper bound for  $S$ .

Claim: If  $L$  is an upper bound for  $S$ , then  $L \geq \sup A + \sup B$ . Suppose toward a contradiction that  $L$  is an upper bound for  $S$  but  $L < \sup A + \sup B$ . Set

$$\varepsilon = \frac{1}{2}(\sup A + \sup B - L) > 0.$$

As  $\sup A - \varepsilon < \sup A$ , then  $\sup A - \varepsilon$  cannot be an upper bound for  $A$ , and so there exists  $a_0 \in A$  such that

$$a_0 > \sup A - \varepsilon.$$

Similarly, there exists  $b_0 \in B$  such that

$$b_0 > \sup B - \varepsilon.$$

Adding these inequalities and using the definition of  $\varepsilon$ ,

$$a_0 + b_0 > \sup A + \sup B - 2\varepsilon = L.$$

This contradicts that  $L$  is an upper bound for  $S$ , so  $L \geq \sup A + \sup B$ . Therefore  $\sup S = \sup A + \sup B$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 6

---

### Problem Statement

### Homework 3, Problem 6

Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded sets. Prove or disprove the following statements:

- (a) If  $a \leq b$  for all  $a \in A$  and for all  $b \in B$ , then  $\sup A \leq \inf B$ .
- (b) If  $a < b$  for all  $a \in A$  and for all  $b \in B$ , then  $\sup A < \inf B$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. In part (a), why is each  $b \in B$  an upper bound for  $A$ ?

---

2. Why does that imply  $\sup A \leq b$  for all  $b \in B$ ?

---

3. Why is  $\sup A$  then a lower bound for  $B$ ?

---

4. In part (b), why might strict pointwise inequalities fail to create a strict gap between  $\sup$  and  $\inf$ ?

---

5. What counterexample has  $\sup A = \inf B$  but  $a < b$  for all choices?

---

**Part (a).**

**Fix  $b \in B$ . Use the hypothesis to show  $b$  is an upper bound for  $A$ :**

---

**Conclude  $\sup A \leq b$  for all  $b \in B$ :**

---

**Use the definition of  $\inf B$  to get  $\sup A \leq \inf B$ :**

---

**Part (b).**

**Give the counterexample  $A = \{1 - \frac{1}{n} : n \in \mathbb{N}\}$ ,  $B = \{1\}$ :**

---

**Verify nonempty and bounded:**

---



**Verify  $a < b$  for all  $a \in A, b \in B$ , but  $\sup A = \inf B$ :**

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Prove: As  $A, B \subseteq \mathbb{R}$  are nonempty and bounded, both  $\sup A, \inf B \in \mathbb{R}$  exist. Let  $b \in B$ . As  $a \leq b$  for all  $a \in A$ ,  $b$  is an upper bound for  $A$ , and so

$$\sup A \leq b.$$

This is true for all  $b \in B$ . Therefore  $\sup A$  is a lower bound for  $B$ , and so

$$\sup A \leq \inf B.$$

(b) Disprove: Consider the sets

$$A = \left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\}, \quad B = \{1\}.$$

Note that  $\inf B = \min B = 1$ , and recall from lecture that  $\sup A = 1$ .

Altogether:

- $A, B$  are nonempty.
- $A, B$  are bounded: 0 is a lower bound and 1 is an upper bound for both  $A$  and  $B$ .
- $a < b$  for all  $a \in A$  and  $b \in B$ , since  $1 - \frac{1}{n} < 1$  for all  $n \geq 1$ .
- $\sup A \not< \inf B$ , since  $\sup A = 1 = \inf B$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Homework 4

---

## Problem 1

---

### Problem Statement

### Homework 4, Problem 1

Let  $\{a_n\}_{n \geq 1}$  be a sequence.

- (a) Prove that if  $\{a_n\}_{n \geq 1}$  converges to  $a$ , then the sequence  $\{|a_n|\}_{n \geq 1}$  converges to  $|a|$ . (Hint:  $||x| - |y|| \leq |x - y|$  for any  $x, y \in \mathbb{R}$ .)
- (b) Provide an example where  $\{|a_n|\}_{n \geq 1}$  converges, but  $\{a_n\}_{n \geq 1}$  diverges.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does  $a_n \rightarrow a$  mean in  $\varepsilon$ - $N$  language?

---

2. How does the reverse triangle inequality compare  $\left| |a_n| - |a| \right|$  with  $|a_n - a|$ ?

---

3. Why does the same  $N$  work for  $|a_n| \rightarrow |a|$ ?

---

4. What is a simple sign-oscillating sequence whose absolute values are constant?

---

5. Why does that sequence fail to converge?

---

**Part (a).**

**Fix  $\varepsilon > 0$ . Since  $a_n \rightarrow a$ , choose  $N$  with:**

---

**Apply  $\left||x| - |y|\right| \leq |x - y|$  with  $x = a_n$ ,  $y = a$ :**

---

**Conclude  $|a_n| \rightarrow |a|$ :**

---

**Part (b).**

**Choose  $a_n = (-1)^n$ :**

---

**Show  $|a_n| = 1$  converges, while  $a_n$  diverges:**

---

Use this page for parts (a) and (b).

### Official Solution (Professor)

(a) Fix  $\varepsilon > 0$ . As  $\lim_{n \rightarrow \infty} a_n = a$ , there exists  $N \in \mathbb{N}$  so that

$$|a_n - a| < \varepsilon \quad \text{for all } n \geq N.$$

By the reverse triangle inequality, we have

$$||a_n| - |a|| \leq |a_n - a| < \varepsilon \quad \text{for all } n \geq N.$$

So  $\lim_{n \rightarrow \infty} |a_n| = |a|$ .

(b) By Lecture 10, the sequence  $\{a_n\}_{n \geq 1}$  where  $a_n = (-1)^n$  diverges. On the other hand,  $|a_n| = 1$  for all  $n \geq 1$ , and so  $\{|a_n|\}_{n \geq 1}$  converges to 1.

What was the key definition?:

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What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

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### Problem Statement

### Homework 4, Problem 2

(Squeeze theorem). Prove that if  $\{a_n\}_{n \geq 1}$ ,  $\{b_n\}_{n \geq 1}$ , and  $\{c_n\}_{n \geq 1}$  are sequences such that

$$a_n \leq b_n \leq c_n \quad \text{for all } n \geq 1$$

and

$$\lim_{n \rightarrow \infty} a_n = L = \lim_{n \rightarrow \infty} c_n,$$

then  $\{b_n\}_{n \geq 1}$  converges and  $\lim_{n \rightarrow \infty} b_n = L$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does convergence of  $a_n$  to  $L$  give for a fixed  $\varepsilon > 0$ ?

---

2. What does convergence of  $c_n$  to  $L$  give?

---

3. Why should  $N = \max\{N_1, N_2\}$  be used?

---

4. How do  $a_n \leq b_n \leq c_n$  and the two convergence estimates trap  $b_n$ ?

---

5. Why is  $L - \varepsilon < b_n < L + \varepsilon$  equivalent to  $|b_n - L| < \varepsilon$ ?

---

**Fix  $\varepsilon > 0$ . Choose  $N_1$  so  $|a_n - L| < \varepsilon$  for  $n \geq N_1$ :**

---

**Choose  $N_2$  so  $|c_n - L| < \varepsilon$  for  $n \geq N_2$ :**

---

**Let  $N = \max\{N_1, N_2\}$ , then for  $n \geq N$  write the full chain:**

---

**Conclude  $|b_n - L| < \varepsilon$ :**

---

**State  $b_n \rightarrow L$ :**

---

Use this page for the complete proof.

### Official Solution (Professor)

Fix  $\varepsilon > 0$ . As  $\lim_{n \rightarrow \infty} a_n = L$ , there exists  $N_1 \in \mathbb{N}$  so that

$$|a_n - L| < \varepsilon \quad \text{for all } n \geq N_1.$$

Similarly, as  $\lim_{n \rightarrow \infty} c_n = L$ , there exists  $N_2 \in \mathbb{N}$  so that

$$|c_n - L| < \varepsilon \quad \text{for all } n \geq N_2.$$

Set  $N = \max\{N_1, N_2\}$ . Then, for  $n \geq N$ ,

$$b_n - L \leq c_n - L < \varepsilon \quad \text{and} \quad b_n - L \geq a_n - L > -\varepsilon.$$

Together, we conclude

$$|b_n - L| < \varepsilon \quad \text{for all } n \geq N.$$

So  $\lim_{n \rightarrow \infty} b_n = L$ , as desired.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

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#### Problem Statement

#### Homework 4, Problem 3

Suppose that the sequence  $\{a_n\}_{n \geq 1}$  converges.

- (a) Prove that if  $a_n \geq m$  for all but finitely many  $n$ , then  $\lim_{n \rightarrow \infty} a_n \geq m$ .
- (b) Prove that if  $a_n \leq M$  for all but finitely many  $n$ , then  $\lim_{n \rightarrow \infty} a_n \leq M$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does "for all but finitely many  $n$ " mean in terms of an  $N_1$ ?

---

2. Why is it useful to evaluate the convergent sequence at  $N = \max\{N_1, N_2\}$ ?

---

3. In part (a), how does  $a - a_N$  get controlled by  $\varepsilon$ ?

---

4. Why does  $a - m > -\varepsilon$  for all  $\varepsilon > 0$  imply  $a \geq m$ ?

---

5. How is part (b) the symmetric argument?

---

**Part (a).**

**Let  $a = \lim a_n$ . Choose  $N_1$  so  $a_n \geq m$  for all  $n \geq N_1$ :**

---

**For  $\varepsilon > 0$ , choose  $N_2$  so  $|a_n - a| < \varepsilon$  for all  $n \geq N_2$ :**

---

**Set  $N = \max\{N_1, N_2\}$  and show  $a - m = (a - a_N) + (a_N - m) > -\varepsilon$ :**

---

**Conclude  $a \geq m$ :**

---

**Part (b).**

**Repeat with  $a_n \leq M$  to get  $a - M < \varepsilon$  for all  $\varepsilon > 0$ :**

---

Use this page for both parts.

### Official Solution (Professor)

(a) Set  $a = \lim_{n \rightarrow \infty} a_n$ . Let  $N_1 \in \mathbb{N}$  be larger than the largest index  $n$  for which  $a_n \not\geq m$ . Then

$$a_n \geq m \quad \text{for all } n \geq N_1.$$

Fix  $\varepsilon > 0$ . As  $\lim_{n \rightarrow \infty} a_n = a$ , there exists  $N_2 \in \mathbb{N}$  so that

$$|a_n - a| < \varepsilon \quad \text{for all } n \geq N_2.$$

Set  $N = \max\{N_1, N_2\}$ . Then

$$a - m = (a - a_N) + (a_N - m) > -\varepsilon + 0.$$

As  $\varepsilon > 0$  was arbitrary, we must have  $a - m \geq 0$ . Otherwise, picking  $\varepsilon = m - a > 0$  would yield a contradiction.

(b) Set  $a = \lim_{n \rightarrow \infty} a_n$ . Let  $N_1 \in \mathbb{N}$  be larger than the largest index  $n$  for which  $a_n \not\leq M$ . Then

$$a_n \leq M \quad \text{for all } n \geq N_1.$$

Fix  $\varepsilon > 0$ . As  $\lim_{n \rightarrow \infty} a_n = a$ , there exists  $N_2 \in \mathbb{N}$  so that

$$|a_n - a| < \varepsilon \quad \text{for all } n \geq N_2.$$

Set  $N = \max\{N_1, N_2\}$ . Then

$$a - M = (a - a_N) + (a_N - M) < \varepsilon + 0.$$

As  $\varepsilon > 0$  was arbitrary, we must have  $a - M \leq 0$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

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### Problem Statement

### Homework 4, Problem 4

Suppose that the sequence  $\{a_n\}_{n \geq 1}$  converges. Prove or disprove the following statements:

- (a) If  $\lim_{n \rightarrow \infty} a_n \geq m$ , then there exists  $N \in \mathbb{N}$  so that  $a_n \geq m$  for all  $n \geq N$ .
- (b) If  $\lim_{n \rightarrow \infty} a_n > m$ , then there exists  $N \in \mathbb{N}$  so that  $a_n > m$  for all  $n \geq N$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why is the weak inequality in part (a) suspicious?

---

2. What counterexample converges to 0 from below?

---

3. Why does that counterexample never satisfy  $a_n \geq 0$ ?

---

4. In part (b), why does  $\lim a_n > m$  give a positive gap?

---

5. What choice of  $\varepsilon$  keeps  $a_n$  eventually above  $m$ ?

---

**Part (a).**

**Disprove using  $a_n = -1/n$  and  $m = 0$ :**

---

**Verify  $\lim a_n = 0 \geq 0$  but  $a_n < 0$  for every  $n$ :**

---

**Part (b).**

**Let  $a = \lim a_n$ , assume  $a > m$ , and set  $\varepsilon = a - m$ :**

---

**Use convergence to get  $|a_n - a| < \varepsilon$  eventually:**

---

**Show  $a_n - m = (a_n - a) + (a - m) > 0$ :**

---

Use this page for both parts.

### Official Solution (Professor)

(a) Disprove. Consider the sequence  $\{a_n\}_{n \geq 1}$  with

$$a_n = -\frac{1}{n}.$$

From Lecture 10,  $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ , and so  $a = \lim_{n \rightarrow \infty} a_n = 0$  by the limit law for multiplication by a constant. So  $m = 0$  satisfies

$$a = 0 \geq 0 = m.$$

On the other hand,

$$a_n = -\frac{1}{n} \not\geq 0 = m$$

for all  $n \in \mathbb{N}$ , and so no such  $N$  can exist.

(b) Prove. Suppose  $a > m$  and set

$$\varepsilon = a - m.$$

Then  $\varepsilon > 0$ , and so there exists  $N \in \mathbb{N}$  such that

$$|a_n - a| < \varepsilon \quad \text{for all } n \geq N.$$

Then, for  $n \geq N$ ,

$$a_n - m = (a_n - a) + (a - m) > -\varepsilon + (a - m) = 0.$$

Therefore  $a_n > m$  for all  $n \geq N$ , as desired.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

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## Problem 5

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### Problem Statement

### Homework 4, Problem 5

Prove that if  $\{a_n\}_{n \geq 1}$  is a sequence of positive numbers that converges to  $a > 0$ , then the sequence  $\{\sqrt{a_n}\}_{n \geq 1}$  converges to  $\sqrt{a}$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What algebraic identity rationalizes  $\sqrt{a_n} - \sqrt{a}$ ?

---

2. Why is the denominator  $\sqrt{a_n} + \sqrt{a}$  bounded below by  $\sqrt{a}$ ?

---

3. What tolerance for  $|a_n - a|$  should be used to make  $|\sqrt{a_n} - \sqrt{a}| < \varepsilon$ ?

---

4. Where is the assumption  $a > 0$  used?

---

5. Where is the assumption  $a_n > 0$  used?

---

**Fix  $\varepsilon > 0$ . Since  $a_n \rightarrow a$ , choose  $N$  so  $|a_n - a| < \sqrt{a} \varepsilon$ :**

---

**For  $n \geq N$ , write  $|\sqrt{a_n} - \sqrt{a}| = \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}}$ :**

---

**Use  $\sqrt{a_n} + \sqrt{a} \geq \sqrt{a}$ :**

---

**Conclude  $|\sqrt{a_n} - \sqrt{a}| < \varepsilon$ :**

---

Use this page for the complete proof.

### Official Solution (Professor)

Fix  $\varepsilon > 0$ . As  $\lim_{n \rightarrow \infty} a_n = a$  and  $a > 0$ , there exists  $N \in \mathbb{N}$  so that

$$|a_n - a| < \sqrt{a} \varepsilon \quad \text{for all } n \geq N.$$

Then, for  $n \geq N$ ,

$$|\sqrt{a_n} - \sqrt{a}| = \frac{|(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a})|}{\sqrt{a_n} + \sqrt{a}} = \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}} \leq \frac{|a_n - a|}{\sqrt{a}} < \frac{\sqrt{a} \varepsilon}{\sqrt{a}} = \varepsilon.$$

In the inequality above, we used  $\sqrt{a_n} + \sqrt{a} \geq \sqrt{a}$  since  $a_n > 0$ . Therefore  $\lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{a}$ , as desired.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 6

---

### Problem Statement

### Homework 4, Problem 6

Consider the following sequence:

$$a_1 = \sqrt{2} \quad \text{and} \quad a_{n+1} = \sqrt{2 + a_n} \quad \text{for all } n \geq 1.$$

- (a) Prove that the sequence  $\{a_n\}_{n \geq 1}$  is bounded.
- (b) Prove that the sequence is increasing.
- (c) Conclude that the sequence converges, and find its limit.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What upper and lower bounds should be proved by induction?

---

2. Why does  $a_n < 2$  imply  $a_{n+1} < 2$ ?

---

3. Why is comparing  $a_n \leq a_{n+1}$  equivalent to comparing squares?

---

4. How does  $(a_n - 2)(a_n + 1) < 0$  prove monotonicity?

---

5. After convergence, what equation does the limit satisfy?

---

**Part (a).**

**Prove  $\sqrt{2} \leq a_n < 2$  by induction: base case:**

---

**Inductive step: from  $\sqrt{2} \leq a_n < 2$ , show  $\sqrt{2} \leq a_{n+1} < 2$ :**

---

**Part (b).**

**Use  $a_n > 0$  and compare  $a_n \leq \sqrt{2 + a_n}$  by squaring:**

---

**Factor  $a_n^2 - a_n - 2 = (a_n - 2)(a_n + 1)$ :**

---

**Part (c).**

**Invoke monotone convergence. Let  $a = \lim a_n$ :**

---

Use Problem 5 and limit laws to pass to the limit in  $a_{n+1} = \sqrt{2 + a_n}$ :

---

Solve  $a = \sqrt{2 + a}$  and choose the valid root:

---

Use this page for parts (a) and (b).

Use this page for part (c).

### Official Solution (Professor)

(a) We will prove  $\sqrt{2} \leq a_n < 2$  for all  $n \in \mathbb{N}$  by induction.

Base case: For  $n = 1$ , the statement  $\sqrt{2} \leq \sqrt{2} < 2$  is true.

Inductive step: Suppose  $\sqrt{2} \leq a_n < 2$  for some  $n \in \mathbb{N}$ . Then

$$a_{n+1} = \sqrt{2 + a_n} > \sqrt{2 + 0} = \sqrt{2},$$

and

$$a_{n+1} = \sqrt{2 + a_n} < \sqrt{2 + 2} = 2.$$

Together,  $\sqrt{2} \leq a_{n+1} < 2$ . By induction,  $\sqrt{2} \leq a_n < 2$  for all  $n \in \mathbb{N}$ .

(b) Fix  $n \in \mathbb{N}$ . Then

$$a_n \leq a_{n+1} = \sqrt{2 + a_n} \iff a_n^2 \leq 2 + a_n \iff 0 \geq a_n^2 - a_n - 2 = (a_n - 2)(a_n + 1),$$

where the first equivalence uses  $a_n > 0$  from part (a). By part (a),  $a_n - 2 < 0$  and  $a_n + 1 > 0$ , so  $(a_n - 2)(a_n + 1) < 0$ . Working backwards,  $a_n \leq a_{n+1}$ .

(c) As  $\{a_n\}_{n \geq 1}$  is increasing and bounded, it converges by Lecture 11. By part (a) and Problem 3,

$$a = \lim_{n \rightarrow \infty} a_n \text{ satisfies } \sqrt{2} \leq a \leq 2.$$

By the limit law for addition and Problem 5,

$$\lim_{n \rightarrow \infty} (2 + a_n) = 2 + a > 0, \quad \lim_{n \rightarrow \infty} \sqrt{2 + a_n} = \sqrt{2 + a}.$$

Taking the limit  $n \rightarrow \infty$  in

$$a_{n+1} = \sqrt{2 + a_n},$$

we see that  $a$  satisfies

$$a = \sqrt{2 + a}.$$

Thus

$$a^2 = 2 + a \implies 0 = a^2 - a - 2 = (a - 2)(a + 1),$$

so  $a = 2$  or  $a = -1$ . The case  $a = -1$  is not possible, since  $a \geq \sqrt{2}$ . Therefore  $a = 2$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---



## Homework 5

---

## Problem 1

---

### Problem Statement

### Homework 5, Problem 1

Let  $\{a_n\}_{n \geq 1}$  be a sequence of positive real numbers. Prove that  $\lim_{n \rightarrow \infty} a_n = +\infty$  if and only if  $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What is the definition of  $a_n \rightarrow +\infty$ ?

---

2. Why is positivity of  $a_n$  essential before taking reciprocals?

---

3. In the forward direction, what should  $M$  be in terms of  $\varepsilon$ ?

---

4. In the reverse direction, what should  $\varepsilon$  be in terms of  $M$ ?

---

5. Where does the inequality reversal happen?

---

**Forward direction.**

**Fix  $\varepsilon > 0$ . Use  $a_n \rightarrow +\infty$  with  $M =$**

---

**For all  $n \geq N$ , take reciprocals to obtain:**

---

**Conclude  $\left| \frac{1}{a_n} - 0 \right| < \varepsilon$ :**

---

**Reverse direction.**

**Fix  $M > 0$ . Use  $\frac{1}{a_n} \rightarrow 0$  with  $\varepsilon =$**

---

**Use positivity to convert  $\frac{1}{a_n} < \frac{1}{M}$  into:**

---

State the definition this verifies:

---

Use this page for both directions.



### Official Solution (Professor)

$\Rightarrow$  Fix  $\varepsilon > 0$ . As  $\lim a_n = +\infty$ , then there exists  $N$  so that

$$a_n > \frac{1}{\varepsilon} \quad \text{for all } n \geq N.$$

So, for  $n \geq N$  we have

$$-\varepsilon < 0 < \frac{1}{a_n} < \varepsilon \quad \implies \quad \left| \frac{1}{a_n} - 0 \right| < \varepsilon.$$

Therefore  $\lim_{a_n} \frac{1}{a_n} = 0$ .

$\Leftarrow$  Fix  $M > 0$ . As  $\lim \frac{1}{a_n} = 0$ , then there exists  $N$  so that

$$\left| \frac{1}{a_n} - 0 \right| < \frac{1}{M} \quad \text{for all } n \geq N.$$

Note that  $\left| \frac{1}{a_n} - 0 \right| = \frac{1}{a_n} > 0$ . So

$$a_n > M \quad \text{for all } n \geq N.$$

Therefore  $\lim a_n = +\infty$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

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### Problem Statement

### Homework 5, Problem 2

Let  $\{a_n\}_{n \geq 1}$  be a sequence. Prove that if  $\{a_{k_n}\}_{n \geq 1}$  is a subsequence of  $\{a_n\}_{n \geq 1}$  that converges to  $L \in \mathbb{R}$ , then

$$\liminf_{n \rightarrow \infty} a_n \leq L \leq \limsup_{n \rightarrow \infty} a_n.$$

(Hint: As  $\{a_n\}_{n \geq 1}$  is not necessarily bounded, we have to consider the cases  $\liminf_{n \rightarrow \infty} a_n = \pm\infty$  and  $\limsup_{n \rightarrow \infty} a_n = \pm\infty$ . However, the proof is much easier in these cases.)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What do  $\limsup a_n$  and  $\liminf a_n$  measure using tail suprema/infima?

---

2. Why does  $k_n \geq n$  imply that a fixed tail eventually contains the subsequence terms?

---

3. Why is the case  $\limsup a_n = +\infty$  immediate?

---

4. Why is the case  $\limsup a_n = -\infty$  impossible when  $L \in \mathbb{R}$ ?

---

5. How do you get  $a_{k_n} \leq \sup\{a_m : m \geq N\}$  for all large  $n$ ?

---

**Prove  $L \leq \limsup a_n$ .**

**Separate the easy cases  $\limsup a_n = +\infty$  and  $\limsup a_n = -\infty$ :**

---

**Assume  $\limsup a_n \in \mathbb{R}$ . For fixed  $N$ , compare  $a_{k_n}$  to the tail supremum for all large  $n$ :**

---

**Pass  $n \rightarrow \infty$  to get  $L \leq \sup\{a_m : m \geq N\}$ :**

---

**Pass  $N \rightarrow \infty$  to obtain  $L \leq \limsup a_n$ :**

---

**Prove  $\liminf a_n \leq L$ .**

**Repeat the same argument using tail infima and reverse inequalities:**

---

Use this page for the lim sup inequality.



Use this page for the  $\liminf$  inequality and endpoint cases.

### Official Solution (Professor)

Let's start with the second inequality:

$$L \leq \limsup_{n \rightarrow \infty} a_n.$$

Case:  $\limsup a_n = +\infty$ . Then the statement is automatically true, since every real number  $L$  satisfies  $L < +\infty$ .

Case:  $\limsup a_n = -\infty$ . As  $\liminf a_n \leq \limsup a_n$ , then we must have  $\liminf a_n = -\infty = \limsup a_n$ . This means that  $\lim a_n = -\infty$  by Lecture 14.

Fix  $m < 0$ . As  $\lim a_n = -\infty$ , there exists  $N$  so that

$$a_n < m \quad \text{for all } n \geq N.$$

As  $k_n \geq n$  for all  $n$ , this implies

$$a_{k_n} < m \quad \text{for all } n \geq N.$$

Therefore  $L = \lim a_{k_n} = -\infty$ . This contradicts that  $L \in \mathbb{R}$ , and so this case is impossible.

Case:  $\limsup a_n \in \mathbb{R}$ . For  $N \geq 1$ , notice that

$$a_{k_n} \leq \sup\{a_n : n \geq N\} \quad \text{for all } n \geq N.$$

We know that the sequence  $\{a_{k_n}\}$  converges to  $L$ . So, by HW4#3, we have

$$L = \lim a_{k_n} \leq \sup\{a_n : n \geq N\}.$$

On the other hand,  $v_N = \sup\{a_n : n \geq N\}$  is a sequence that converges to  $\limsup a_n$ . So, using HW4#3 again, we conclude that

$$L \leq \lim_{N \rightarrow \infty} \sup\{a_n : n \geq N\} = \limsup a_n.$$

Exercise: Using similar steps, prove that

$$\liminf_{n \rightarrow \infty} a_n \leq L.$$

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

---

#### Problem Statement

#### Homework 5, Problem 3

Let  $\{a_n\}_{n \geq 1}$  and  $\{b_n\}_{n \geq 1}$  be two bounded sequences. Prove that

$$\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n.$$

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why does boundedness guarantee finite tail suprema for  $a_n$ ,  $b_n$ , and  $a_n + b_n$ ?

---

2. For a fixed  $N$ , what upper bounds do the tail suprema give for  $a_n$  and  $b_n$ ?

---

3. Why does adding two pointwise upper bounds produce an upper bound for  $a_n + b_n$ ?

---

4. How does an upper bound for every tail term become an inequality of suprema?

---

5. Which limit law finishes the proof after passing  $N \rightarrow \infty$ ?

Fix  $N \geq 1$ . Write the two tail upper bounds for all  $n \geq N$ :

---

Add them to get an upper bound for  $a_n + b_n$ :

---

Take the supremum over  $n \geq N$  to obtain:

---

Let  $N \rightarrow \infty$  and identify each term with a limsup:

---

State the final inequality:

---



Use this page for the complete proof.

### Official Solution (Professor)

Fix  $N \geq 1$ . For  $n \geq N$  we have

$$a_n \leq \sup\{a_n : n \geq N\}, \quad b_n \leq \sup\{b_n : n \geq N\},$$

and so

$$a_n + b_n \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

In other words,  $\sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}$  is an upper bound for the set  $\{a_n + b_n : n \geq N\}$ . Therefore

$$\sup\{a_n + b_n : n \geq N\} \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

Altogether, we have shown that for any  $N \geq 1$  we have

$$\sup\{a_n + b_n : n \geq N\} \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

As  $\{a_n\}$  is a bounded sequence, we know from lecture that the sequence  $v_N = \sup\{a_n : n \geq N\}$  converges to  $v = \limsup a_n \in \mathbb{R}$ . Similarly, as  $\{b_n\}$  is bounded, the sequence  $\sup\{b_n : n \geq N\}$  converges to  $\limsup b_n$  as  $N \rightarrow \infty$ . By the inequality above, we see that the sequence  $\sup\{a_n + b_n : n \geq N\}$  is also bounded above. Therefore  $\limsup(a_n + b_n)$  is a finite number, and  $\sup\{a_n + b_n : n \geq N\}$  converges to  $\limsup(a_n + b_n)$  as  $N \rightarrow \infty$ .

By the limit laws, the sequence

$$\sup\{a_n + b_n : n \geq N\} - \sup\{a_n : n \geq N\} - \sup\{b_n : n \geq N\}$$

converges as  $N \rightarrow \infty$ . We also know that it is  $\leq 0$  for all  $N \geq 1$ . Therefore, by HW4#3, the limit is also  $\leq 0$ :

$$\limsup(a_n + b_n) - \limsup a_n - \limsup b_n \leq 0$$

and hence

$$\limsup_{n \rightarrow \infty}(a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n.$$

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

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### Problem Statement

### Homework 5, Problem 4

Let  $\{a_n\}_{n \geq 1}$  be a sequence. Prove that  $\limsup_{n \rightarrow \infty} |a_n| = 0$  if and only if  $\lim_{n \rightarrow \infty} a_n = 0$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why is  $\liminf |a_n| \geq 0$  automatic?

---

2. How does  $\liminf |a_n| = \limsup |a_n| = 0$  imply  $|a_n| \rightarrow 0$ ?

---

3. Why is  $|a_n| \rightarrow 0$  exactly the same as  $a_n \rightarrow 0$ ?

---

4. In the reverse direction, which earlier homework result gives  $|a_n| \rightarrow 0$ ?

---

5. How does Lecture 14 connect ordinary limits with  $\liminf$  and  $\limsup$ ?

---

**Forward direction.**

**Assume**  $\limsup |a_n| = 0$ . **Prove**  $\liminf |a_n| \geq 0$ :

---

Use  $\liminf |a_n| \leq \limsup |a_n|$  **to conclude**:

---

Invoke the liminf/limsup convergence criterion to get  $|a_n| \rightarrow 0$ :

---

**Translate**  $|a_n| \rightarrow 0$  **into**  $a_n \rightarrow 0$ :

---

**Reverse direction.**

**Assume**  $a_n \rightarrow 0$ . **Use** HW4#1 **to get**:

---



Use Lecture 14 to identify  $\limsup |a_n|$ :

---

Use this page for both directions.

### Official Solution (Professor)

$\Rightarrow$  Suppose  $\limsup |a_n| = 0$ .

Claim:  $\liminf |a_n| \geq 0$ . As  $|a_n| \geq 0$  for all  $n$ , we have

$$\inf_{n \geq N} |a_n| \geq 0 \quad \text{for all } N \geq 1.$$

Therefore

$$\lim_{N \rightarrow \infty} \inf_{n \geq N} |a_n| \geq 0$$

by HW4#3.

Altogether, we have

$$0 \leq \liminf |a_n| \leq \limsup |a_n| = 0.$$

Therefore

$$\liminf |a_n| = 0 = \limsup |a_n|.$$

By Lecture 10, this means that

$$\lim |a_n| = 0.$$

In other words: for all  $\varepsilon > 0$  there exists  $N$  so that

$$||a_n| - 0| < \varepsilon \quad \text{for all } n \geq N.$$

This is the same as: for all  $\varepsilon > 0$  there exists  $N$  so that

$$|a_n - 0| < \varepsilon \quad \text{for all } n \geq N.$$

So  $\lim a_n = 0$ .

$\Leftarrow$  Suppose  $\lim a_n = 0$ . By HW4#1, this implies

$$\lim |a_n| = 0.$$

By Lecture 14, we have

$$\liminf |a_n| = 0 = \limsup |a_n|.$$

So  $\limsup |a_n| = 0$ , as desired.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Homework 6

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## Problem 1

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### Problem Statement

### Homework 6, Problem 1

Prove that the series

$$\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$$

converges, and find its value. (Hint: You computed the partial sums on Homework 1.)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**



1. What does it mean for a series to converge in terms of partial sums?

---

2. What partial fraction decomposition should be used?

---

3. Which terms cancel in the  $N$ th partial sum?

---

4. What explicit formula for  $S_N$  should remain after cancellation?

---

5. What limit gives the value of the series?

---

Define the  $N$ th partial sum  $S_N =$

---

Rewrite the summand using partial fractions:

---

Write the telescoping sum and identify the surviving first and last terms:

---

Compute  $\lim_{N \rightarrow \infty} S_N$ :

---

State convergence and the value of the series:

---

Use this page for the complete proof.

**Student-Derived Solution (No official solution available)**

Let

$$a_n = \frac{1}{(3n-2)(3n+1)}$$

and let  $S_N = \sum_{n=1}^N a_n$ . Using partial fractions,

$$\frac{1}{(3n-2)(3n+1)} = \frac{1}{3} \left( \frac{1}{3n-2} - \frac{1}{3n+1} \right).$$

Hence

$$S_N = \frac{1}{3} \sum_{n=1}^N \left( \frac{1}{3n-2} - \frac{1}{3n+1} \right).$$

Writing out the terms gives

$$S_N = \frac{1}{3} \left[ \left(1 - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{7}\right) + \cdots + \left(\frac{1}{3N-2} - \frac{1}{3N+1}\right) \right].$$

All intermediate terms cancel, so

$$S_N = \frac{1}{3} \left( 1 - \frac{1}{3N+1} \right) = \frac{N}{3N+1}.$$

Therefore

$$\lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} \frac{N}{3N+1} = \frac{1}{3}.$$

Since the sequence of partial sums converges, the series converges, and its value is

$$\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)} = \frac{1}{3}.$$

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

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## Problem 2

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### Problem Statement

### Homework 6, Problem 2

Let  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} b_n$  be convergent series.

(a) Prove that for any  $k \in \mathbb{R}$ , the series  $\sum_{n=1}^{\infty} ka_n$  converges and

$$\sum_{n=1}^{\infty} ka_n = k \sum_{n=1}^{\infty} a_n.$$

(b) Prove that the series  $\sum_{n=1}^{\infty} (a_n + b_n)$  converges and

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**



1. How is the sum of a convergent series defined using partial sums?

---

2. In part (a), what are the partial sums of  $\sum ka_n$  in terms of the partial sums of  $\sum a_n$ ?

---

3. Which sequence limit law proves convergence in part (a)?

---

4. In part (b), how do the partial sums of  $\sum(a_n + b_n)$  split?

---

5. Which limit law gives the final identity for the sum?

---

Part (a).

Let  $s_n = \sum_{j=1}^n a_j$ . The partial sums of  $\sum ka_n$  are:

---

Use  $s_n \rightarrow \sum a_n$  and the constant multiple limit law:

---

Translate the limit of partial sums into the series identity:

---

Part (b).

Let  $s_n = \sum_{j=1}^n a_j$  and  $t_n = \sum_{j=1}^n b_j$ . The partial sums are:

---

Use the addition limit law and state the resulting sum:

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Let  $s_n = \sum_{j=1}^n a_j$  denote the  $n$ th partial sum. As  $\sum a_n$  converges, we know that the sequence  $\{s_n\}$  converges.

Fix  $k \in \mathbb{R}$ . Then the partial sums  $\sum_{j=1}^n ka_j$  are equal to  $ks_n$ . By the limit law for multiplication by a constant, we conclude that the sequence  $\{ks_n\}_{n \geq 1}$  converges and  $\lim(ks_n) = k \lim s_n$ . By definition, this is exactly what it means for the series  $\sum ka_n$  to converge to the number  $k \sum a_n$ .

(b) Let  $s_n = \sum_{j=1}^n a_j$  and  $t_n = \sum_{j=1}^n b_j$  denote the  $n$ th partial sums. As  $\sum a_n$  and  $\sum b_n$  converge, we know that the sequences  $\{s_n\}$  and  $\{t_n\}$  converge.

Note that the partial sums  $\sum_{j=1}^n (a_j + b_j)$  are equal to  $s_n + t_n$ . By the limit law for addition, we conclude that the sequence  $\{s_n + t_n\}_{n \geq 1}$  converges and  $\lim(s_n + t_n) = \lim s_n + \lim t_n$ . By definition, this is exactly what it means for the series  $\sum(a_n + b_n)$  to converge to the number  $\sum a_n + \sum b_n$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

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#### Problem Statement

#### Homework 6, Problem 3

(a) Prove that

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(Hint:  $\frac{k-1}{2^{k+1}} = \frac{k}{2^k} - \frac{k+1}{2^{k+1}}$ .)

(b) Use part (a) to compute

$$\sum_{n=1}^{\infty} \frac{n}{2^n}.$$

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What identity turns the summand in part (a) into a telescoping difference?

---

2. What formula should the  $n$ th partial sum satisfy?

---

3. Why does  $\frac{n}{2^n} \rightarrow 0$ ?

---

4. In part (b), how do you rewrite  $\frac{n}{2^n}$  using  $\frac{n-1}{2^{n+1}}$  and  $\frac{1}{2^n}$ ?

---

5. Which geometric series value is needed?

---

Part (a).

Define  $s_n = \sum_{k=1}^n \frac{k-1}{2^{k+1}}$ . Prove by induction that  $s_n =$

---

Base case  $n = 1$ :

---

Inductive step using  $\frac{k-1}{2^{k+1}} = \frac{k}{2^k} - \frac{k+1}{2^{k+1}}$ :

---

Pass to the limit using  $\frac{n}{2^n} \rightarrow 0$ :

---

Part (b).

Rewrite  $\sum_{k=1}^n \frac{k}{2^k}$  as  $2 \sum \frac{k-1}{2^{k+1}} + \sum \frac{1}{2^k}$ :

---

Use part (a) and the geometric series to compute the value:

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Let  $s_n = \sum_{k=1}^n \frac{k-1}{2^{k+1}}$ .

We will prove that

$$s_n = \frac{1}{2} - \frac{n+1}{2^{n+1}} \quad \text{for } n \geq 1$$

by induction. For the base case, we note that

$$s_1 = \frac{0}{4} = \frac{1}{2} - \frac{2}{4}.$$

For the inductive step, assume that  $s_n = \frac{1}{2} - \frac{n+1}{2^{n+1}}$  for some  $n \geq 1$ . Then

$$s_{n+1} = s_n + \frac{n}{2^{n+2}} = \frac{1}{2} - \frac{n+1}{2^{n+1}} + \frac{n+1}{2^{n+1}} - \frac{n+2}{2^{n+2}} = \frac{1}{2} - \frac{n+2}{2^{n+2}}.$$

Together, we conclude that our formula for  $s_n$  holds for all  $n \geq 1$  by induction.

Next, we claim that  $\lim_{n \rightarrow \infty} \frac{n}{2^n} = 0$ . Notice that  $n \leq (3/2)^n$  for all  $n \geq 1$ , and so

$$0 \leq \frac{n}{2^n} \leq \frac{(3/2)^n}{2^n} = \left(\frac{3}{4}\right)^n.$$

The right-hand side converges to zero as  $n \rightarrow \infty$ . Therefore,  $\frac{n}{2^n}$  also converges to zero by the squeeze theorem.

Altogether, we conclude

$$\lim s_n = \lim \left( \frac{1}{2} - \frac{n+1}{2^{n+1}} \right) = \frac{1}{2}.$$

(In the last equality, we noted that  $\frac{n+1}{2^{n+1}}$  is a subsequence of  $\frac{n}{2^n}$ , and therefore also converges to zero.) By definition of a convergent series, we conclude that

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(b) Note that

$$\sum_{k=1}^n \frac{k}{2^k} = 2 \sum_{k=1}^n \frac{k-1}{2^{k+1}} + \sum_{k=1}^n \frac{1}{2^k}.$$

By part (a),  $\sum_{k=1}^{\infty} \frac{k-1}{2^{k+1}}$  converges to  $\frac{1}{2}$ , and thus  $2 \sum_{k=1}^{\infty} \frac{k-1}{2^{k+1}}$  converges to 1 by problem 2. From lecture, we know that the geometric series  $\sum_{k=1}^{\infty} \frac{1}{2^k}$  converges to  $\frac{1}{1-(1/2)} - 1 = 1$ . (Notice that we are starting at  $k = 1$ , not  $k = 0$ .) Together, we conclude  $\sum_{k=1}^{\infty} \frac{k}{2^k}$  converges to  $1 + 1 = 2$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---



## Problem 4

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### Problem Statement

### Homework 6, Problem 4

Prove that if  $\sum_{n=1}^{\infty} a_n$  converges absolutely and  $\{b_n\}_{n \geq 1}$  is a bounded sequence, then the series  $\sum_{n=1}^{\infty} a_n b_n$  also converges. (Hint: Use the Cauchy criterion.)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does absolute convergence of  $\sum a_n$  say about  $\sum |a_n|$ ?

---

2. What bound does boundedness of  $\{b_n\}$  provide?

---

3. What is the Cauchy criterion for convergence of a series?

---

4. How can  $|\sum_{k=m}^n a_k b_k|$  be controlled by  $\sum_{k=m}^n |a_k|$ ?

---

5. Why is  $M + 1$  used in the denominator of the tail estimate?

---

Choose  $M \geq 0$  such that  $|b_n| \leq M$  for all  $n$ :

---

Use the Cauchy criterion for  $\sum |a_n|$  with tolerance  $\varepsilon/(M+1)$ :

---

For  $n \geq m \geq N$ , estimate  $|\sum_{k=m}^n a_k b_k|$ :

---

Conclude the Cauchy criterion for  $\sum a_n b_n$ :

---

Use this page for the complete proof.

### Student-Derived Solution (No official solution available)

Since  $\sum_{n=1}^{\infty} a_n$  converges absolutely, the series

$$\sum_{n=1}^{\infty} |a_n|$$

converges. By the Cauchy criterion for series, given  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$\sum_{k=m}^n |a_k| < \frac{\varepsilon}{M+1} \quad \text{for all } n \geq m \geq N,$$

where  $M \geq 0$  is chosen so that

$$|b_n| \leq M \quad \text{for all } n \in \mathbb{N}.$$

Such an  $M$  exists because  $\{b_n\}$  is bounded.

Now let  $n \geq m \geq N$ . Then

$$\left| \sum_{k=m}^n a_k b_k \right| \leq \sum_{k=m}^n |a_k b_k| = \sum_{k=m}^n |a_k| |b_k| \leq M \sum_{k=m}^n |a_k|.$$

Therefore

$$\left| \sum_{k=m}^n a_k b_k \right| < M \cdot \frac{\varepsilon}{M+1} < \varepsilon.$$

Thus, for every  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$\left| \sum_{k=m}^n a_k b_k \right| < \varepsilon \quad \text{for all } n \geq m \geq N.$$

By the Cauchy criterion for series,  $\sum_{n=1}^{\infty} a_n b_n$  converges.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---



## Problem 5

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### Problem Statement

### Homework 6, Problem 5

- (a) Prove that if  $\sum_{n=1}^{\infty} a_n$  converges absolutely, then the series  $\sum_{n=1}^{\infty} a_n^2$  also converges.
- (b) Provide an example where  $\sum_{n=1}^{\infty} a_n$  diverges and  $\sum_{n=1}^{\infty} a_n^2$  converges.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why does absolute convergence imply  $a_n \rightarrow 0$ ?

---

2. Why does  $a_n \rightarrow 0$  imply  $|a_n| < 1$  eventually?

---

3. For  $|x| < 1$ , why is  $x^2 \leq |x|$ ?

---

4. Which comparison proves the tail of  $\sum a_n^2$  converges?

---

5. What standard example makes  $\sum a_n$  diverge but  $\sum a_n^2$  converge?

---

Part (a).

From absolute convergence, write the convergent positive series:

---

Use convergence of  $\sum a_n$  to get  $a_n \rightarrow 0$ , then choose  $N$  with  $|a_n| < 1$ :

---

For  $n \geq N$ , compare  $a_n^2$  with  $|a_n|$ :

---

Use the comparison test and finite initial terms:

---

Part (b).

Choose  $a_n =$  and cite the harmonic and  $p$ -series facts:

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Note that  $x^2 \leq |x|$  for any  $x \in [-1, 1]$ . (I recommend that you draw the graphs of  $x^2$  and  $|x|$  to check that this is true.)

As  $\sum a_n$  converges, then we have  $\lim a_n = 0$ , and so there exists an  $N$  so that  $|a_n| < 1$  for all  $n \geq N$ . By the previous paragraph, this means that  $a_n^2 \leq |a_n|$  for all  $n \geq N$ .

As  $\sum |a_n|$  converges, then  $\sum_{n=N}^{\infty} a_n^2$  also converges by the comparison test. Adding the first  $N - 1$  terms back in, we conclude  $\sum_{n=1}^{\infty} a_n^2$  converges as well.

(b) Consider  $a_n = \frac{1}{n}$ . In lecture, we proved that  $\sum \frac{1}{n}$  diverges and  $\sum \frac{1}{n^2}$  converges.



What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 6

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### Problem Statement

### Homework 6, Problem 6

For each of the following series, determine if it converges and prove your answer.

$$(a) \sum_{n=2}^{\infty} \frac{1}{[n + (-1)^n]^2} \quad (b) \sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n}) \quad (c) \sum_{n=1}^{\infty} (-1)^n$$

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. In part (a), what lower bound can you put on  $n + (-1)^n$  for  $n \geq 2$ ?

---

2. Which convergent  $p$ -series controls part (a)?

---

3. In part (b), what is the  $N$ th partial sum after telescoping?

---

4. Why does that partial sum sequence diverge to  $+\infty$ ?

---

5. In part (c), what necessary condition for convergence fails?

---

Part (a).

For  $n \geq 2$ , show  $n + (-1)^n \geq n - 1$ , then compare:

---

Use the comparison test with  $\sum 1/(n-1)^2$ :

---

Part (b).

Let  $s_n = \sum_{k=1}^n (\sqrt{k+1} - \sqrt{k})$ . Prove by induction/telescoping that  $s_n =$

---

Show  $s_n \rightarrow +\infty$ , so the series diverges:

---

Part (c).

Suppose the series converged. Then its terms would satisfy:

---

Explain why  $(-1)^n$  does not satisfy this:

---

Use this page for parts (a) and (b).



Use this page for part (c) and final conclusions.

### Official Solution (Professor)

(a) We will prove that this series converges. For  $n \geq 2$ , we have

$$\frac{1}{[n + (-1)^n]^2} \leq \frac{1}{(n-1)^2}.$$

From lecture, we know that the series

$$\sum_{n=2}^{\infty} \frac{1}{(n-1)^2} = \sum_{k=1}^{\infty} \frac{1}{k^2}$$

converges. Therefore our series  $\sum \frac{1}{[n+(-1)^n]^2}$  also converges by the comparison test.

(b) We will prove that this series diverges. Let

$$s_n = \sum_{k=1}^n (\sqrt{k+1} - \sqrt{k})$$

denote the  $n$ th partial sum.

Claim:  $s_n = \sqrt{n+1} - 1$  for all  $n \geq 1$ . We will argue using induction.

- Base case: For  $n = 1$ , we have

$$s_1 = \sqrt{2} - \sqrt{1} = \sqrt{2} - 1.$$

- Inductive step: Suppose that  $s_n = \sqrt{n+1} - 1$  for some  $n \geq 1$ . Then the formula for  $s_{n+1}$  holds, since

$$s_{n+1} = s_n + (\sqrt{n+2} - \sqrt{n+1}) = (\sqrt{n+1} - 1) + (\sqrt{n+2} - \sqrt{n+1}) = \sqrt{n+2} - 1.$$

By induction, we conclude that  $s_n = \sqrt{n+1} - 1$  holds for all  $n \geq 1$ .

Notice that the sequence  $s_n = \sqrt{n+1} - 1$  diverges to  $+\infty$ . Indeed, given  $M > 0$ , pick  $N \in \mathbb{N}$  so that  $N > (M+1)^2$ , and then for any  $n \geq N$  we have

$$\sqrt{n+1} - 1 \geq \sqrt{N} - 1 > \sqrt{(M+1)^2} - 1 = M.$$

As the sequence  $\{s_n\}$  diverges, then the series  $\sum(\sqrt{n+1} - \sqrt{n})$  diverges.

(c) We will prove that this series diverges. Suppose towards a contradiction that  $\sum(-1)^n$  converges. Then  $\lim(-1)^n = 0$  by Lecture 16. However, we know that the sequence  $\{(-1)^n\}$  diverges by Lecture 10, and so this is a contradiction.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Homework 7

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## Problem 1

---

### Problem Statement

### Homework 7, Problem 1

For each of the following series, determine if it converges and prove your answer.

$$(a) \sum_{n=2}^{\infty} \frac{1}{\log n} \quad (b) \sum_{n=1}^{\infty} \frac{n-1}{n^2} \quad (c) \sum_{n=1}^{\infty} \frac{n^2}{n!}$$

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Which convergence test is adapted to  $\sum 1/\log n$ , and why?

---

2. In part (a), why do the dyadic terms fail the term test?

---

3. In part (b), how does  $(n-1)/n^2$  compare to  $1/(2n)$ ?

---

4. In part (c), why is the ratio test natural for  $n^2/n!$ ?

---

5. Which parts converge and which diverge?

---



Part (a).

Reindex with  $a_k = 1/\log(k+1)$  and state the dyadic criterion:

---

Estimate  $2^n a_{2^n}$  from below and explain why the terms do not go to zero:

---

Part (b).

For  $n \geq 2$ , prove  $\frac{n-1}{n^2} \geq \frac{1}{2n}$ :

---

Use comparison with the harmonic series:

---

Part (c).

Set  $a_n = \frac{n^2}{n!}$ . Compute  $a_{n+1}/a_n$ :

---

Apply the ratio test:

---

Use this page for parts (a) and (b).

Use this page for part (c) and final conclusions.

### Official Solution (Professor)

(a) We'll use the dyadic criterion. Consider the series  $\sum_{k=1}^{\infty} a_k$  with  $a_k = \frac{1}{\log(k+1)}$  for  $k \geq 1$ . (Strictly speaking, in order to apply the dyadic criterion we need a series that starts at  $n = 1$ , so we reindexed.)

Then  $\{a_k\}_{k \geq 1}$  is a decreasing sequence of nonnegative numbers. Therefore, by the dyadic criterion,  $\sum_{k=1}^{\infty} a_k$  converges if and only if the series  $\sum_{n=0}^{\infty} 2^n a_{2^n}$  converges. Note that

$$2^n a_{2^n} = \frac{2^n}{\log(2^n + 1)} \geq \frac{2^n}{\log(2^{n+1})} = \frac{1}{\log 2} \cdot \frac{2^n}{n+1}.$$

Now, the sequence  $\frac{2^n}{n+1}$  diverges to  $+\infty$ , and so it cannot converge to 0. Multiplying this by the constant  $\frac{1}{\log 2}$ , we see that  $2^n a_{2^n}$  also cannot converge to 0. Therefore the series  $\sum 2^n a_{2^n}$  diverges, and thus  $\sum a_k$  diverges as well.

(b) We'll use the comparison test. For  $n \geq 2$ , we have

$$n-1 \geq n - \frac{n}{2} = \frac{n}{2} \quad \implies \quad \frac{n-1}{n^2} \geq \frac{n/2}{n^2} = \frac{1}{2n}.$$

From lecture we know that the series  $\sum \frac{1}{n}$  diverges, and thus  $\sum_{n=2}^{\infty} \frac{1}{2n}$  diverges as well. Therefore, by the comparison test,  $\sum_{n=2}^{\infty} \frac{n-1}{n^2}$  also diverges. Note that  $\frac{n-1}{n^2}$  is equal to zero for  $n = 1$ , so  $\sum_{n=2}^{\infty} \frac{n-1}{n^2}$  and  $\sum_{n=1}^{\infty} \frac{n-1}{n^2}$  are the same series.

(c) We'll use the ratio test. Set  $a_n = \frac{n^2}{n!}$ . Then

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2 \cdot n!}{n^2 \cdot (n+1)!} = \frac{n+1}{n^2} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore

$$\limsup \left| \frac{a_{n+1}}{a_n} \right| = \lim \frac{a_{n+1}}{a_n} = 0 < 1,$$

and so the series converges by the ratio test.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

---

### Problem Statement

### Homework 7, Problem 2

For each of the following series, determine if it converges and prove your answer.

$$(a) \sum_{n=1}^{\infty} \frac{n^4}{4^n} \quad (b) \sum_{n=1}^{\infty} \frac{n!}{n^4 + 3} \quad (c) \sum_{n=1}^{\infty} \frac{2^n}{n!}$$

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why does a polynomial-over-exponential series suggest the ratio test?

---

2. In part (b), what necessary condition for convergence should be checked?

---

3. How does factorial growth compare to  $n^4 + 3$ ?

---

4. In part (c), what ratio do successive terms have?

---

5. Which of these three series fail because the terms do not go to 0?

---

**Part (a).**

Set  $a_n = \frac{n^4}{4^n}$ . Compute  $a_{n+1}/a_n$  and its limit:

---

Apply the ratio test:

---

**Part (b).**

Show  $\frac{n^4+3}{n!} \rightarrow 0$ , for example using  $n! \geq n(n-1)(n-2)(n-3)(n-4)$ :

---

Use HW5#1 to conclude  $\frac{n!}{n^4+3} \rightarrow +\infty$ , hence the series diverges:

---

**Part (c).**

Set  $a_n = \frac{2^n}{n!}$ . Compute  $a_{n+1}/a_n$  and apply the ratio test:

---

Use this page for parts (a) and (b).

Use this page for part (c) and final conclusions.

### Official Solution (Professor)

(a) We'll use the ratio test. Set  $a_n = \frac{n^4}{4^n}$ . Then

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^4 \cdot 4^n}{n^4 \cdot 4^{n+1}} = \frac{(n+1)^4}{4n^4} \longrightarrow \frac{1}{4} \quad \text{as } n \rightarrow \infty.$$

Therefore  $\limsup \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{4} < 1$ , and so the series converges by the ratio test.

(b) This series diverges. In brief, the sequence  $\frac{n!}{n^4+3}$  diverges to  $+\infty$ , and so it cannot converge to zero. More concretely, notice that for  $n \geq 5$  we have

$$n! \geq n(n-1)(n-2)(n-3)(n-4),$$

and so

$$\frac{n^4+3}{n!} \leq \frac{n^4+3}{n(n-1)(n-2)(n-3)(n-4)} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore the sequence  $\frac{n!}{n^4+3} \rightarrow +\infty$  by HW5#1. In particular, there must exist an  $N$  so that  $\frac{n!}{n^4+3} \geq 1$  for all  $n \geq N$ , and so  $\frac{n!}{n^4+3}$  cannot converge to 0.

(c) We'll use the ratio test. Set  $a_n = \frac{2^n}{n!}$ . Then

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1} \cdot n!}{(n+1)! \cdot 2^n} = \frac{2}{n+1} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore  $\limsup \left| \frac{a_{n+1}}{a_n} \right| = 0 < 1$ , and so the series converges by the ratio test.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

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#### Problem Statement

#### Homework 7, Problem 3

For each of the following series, determine if it converges and prove your answer.

$$(a) \sum_{n=1}^{\infty} \frac{1}{n^n} \quad (b) \sum_{n=1}^{\infty} \frac{n!}{n^n} \quad (c) \sum_{n=1}^{\infty} \frac{(-1)^n n!}{2^n}$$



Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why does  $\frac{1}{n^n}$  suggest the root test?

---

2. What lecture result gives  $(n!)^{1/n}/n \rightarrow 1/e$ ?

---

3. Why does  $1/e < 1$  prove convergence in part (b)?

---

4. In part (c), why should you check whether the terms go to 0?

---

5. How does HW5#1 convert  $2^n/n! \rightarrow 0$  into  $n!/2^n \rightarrow +\infty$ ?

---

**Part (a).**

**Set  $a_n = 1/n^n$ . Compute  $|a_n|^{1/n}$ :**

---

**Apply the root test:**

---

**Part (b).**

**Set  $a_n = n!/n^n$ . Compute  $|a_n|^{1/n}$ :**

---

**Use Lecture 15 to identify the limit and apply the root test:**

---

**Part (c).**

**Use Problem 2(c) to know  $\frac{2^n}{n!} \rightarrow 0$ :**

---

Apply HW5#1 to show  $\frac{n!}{2^n} \rightarrow +\infty$ , then conclude the terms do not go to 0:

---

Use this page for parts (a) and (b).

Use this page for part (c) and final conclusions.

### Official Solution (Professor)

(a) We'll use the root test. Set  $a_n = \frac{1}{n^n}$ . Then

$$|a_n|^{1/n} = \frac{1}{n} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore  $\limsup |a_n|^{1/n} = 0 < 1$ , and so the series converges by the root test.

(b) We'll use the root test. Set  $a_n = \frac{n!}{n^n}$ , and consider the sequence

$$|a_n|^{1/n} = \left( \frac{n!}{n^n} \right)^{1/n} = \frac{(n!)^{1/n}}{n}.$$

In Lecture 15 we showed that this sequence converges to  $1/e$ . Therefore

$$\limsup |a_n|^{1/n} = 1/e < 1,$$

and so the series converges by the root test.

(c) This series diverges, because the sequence  $\frac{(-1)^n n!}{2^n}$  does not converge to 0. Specifically, we know that the sequence  $\frac{2^n}{n!}$  converges to 0. (Indeed, we proved in problem 2c that the series  $\sum \frac{2^n}{n!}$  converges, and so its terms must converge to 0.) So, by HW5#1, the sequence  $\frac{n!}{2^n}$  diverges to  $+\infty$ . In particular, there exists an  $N$  so that  $\frac{n!}{2^n} \geq 1$  for all  $n \geq N$ . Therefore  $\left| \frac{(-1)^n n!}{2^n} \right| \geq 1$  for all  $n \geq N$ , and so the terms  $\frac{(-1)^n n!}{2^n}$  in our series cannot converge to 0.



What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

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## Problem 4

---

### Problem Statement

### Homework 7, Problem 4

Consider the series

$$\sum_{n=1}^{\infty} 2^{(-1)^n - n} = \frac{1}{4} + \frac{1}{2} + \frac{1}{16} + \frac{1}{8} + \frac{1}{64} + \frac{1}{32} + \cdots.$$

- (a) Show that the ratio test is inconclusive.
- (b) Prove that the series converges using the root test.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What are the first few terms of  $a_{n+1}/a_n$ ?

---

2. Why does the ratio sequence have two subsequential limits?

---

3. What does it mean for the ratio test to be inconclusive?

---

4. Why does the root test handle  $2^{(-1)^n - n}$  better than the ratio test?

---

5. What is  $\lim |a_n|^{1/n}$ ?

---

**Part (a).**

Set  $a_n = 2^{(-1)^n - n}$ . Compute the first few ratios  $|a_{n+1}/a_n|$ :

---

Identify the liminf and limsup of the ratio sequence:

---

State why this makes the ratio test inconclusive:

---

**Part (b).**

Compute  $|a_n|^{1/n} = 2^{((-1)^n/n) - 1}$ :

---

Use  $(-1)^n/n \rightarrow 0$  to find the root-test limit:

---

Conclude convergence by the root test:

---

Use this page for part (a).



Use this page for part (b).

### Official Solution (Professor)

(a) Set  $a_n = 2^{(-1)^n - n}$ . Then the sequence  $\left| \frac{a_{n+1}}{a_n} \right|$  is

$$\frac{1/2}{1/4} = 2, \quad \frac{1/16}{1/2} = \frac{1}{8}, \quad \frac{1/8}{1/16} = 2, \quad \frac{1/64}{1/8} = \frac{1}{8}, \dots$$

In other words, the even terms of this sequence are  $\frac{1}{8}$  and the odd terms are 2. Therefore

$$\liminf \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{8}, \quad \limsup \left| \frac{a_{n+1}}{a_n} \right| = 2,$$

and so

$$\liminf \left| \frac{a_{n+1}}{a_n} \right| \leq 1 \leq \limsup \left| \frac{a_{n+1}}{a_n} \right|.$$

(b) Set  $a_n = 2^{(-1)^n - n}$ . Then

$$|a_n|^{1/n} = 2^{\frac{(-1)^n}{n} - 1}.$$

Since  $\frac{(-1)^n}{n} \rightarrow 0$ , we get

$$\lim_{n \rightarrow \infty} |a_n|^{1/n} = \frac{1}{2} < 1.$$

Therefore the series converges by the root test.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 5

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### Problem Statement

### Homework 7, Problem 5

Below are two series  $\sum_{n=1}^{\infty} a_n$  and the numbers  $s \in \mathbb{R}$  to which they converge. For each of these series, what is the set of all numbers  $\tilde{s} \in \mathbb{R}$  such that there exists a rearrangement that satisfies  $\sum_{n=1}^{\infty} \tilde{a}_n = \tilde{s}$ ? Prove your answer.

$$(a) \sum_{n=1}^{\infty} \frac{(-1)^n}{n} = -\log 2 \quad (b) \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$

(Hint: Your proof should be quite short, because we already did most of the work in lecture.)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What is the difference between conditional and absolute convergence for rearrangements?

---

2. Why is the alternating harmonic series conditionally convergent?

---

3. What does Riemann's rearrangement theorem say about possible sums?

---

4. Why does  $\sum (-1)^n/n^2$  converge absolutely?

---

5. What theorem says every rearrangement of an absolutely convergent series has the same sum?

---

**Part (a).**

Show  $\sum (-1)^n/n$  converges but not absolutely:

---

Apply Riemann's rearrangement result with  $\alpha = \tilde{s} = \beta$ :

---

Conclude the set of possible rearranged sums is:

---

**Part (b).**

Show  $\sum (-1)^n/n^2$  converges absolutely:

---

Apply the theorem on rearrangements of absolutely convergent series:

---



Conclude the set of possible rearranged sums is:

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) We claim that the set is  $\mathbb{R}$ . Note that:

- $\sum \frac{(-1)^n}{n}$  converges: by the alternating series test, since  $\{\frac{1}{n}\}$  is a decreasing sequence that converges to 0.
- $\sum \frac{(-1)^n}{n}$  does not converge absolutely: since  $\left|\frac{(-1)^n}{n}\right| = \frac{1}{n}$ , and we know  $\sum \frac{1}{n}$  diverges by Lecture 11.

Fix  $\tilde{s} \in \mathbb{R}$ . Applying the Riemann rearrangement result (with  $\alpha = \tilde{s} = \beta$ ), we deduce that there exists a rearrangement  $\sum \tilde{a}_n$  that converges to  $\tilde{s}$ . This works for any  $\tilde{s} \in \mathbb{R}$ , and so  $\mathbb{R}$  is the desired set.

(b) We claim that the set is  $\{-\frac{\pi^2}{12}\}$ . Note that  $\sum \frac{(-1)^n}{n^2}$  converges absolutely, since  $\left|\frac{(-1)^n}{n^2}\right| = \frac{1}{n^2}$  and we know  $\sum \frac{1}{n^2}$  converges by Lecture 16. Suppose that  $\sum \tilde{a}_n$  is a rearrangement that converges to some  $\tilde{s} \in \mathbb{R}$ . Applying the second theorem from Lecture 20, we deduce that  $\tilde{s} = -\frac{\pi^2}{12}$ . This works for any rearrangement, so  $-\frac{\pi^2}{12}$  is the only value that can be attained.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 6

---

### Problem Statement

### Homework 7, Problem 6

Prove or disprove the following statements:

- (a) If  $\sum_{n=1}^{\infty} a_n$  converges, then  $\sum_{n=1}^{\infty} a_{2n}$  converges.
- (b) If  $\sum_{n=1}^{\infty} a_n$  converges absolutely, then  $\sum_{n=1}^{\infty} a_{2n}$  converges absolutely.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**



1. Why can ordinary convergence depend on cancellation between odd and even terms?

---

2. What standard alternating example disproves part (a)?

---

3. What is  $a_{2n}$  for that example?

---

4. Why does absolute convergence survive passing to a subseries?

---

5. How does the Cauchy criterion prove part (b)?

---

Part (a).

Choose  $a_n =$  so that  $\sum a_n$  is alternating harmonic:

---

Compute  $a_{2n}$  and identify the divergent series:

---

State the counterexample conclusion:

---

Part (b).

Assume  $\sum |a_n|$  converges. Use the Cauchy criterion to choose  $N$ :

---

For  $n \geq m \geq N$ , compare  $\sum_{k=m}^n |a_{2k}|$  to a tail of  $\sum |a_j|$ :

---

**Conclude  $\sum |a_{2n}|$  converges:**

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Disprove: Consider the series  $a_n = \frac{(-1)^n}{n}$ . Then:

- $\sum a_n$  converges: by the alternating series test, since  $\frac{1}{n}$  is a decreasing sequence that converges to 0.
- $\sum a_{2n} = \sum \frac{1}{2n}$  diverges: This is  $\frac{1}{2}$  times the series  $\sum \frac{1}{n}$ , which we know diverges by lecture.

(b) Prove: We will use the Cauchy criterion. Fix  $\varepsilon > 0$ . As  $\sum |a_n|$  converges, then there exists an  $N$  so that

$$\sum_{k=m}^n |a_k| < \varepsilon \quad \text{for all } n \geq m \geq N.$$

Then for  $n \geq m \geq N$ , we have

$$\sum_{k=m}^n |a_{2k}| \leq \sum_{j=2m}^{2n} |a_j| < \varepsilon.$$

(Note that first sum above consists of only even terms  $|a_{2m}| + |a_{2m+2}| + \cdots + |a_{2n}|$ , which differs from the second sum  $|a_{2m}| + |a_{2m+1}| + |a_{2m+2}| + \cdots + |a_{2n}|$  by only the odd terms, each of which is nonnegative.) Therefore  $\sum |a_{2n}|$  converges by the Cauchy criterion.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Homework 8

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## Problem 1

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### Problem Statement

### Homework 8, Problem 1

Consider the metric space  $(\mathbb{R}, d)$  with the usual metric  $d(x, y) = |x - y|$ . For each of the following sets  $A \subseteq \mathbb{R}$ , find its interior and prove your answer.

- (a)  $A = [0, 1)$
- (b)  $A = \mathbb{Z}$
- (c)  $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does it mean for  $x$  to be an interior point of  $A$ ?

---

2. Why is every point of  $(0, 1)$  interior to  $[0, 1)$ ?

---

3. Why is 0 not an interior point of  $[0, 1)$ ?

---

4. Why can no open interval in  $\mathbb{R}$  be contained in  $\mathbb{Z}$ ?

---

5. How does density of irrational numbers show  $\{x \in \mathbb{Q} : 0 < x < 1\}^\circ = \emptyset$ ?

---

**Part (a).**

**Show  $(0, 1) \subseteq A^\circ$  by choosing  $r = \frac{1}{2} \min\{x, 1 - x\}$ :**

---

**Show  $0 \notin A^\circ$  by finding a point in  $(-r, r) \setminus [0, 1]$ :**

---

**Conclude  $A^\circ = (0, 1)$ :**

---

**Part (b).**

**Fix  $n \in \mathbb{Z}$  and  $r > 0$ . Find a noninteger point in  $(n - r, n + r)$ :**

---

**Conclude  $\mathbb{Z}^\circ = \emptyset$ :**

---

**Part (c).**

**Fix  $a \in A$  and  $r > 0$ . Use density of  $\mathbb{R} \setminus \mathbb{Q}$  to find  $y \in (a - r, a + r) \setminus A$ :**

---

Use this page for parts (a) and (b).

Use this page for part (c) and final conclusions.



### Official Solution (Professor)

(a) Claim:  $A^\circ = (0, 1)$ . As  $A^\circ$  is the largest open subset of  $A$ , and  $(0, 1)$  is one open subset, we must have  $(0, 1) \subseteq A^\circ \subseteq [0, 1)$ . Note that 0 is not an interior point: given  $r > 0$  we have  $(-r, r) \not\subseteq [0, 1)$ , since  $-\frac{r}{2}$  is a point in  $(-r, r)$  that is not in  $[0, 1)$ .

(b) Claim:  $A^\circ = \emptyset$ . Fix  $n \in \mathbb{Z}$ ; we will show that  $n$  is not an interior point. Given  $r > 0$  we have  $(n - r, n + r) \not\subseteq \mathbb{Z}$ , since  $\min\{n + \frac{r}{2}, n + \frac{1}{2}\}$  is in  $(n - r, n + r)$  but not in  $\mathbb{Z}$ .

(c) Claim:  $A^\circ = \emptyset$ . Fix  $a \in A$ ; we will show that  $a$  is not an interior point. As the irrational numbers are dense in  $\mathbb{R}$ , then for any  $r > 0$  we can find a point  $y \in \mathbb{Q}^c$  between  $a - r$  and  $a$ . This  $y$  is a point in  $(a - r, a + r)$  that is not in  $A$ , and so  $(a - r, a + r) \not\subseteq A$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

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### Problem Statement

### Homework 8, Problem 2

Recall that  $d_1$  and  $d_\infty$  are both metrics on  $\mathbb{R}^2$ .

- (a) Prove that  $d_\infty(x, y) \leq d_1(x, y) \leq 2d_\infty(x, y)$  for all  $x, y \in \mathbb{R}^2$ .
- (b) Prove that for any  $x \in \mathbb{R}^2$  and  $r > 0$  we have

$$\{y \in \mathbb{R}^2 : d_\infty(x, y) < r/2\} \subseteq \{y \in \mathbb{R}^2 : d_1(x, y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x, y) < r\}.$$

I recommend that you draw a picture of these three sets.

- (c) Prove that a set  $A \subseteq \mathbb{R}^2$  is open in the metric space  $(\mathbb{R}^2, d_1)$  if and only if it is open in the metric space  $(\mathbb{R}^2, d_\infty)$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What are the formulas for  $d_1(x, y)$  and  $d_\infty(x, y)$  in coordinates?

---

2. Why is the maximum of two nonnegative numbers at most their sum?

---

3. Why is the sum of two nonnegative numbers at most twice their maximum?

---

4. How do metric inequalities translate into ball containments?

---

5. How do the ball containments prove the two metrics define the same open sets?

---

**Part (a).**

**Write  $x = (x_1, x_2)$ ,  $y = (y_1, y_2)$ , and expand  $d_1, d_\infty$ :**

---

**Prove  $d_\infty \leq d_1$ :**

---

**Prove  $d_1 \leq 2d_\infty$ :**

---

**Part (b).**

**If  $d_\infty(x, y) < r/2$ , use part (a) to prove  $d_1(x, y) < r$ :**

---

**If  $d_1(x, y) < r$ , use part (a) to prove  $d_\infty(x, y) < r$ :**

---

**Part (c).**

**For  $d_\infty$ -open  $\Rightarrow d_1$ -open, use  $B_r^1(a) \subseteq B_r^\infty(a)$ :**

---

**For  $d_1$ -open  $\Rightarrow d_\infty$ -open, use  $B_{r/2}^\infty(a) \subseteq B_r^1(a)$ :**

---



Use this page for parts (a) and (b).

Use this page for part (c).

### Official Solution (Professor)

(a) Fix  $x = (x_1, x_2)$  and  $y = (y_1, y_2)$  in  $\mathbb{R}^2$ . Then

$$|x_1 - y_1| \leq |x_1 - y_1| + |x_2 - y_2| = d_1(x, y),$$

$$|x_2 - y_2| \leq |x_1 - y_1| + |x_2 - y_2| = d_1(x, y),$$

and so

$$\max\{|x_1 - y_1|, |x_2 - y_2|\} \leq d_1(x, y).$$

This proves the first inequality:  $d_\infty(x, y) \leq d_1(x, y)$ . On the other hand, note that

$$|x_1 - y_1| \leq \max\{|x_1 - y_1|, |x_2 - y_2|\} = d_\infty(x, y),$$

$$|x_2 - y_2| \leq \max\{|x_1 - y_1|, |x_2 - y_2|\} = d_\infty(x, y),$$

and so

$$|x_1 - y_1| + |x_2 - y_2| \leq 2d_\infty(x, y).$$

This proves that  $d_1(x, y) \leq 2d_\infty(x, y)$ .

(b) Let's start with the first containment:

$$\{y \in \mathbb{R}^2 : d_\infty(x, y) < r/2\} \subseteq \{y \in \mathbb{R}^2 : d_1(x, y) < r\}.$$

Fix  $y \in \mathbb{R}^2$  so that  $d_\infty(x, y) < r/2$ . Then by the second inequality in part (a) we have

$$d_1(x, y) \leq 2d_\infty(x, y) < 2 \cdot \frac{r}{2} = r,$$

and so  $d_1(x, y) < r$ .

Next, we turn to the second containment:

$$\{y \in \mathbb{R}^2 : d_1(x, y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x, y) < r\}.$$

Fix  $y \in \mathbb{R}^2$  so that  $d_1(x, y) < r$ . Then by the first inequality in part (a), we have

$$d_\infty(x, y) \leq d_1(x, y) < r,$$

and so  $d_\infty(x, y) < r$ .

(c)  $\Leftarrow$  Suppose that  $A$  is open in  $(\mathbb{R}^2, d_\infty)$ . Fix  $a \in A$ . Then there exists  $r > 0$  so that

$$B_r^\infty(a) = \{x \in \mathbb{R}^2 : d_\infty(x, a) < r\} \subseteq A.$$

Note that  $B_r^1(a) \subseteq B_r^\infty(a)$  by part (b), and so  $B_r^1(a) \subseteq B_r^\infty(a) \subseteq A$ . As  $a \in A$  was arbitrary, we conclude that  $A$  is open in  $(\mathbb{R}^2, d_1)$ .

$\Rightarrow$  Suppose that  $A$  is open in  $(\mathbb{R}^2, d_1)$ . Fix  $a \in A$ . Then there exists  $r > 0$  so that  $B_r^1(a) \subseteq A$ . Note that  $B_{r/2}^\infty(a) \subseteq B_r^1(a)$  by part (b), and so  $B_{r/2}^\infty(a) \subseteq B_r^1(a) \subseteq A$ . As  $a \in A$  was arbitrary, we conclude that  $A$  is open in  $(\mathbb{R}^2, d_\infty)$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

---

#### Problem Statement

#### Homework 8, Problem 3

Let  $X$  be a nonempty set, and recall that the discrete metric  $d$  is a metric on  $X$ . Prove that every set  $A \subseteq X$  is open in the metric space  $(X, d)$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What is the definition of the discrete metric?

---

2. What is  $B_{1/2}(a)$  in the discrete metric?

---

3. Why is  $\{a\} \subseteq A$  whenever  $a \in A$ ?

---

4. How does this prove every point of  $A$  is interior?

---

5. Why does the proof also work when  $A = \emptyset$ ?

---



**Fix  $A \subseteq X$  and  $a \in A$ :**

---

**Compute  $B_{1/2}(a) = \{x \in X : d(x, a) < 1/2\}$ :**

---

**Use  $B_{1/2}(a) = \{a\} \subseteq A$  to show  $a$  is interior:**

---

**Conclude  $A$  is open because  $a \in A$  was arbitrary:**

---

Use this page for the complete proof.

### Official Solution (Professor)

Fix  $A \subseteq X$  and  $a \in A$ . Notice that

$$B_{1/2}(a) = \{x \in X : d(x, a) < 1/2\} = \{a\}.$$

(Indeed,  $d(x, a)$  is either equal to 0 or 1, and so the only time that  $d(x, a) < 1/2$  is when  $d(x, a) = 0$ .) Then  $a \in B_{1/2}(a) \subseteq A$ , and so  $a$  is an interior point for  $A$ . As  $a \in A$  was arbitrary, we conclude that  $A$  is open.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

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### Problem Statement

### Homework 8, Problem 4

Let  $(X, d)$  be any metric space. Define the function  $\tilde{d}: X \times X \rightarrow \mathbb{R}$  by

$$\tilde{d}(x, y) = \min\{d(x, y), 1\}.$$

- (a) Prove that  $\tilde{d}$  is also a metric on  $X$ .
- (b) Prove that a set  $A \subseteq X$  is open in  $(X, d)$  if and only if it is open in  $(X, \tilde{d})$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Which metric axioms are immediate from the definition of  $\tilde{d}$ ?

---

2. Why is definiteness preserved by taking  $\min\{d(x, y), 1\}$ ?

---

3. What are the key cases for the triangle inequality?

---

4. Why do  $d$  and  $\tilde{d}$  agree on distances less than 1?

---

5. Why does shrinking a radius to  $\min\{r, 1/2\}$  transfer openness?

---



Part (a).

Check nonnegativity, definiteness, and symmetry:

---

For the triangle inequality, split into cases depending on whether  $d(x, y) < 1$  and  $d(y, z) < 1$ :

---

Conclude  $\tilde{d}$  is a metric:

---

Part (b).

Assume  $A$  is open in  $(X, \tilde{d})$ . Use  $\tilde{d}(x, a) \leq d(x, a)$  to get a  $d$ -ball inside  $A$ :

---

Assume  $A$  is open in  $(X, d)$ . Set  $r_2 = \min\{r_1, 1/2\}$  and prove the  $\tilde{d}$ -ball is inside the  $d$ -ball:

---

Use this page for part (a).

Use this page for part (b).

## Official Solution (Professor)

(a) There are 4 properties to check.

1.  $\tilde{d}(x, y) \geq 0$  for all  $x, y \in X$ . Note that  $\tilde{d}(x, y)$  is equal to 1 or  $d(x, y)$ . Both of these are nonnegative; we know  $d(x, y) \geq 0$  since  $d$  is a metric on  $X$ .
2.  $\tilde{d}(x, y) = 0 \iff x = y$ . If  $x = y$ , then  $d(x, y) = 0$ , so  $\tilde{d}(x, y) = \min\{0, 1\} = 0$ . Conversely, if  $\tilde{d}(x, y) = 0$ , then  $\tilde{d}(x, y) \neq 1$  and so we must have  $d(x, y) = \tilde{d}(x, y) = 0$ . As  $d$  is a metric, this can only mean that  $x = y$ .
3.  $\tilde{d}(x, y) = \tilde{d}(y, x)$  for all  $x, y \in X$ . As  $d$  is a metric, we know that  $d(x, y) = d(y, x)$ , and so

$$\tilde{d}(x, y) = \min\{d(x, y), 1\} = \min\{d(y, x), 1\} = \tilde{d}(y, x).$$

4.  $\tilde{d}(x, z) \leq \tilde{d}(x, y) + \tilde{d}(y, z)$  for all  $x, y, z \in X$ . We distinguish 4 possible cases:

- $d(x, y) < 1$  and  $d(y, z) < 1$ : Then

$$\tilde{d}(x, z) \leq d(x, z) \leq d(x, y) + d(y, z) = \tilde{d}(x, y) + \tilde{d}(y, z).$$

- $d(x, y) < 1$  and  $d(y, z) \geq 1$ : Then

$$\tilde{d}(x, z) \leq 1 \leq d(x, y) + 1 = \tilde{d}(x, y) + \tilde{d}(y, z).$$

- $d(x, y) \geq 1$  and  $d(y, z) < 1$ : Then

$$\tilde{d}(x, z) \leq 1 \leq 1 + d(y, z) = \tilde{d}(x, y) + \tilde{d}(y, z).$$

- $d(x, y) \geq 1$  and  $d(y, z) \geq 1$ : Then

$$\tilde{d}(x, z) \leq 1 \leq 1 + 1 = \tilde{d}(x, y) + \tilde{d}(y, z).$$

In all cases, the triangle inequality holds.

(b)  $\Leftarrow$  Suppose that  $A$  is open in  $(X, \tilde{d})$ . Fix  $a \in A$ . Then there exists  $r > 0$  so that

$$B_r^{\tilde{d}}(a) = \{x \in X : \tilde{d}(x, a) < r\} \subseteq A.$$

Note that  $B_r^d(a) \subseteq B_r^{\tilde{d}}(a)$ , since

$$x \in B_r^d(a) \implies \tilde{d}(x, a) \leq d(x, a) < r \implies x \in B_r^{\tilde{d}}(a).$$

So  $B_r^d(a) \subseteq B_r^{\tilde{d}}(a) \subseteq A$ . As  $a \in A$  was arbitrary, we conclude that  $A$  is open in  $(X, d)$ .

$\Rightarrow$  Suppose that  $A$  is open in  $(X, d)$ . Fix  $a \in A$ . Then there exists  $r_1 > 0$  so that  $B_{r_1}^d(a) \subseteq A$ . Set  $r_2 = \min\{r_1, \frac{1}{2}\}$ . Note that  $B_{r_2}^{\tilde{d}}(a) \subseteq B_{r_1}^d(a)$ , since

$$x \in B_{r_2}^{\tilde{d}}(a) \implies \tilde{d}(x, a) = \min\{d(x, a), 1\} < r_2 \leq \frac{1}{2}$$

$$\implies \tilde{d}(x, a) = d(x, a) \implies d(x, a) < r_2 \leq r_1.$$

So  $B_{r_2}^{\tilde{d}}(a) \subseteq B_{r_1}^d(a) \subseteq A$ . As  $a \in A$  was arbitrary, we conclude that  $A$  is open in  $(X, \tilde{d})$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 5

---

### Problem Statement

### Homework 8, Problem 5

Let  $(X, d)$  be a metric space,  $A \subseteq X$ , and  $x \in X$ . Prove that  $x$  is in  $A^\circ$  if and only if there exists an open set  $U$  so that  $x \in U \subseteq A$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**



1. What is the definition of  $x \in A^\circ$ ?

---

2. Why is an open ball an open set?

---

3. In the forward direction, what should  $U$  be?

---

4. In the reverse direction, why does openness of  $U$  give a ball around  $x$ ?

---

5. How does  $B_r(x) \subseteq U \subseteq A$  finish the proof?

---

**Forward direction.**

**Assume  $x \in A^\circ$ . Choose  $r > 0$  with  $B_r(x) \subseteq A$ :**

---

**Set  $U =$  and verify open,  $x \in U$ , and  $U \subseteq A$ :**

---

**Reverse direction.**

**Assume  $x \in U \subseteq A$  with  $U$  open. Choose  $r > 0$  with:**

---

**Use  $B_r(x) \subseteq U \subseteq A$  to conclude  $x \in A^\circ$ :**

---

Use this page for both directions.

### Official Solution (Professor)

$\Rightarrow$  Suppose that  $x \in A^\circ$ . Then there exists  $r > 0$  such that  $B_r(x) \subseteq A$ . Notice that  $U = B_r(x)$  is an open set that satisfies  $x \in U \subseteq A$ .

$\Leftarrow$  Suppose that  $x \in U \subseteq A$  for some open set  $U$ . As  $U$  is open, there exists  $r > 0$  so that  $B_r(x) \subseteq U$ . Together we have  $B_r(x) \subseteq U \subseteq A$ , and so  $x \in A^\circ$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 6

---

### Problem Statement

### Homework 8, Problem 6

Let  $(X, d)$  be a metric space and let  $A$  be a nonempty subset of  $X$ . Prove that  $A$  is open if and only if there exists a set  $I$ , positive numbers  $r_i > 0$ , and points  $a_i \in X$  so that

$$A = \bigcup_{i \in I} B_{r_i}(a_i).$$

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**



1. Why should open sets be expressible as unions of balls?

---

2. In the forward direction, what index set  $I$  is the natural choice?

---

3. Why does every point  $a \in A$  belong to its chosen ball  $B_{r_a}(a)$ ?

---

4. Why is each ball  $B_{r_i}(a_i)$  open?

---

5. Which theorem says arbitrary unions of open sets are open?

---

**Forward direction.**

**Assume  $A$  is open. For each  $a \in A$ , choose  $r_a > 0$  with  $B_{r_a}(a) \subseteq A$ :**

---

**Use  $I = A$ , and for each  $i = a \in A$ , set the center  $a_i = a$  and radius  $r_i = r_a$ :**

---

**Prove both inclusions  $A \subseteq \bigcup_{a \in A} B_{r_a}(a)$  and  $\bigcup_{a \in A} B_{r_a}(a) \subseteq A$ :**

---

**Reverse direction.**

**Assume  $A = \bigcup_{i \in I} B_{r_i}(a_i)$ :**

---

**Each ball is open, and arbitrary unions of open sets are open. Conclude:**

---

Use this page for the forward direction.

Use this page for the reverse direction.

### Student-Derived Solution (No official solution available)

We prove both directions.

Suppose first that  $A$  is open. Since  $A$  is nonempty, for each  $a \in A$  choose  $r_a > 0$  such that

$$B_{r_a}(a) \subseteq A.$$

Then

$$A = \bigcup_{a \in A} B_{r_a}(a).$$

Indeed, if  $x \in A$ , then  $x \in B_{r_x}(x)$ , so  $x \in \bigcup_{a \in A} B_{r_a}(a)$ . Conversely, each  $B_{r_a}(a) \subseteq A$ , so the union is contained in  $A$ . Thus  $A$  has the required form, with index set  $I = A$ , and for each index  $i = a \in A$ , center  $a_i = a$  and radius  $r_i = r_a$ .

Conversely, suppose there exist a set  $I$ , positive numbers  $r_i > 0$ , and points  $a_i \in X$  such that

$$A = \bigcup_{i \in I} B_{r_i}(a_i).$$

Each ball  $B_{r_i}(a_i)$  is open in  $(X, d)$ , and an arbitrary union of open sets is open. Therefore  $A$  is open.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Homework 9

---

## Problem 1

---

### Problem Statement

### Homework 9, Problem 1

Consider the metric space  $(\mathbb{R}, d)$  with the usual metric  $d(x, y) = |x - y|$ . For each of the following sets  $A \subseteq \mathbb{R}$ , find its closure, isolated points, and limit points. Prove your answer.

- (a)  $A = [0, 1)$
- (b)  $A = \mathbb{Z}$
- (c)  $A = \{x \in \mathbb{Q} : 0 < x < 1\}$



Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. For each set, which points are obvious members of the closure, and which boundary points require a ball argument?

---

2. How do you prove that a point is not in  $\overline{\mathbb{Z}}$ ? What radius separates it from all integers?

---

3. What is the difference between proving  $a \in A$  is isolated and proving  $a \in A$  is a limit point?

---

4. Where is density of  $\mathbb{Q}$  used in part (c), and why is it not enough just to say “there are rationals nearby”?

---

5. How does the identity  $\overline{A} = (\text{isolated points}) \cup A'$  finish the limit-point computations?

---

**Part (a):**  $A = [0, 1)$

**Closure:** show  $[0, 1) \subseteq \overline{A} \subseteq [0, 1]$ , then prove  $1 \in \overline{A}$  by choosing a point of  $A$  in  $(1 - r, 1 + r)$ :

---

**Isolated points:** fix  $a \in A$ . Given  $r > 0$ , choose a second point of  $A$  inside  $(a - r, a + r)$ :

---

**Limit points:** use the closure/isolated-point relationship to conclude:

---

**Part (b):**  $A = \mathbb{Z}$

**Closure:** for  $x \notin \mathbb{Z}$ , choose  $n \in \mathbb{Z}$  with  $n < x < n + 1$  and radius  $r =$

---

**Isolated points and limit points:** choose the ball around  $n \in \mathbb{Z}$ :

---

**Part (c):**  $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

**Closure:** prove 0, points in  $(0, 1]$ , and no points outside  $[0, 1]$  behave correctly:

---

**Isolated points and limit points:** use density of  $\mathbb{Q}$  to find a second point:

---

Use this page for parts (a) and (b).

Use this page for part (c) and final cleanup.

### Official Solution (Professor)

- (a) Closure:  $\overline{A} = [0, 1]$ . Since  $\overline{A}$  is the smallest closed set containing  $A$ , and  $[0, 1]$  is one such closed set, we must have

$$[0, 1) \subseteq \overline{A} \subseteq [0, 1].$$

Also  $1 \in \overline{A}$ : given  $r > 0$ ,

$$(1 - r, 1 + r) \cap [0, 1) \neq \emptyset,$$

since  $\max\{1 - r/2, 0\} \in (1 - r, 1 + r) \cap [0, 1)$ .

Isolated points:  $\emptyset$ . Fix  $a \in A$ . Given  $r > 0$ , the point

$$\min\left\{a + \frac{r}{2}, \frac{a + 1}{2}\right\}$$

is in  $(a - r, a + r) \cap A$  and is not  $a$ .

Limit points:  $A' = [0, 1]$ . By definition,  $\overline{A}$  is the union of isolated points and limit points, and there are no isolated points.

- (b) Closure:  $\overline{A} = \mathbb{Z}$ . We automatically have  $\mathbb{Z} \subseteq \overline{A}$ . Fix  $x \in \mathbb{Z}^c$ . Let  $n \in \mathbb{Z}$  satisfy  $n < x < n + 1$ . Then

$$r = \min\{x - n, n + 1 - x\}$$

is positive and  $(x - r, x + r) \cap \mathbb{Z} = \emptyset$ . Thus  $x \notin \overline{A}$ .

Isolated points:  $\mathbb{Z}$ . For  $n \in \mathbb{Z}$ ,

$$(n - 1, n + 1) \cap \mathbb{Z} = \{n\}.$$

Limit points:  $A' = \emptyset$ . Every point of  $A$  is isolated.

- (c) Closure:  $\overline{A} = [0, 1]$ . First,  $A \subseteq [0, 1]$ , and  $[0, 1]$  is closed, so  $\overline{A} \subseteq [0, 1]$ . The point 0 is adherent to  $A$ : given  $r > 0$ , choose  $n > \max\{1/r, 1\}$ , so  $1/n \in (-r, r) \cap A$ . Now fix  $x \in (0, 1]$ . By density of  $\mathbb{Q}$  in  $\mathbb{R}$ , for every  $r > 0$  there is  $q \in \mathbb{Q}$  between  $\max\{x - r, 0\}$  and  $x$ , and this  $q \in (x - r, x + r) \cap A$ .

Isolated points:  $\emptyset$ . Given  $x \in A$  and  $r > 0$ , choose  $q \in \mathbb{Q}$  as above. This  $q$  lies in  $(x - r, x + r) \cap A$  and  $q \neq x$ .

Limit points:  $A' = [0, 1]$ . There are no isolated points.



What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

---

### Problem Statement

### Homework 9, Problem 2

Let  $(X, d)$  be a metric space, let  $A$  be a nonempty subset of  $X$ , and let  $U$  be an open subset of  $X$ .

- (a) Prove that  $U \cap \overline{A} \subseteq \overline{U \cap A}$ .
- (b) Prove that  $\overline{U \cap \overline{A}} = \overline{U \cap A}$ . (Hint: Use part (a) and properties of closure from lecture.)
- (c) Conclude that if  $U \cap A = \emptyset$ , then  $U \cap \overline{A} = \emptyset$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why is the assumption that  $U$  is open essential in part (a)?

---

2. After fixing  $x \in U \cap \overline{A}$  and  $r > 0$ , why do we introduce a second radius  $s > 0$ ?

---

3. Why is  $\tilde{r} = \min\{r, s\}$  the right radius to use?

---

4. Which closure property turns part (a) into the first inclusion in part (b)?

---

5. In part (c), why does  $\overline{U \cap \overline{A}} = \emptyset$  force  $U \cap \overline{A} = \emptyset$ ?

---

**Part (a):** prove  $U \cap \overline{A} \subseteq \overline{U \cap A}$ .

**Fix**  $x \in \underline{\hspace{2cm}}$ . **To show**  $x \in \overline{U \cap A}$ , **fix**  $r > 0$  **and aim to find a point in:**

---

**Use openness of  $U$ :** since  $x \in U$ , choose  $s > 0$  such that:

---

**Use**  $x \in \overline{A}$  **with**  $\tilde{r} = \min\{r, s\}$  **to choose**  $y \in$ :

---

**Check the three memberships**  $y \in A$ ,  $y \in U$ , **and**  $y \in B_r(x)$ :

---

Part (b): prove equality of closures.

First inclusion from part (a) and monotonicity of closure:

---

Second inclusion from  $A \subseteq \overline{A}$ :

---

Part (c): empty-set conclusion.

Start from  $U \cap A = \emptyset$ , take closures, use part (b), and then use  $B \subseteq \overline{B}$ :

---

Use this page for parts (a) and (b).



Use this page for part (c) and final cleanup.

### Official Solution (Professor)

- (a) Fix  $x \in U \cap \overline{A}$ . We show  $x \in \overline{U \cap A}$ , i.e., for any  $r > 0$ ,  $B_r(x) \cap U \cap A$  is nonempty. Fix  $r > 0$ . Since  $x \in U$  and  $U$  is open, there exists  $s > 0$  such that  $B_s(x) \subseteq U$ . Set  $\tilde{r} = \min\{r, s\}$ . Since  $x \in \overline{A}$ , the set  $B_{\tilde{r}}(x) \cap A$  is nonempty, so choose  $y \in B_{\tilde{r}}(x) \cap A$ . Then  $y \in A$ . Also  $y \in B_r(x)$ , since  $d(y, x) < \tilde{r} \leq r$ . Finally  $y \in U$ , since  $d(y, x) < \tilde{r} \leq s$ , so  $y \in B_s(x) \subseteq U$ . Hence  $y \in B_r(x) \cap U \cap A$ , as desired.
- (b) Recall that  $B \subseteq C$  implies  $\overline{B} \subseteq \overline{C}$ . Applying this to part (a),

$$\overline{U \cap \overline{A}} \subseteq \overline{U \cap A}.$$

On the other hand,  $A \subseteq \overline{A}$ , so  $U \cap A \subseteq U \cap \overline{A}$ . Therefore

$$\overline{U \cap A} \subseteq \overline{U \cap \overline{A}}.$$

Together,

$$\overline{U \cap \overline{A}} = \overline{U \cap A}.$$

- (c) For any set  $B$ ,  $B \subseteq \overline{B}$ . If  $U \cap A = \emptyset$ , then  $\overline{U \cap A} = \emptyset$ . By part (b),

$$\overline{U \cap \overline{A}} = \emptyset.$$

Therefore  $U \cap \overline{A} = \emptyset$ .

What was the key definition?:

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What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

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### Problem 3

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#### Problem Statement

#### Homework 9, Problem 3

Recall that  $d_2$  is a metric on  $\mathbb{R}^2$ . Consider the subspace

$$Y = \{(x, 0) : x \in \mathbb{R}\}$$

with the induced metric  $d_Y$ .

- (a) Provide an example of a set  $A \subseteq Y$  where  $A$  is open in  $(Y, d_Y)$  but not open in  $(\mathbb{R}^2, d_2)$ .
- (b) Prove that for any  $B \subseteq Y$ , if  $B$  is closed in  $(Y, d_Y)$  then  $B$  is closed in  $(\mathbb{R}^2, d_2)$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does a ball in the subspace  $Y$  look like compared with a ball in  $\mathbb{R}^2$ ?

---

2. Why is the line segment  $(-1, 1) \times \{0\}$  open in  $Y$ ?

---

3. Why does the point  $(0, r/2)$  prove that this set is not open in  $\mathbb{R}^2$ ?

---

4. What lecture result lets us reduce part (b) to proving that  $Y$  is closed in  $\mathbb{R}^2$ ?

---

5. How does the vertical coordinate  $x_2 \neq 0$  give a ball contained in  $Y^c$ ?

---

Part (a): example open in  $Y$ , not open in  $\mathbb{R}^2$ .

Choose  $A =$

---

To prove  $A$  is open in  $Y$ , fix  $x = (x_1, 0) \in A$  and choose  $r =$

---

If  $y \in B_r^Y(x)$ , show  $|y_1| < 1$ , hence  $y \in A$ :

---

To prove  $A$  is not open in  $\mathbb{R}^2$ , fix  $r > 0$  around  $(0, 0)$  and find an off-axis point:

---

Part (b): closedness transfer.

State the lecture corollary being used:

---



Part (b): closedness transfer, continued.

Prove  $Y$  is closed by fixing  $x = (x_1, x_2) \in Y^c$ , so  $x_2 \neq 0$ , and choosing radius  $r =$

---

Show  $B_r^{\mathbb{R}^2}(x) \subseteq Y^c$  using the inequality involving  $|x_2 - y_2|$ :

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Consider

$$A = \{(x_1, 0) : -1 < x_1 < 1\}.$$

First,  $A$  is open in  $(Y, d_Y)$ . Fix  $x = (x_1, 0) \in A$ , and set  $r = 1 - |x_1|$ . Then  $r > 0$ , and

$$B_r^Y(x) = \{y \in Y : d_Y(x, y) < r\} \subseteq A,$$

since

$$y \in B_r^Y(x) \implies r > d_Y(x, y) = \sqrt{|x_1 - y_1|^2 + |0 - 0|^2} = |x_1 - y_1|$$

and therefore

$$|y_1| \leq |x_1| + |y_1 - x_1| < |x_1| + r = 1.$$

Next,  $A$  is not open in  $(\mathbb{R}^2, d_2)$ , because  $(0, 0) \in A$  is not an interior point. For any  $r > 0$ , the point  $(0, r/2)$  lies in  $B_r^{\mathbb{R}^2}((0, 0))$  but not in  $A$ .

(b) By the corollary from Lecture 24, it suffices to show that  $Y$  is closed in  $(\mathbb{R}^2, d_2)$ . Fix  $x = (x_1, x_2) \in Y^c$ . Then  $x_2 \neq 0$ . Set  $r = |x_2|$ . If  $y \in B_r^{\mathbb{R}^2}(x)$ , then

$$r > d_2(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2} \geq |x_2 - y_2|.$$

Hence

$$|y_2| \geq |x_2| - |x_2 - y_2| > |x_2| - r = 0.$$

Thus  $y_2 \neq 0$ , so  $y \in Y^c$ . Therefore  $B_r^{\mathbb{R}^2}(x) \subseteq Y^c$ . Since every point of  $Y^c$  is interior to  $Y^c$ ,  $Y^c$  is open, and  $Y$  is closed.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

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### Problem Statement

### Homework 9, Problem 4

Let  $(X, d)$  be a metric space,  $\{x_n\}_{n \geq 1}$  a sequence in  $X$ , and  $x \in X$ . Prove that the sequence  $\{x_n\}_{n \geq 1}$  converges to  $x$  if and only if for every open set  $U$  that contains  $x$ , we have  $x_n \in U$  for all but finitely many  $n \in \mathbb{N}$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**



1. In the forward direction, how does openness of  $U$  connect an arbitrary neighborhood to a ball around  $x$ ?

---

2. What does “for all but finitely many  $n$ ” mean in terms of choosing an  $N$ ?

---

3. In the reverse direction, why is  $B_r(x)$  the correct open set to test?

---

4. Why is it enough to prove  $x_n \in B_r(x)$  eventually for each  $r > 0$ ?

---

5. What is the main logical difference between the two directions?

---

**Forward direction:** assume  $x_n \rightarrow x$ .

Fix an open set  $U \subseteq X$  with  $x \in U$ . By openness, choose  $r > 0$  such that:

---

By convergence, choose  $N$  such that for all  $n \geq N$ :

---

Conclude  $x_n \in U$  for all but finitely many  $n$ :

---

**Reverse direction:** assume eventual membership in every open  $U \ni x$ .

Fix  $r > 0$ . Apply the hypothesis to the open set:

---

Convert “all but finitely many” into an  $N$  so that for all  $n \geq N$ :

---

Translate  $x_n \in B_r(x)$  into the metric convergence inequality:

---

Conclude because  $r > 0$  was arbitrary:

---

Use this page for both directions.

### Official Solution (Professor)

$\Rightarrow$  Suppose  $x_n \rightarrow x$  in  $(X, d)$  as  $n \rightarrow \infty$ . Fix an open set  $U \subseteq X$  such that  $x \in U$ . Since  $U$  is open, there exists  $r > 0$  such that  $B_r(x) \subseteq U$ . By definition of convergence, there exists  $N$  such that  $d(x_n, x) < r$  for all  $n \geq N$ . Therefore  $x_n \in B_r(x) \subseteq U$  for all  $n \geq N$ .

$\Leftarrow$  Suppose that for every open set  $U$  containing  $x$ , we have  $x_n \in U$  for all but finitely many  $n \in \mathbb{N}$ . Fix  $r > 0$ . Apply the assumption to the open set  $B_r(x)$ . Then  $x_n \in B_r(x)$  for all but finitely many  $n \in \mathbb{N}$ . Let  $N$  be larger than the largest index for which  $x_n \notin B_r(x)$ . Then for every  $n \geq N$ ,  $x_n \in B_r(x)$ , and hence  $d(x_n, x) < r$ . Since  $r > 0$  was arbitrary,  $\{x_n\}_{n \geq 1}$  converges to  $x$  in  $(X, d)$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 5

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### Problem Statement

### Homework 9, Problem 5

Recall that  $d_2$  is a metric on  $\mathbb{R}^2$ . Suppose that  $x^n = (x_1^n, x_2^n)$  is a sequence of points in  $\mathbb{R}^2$ .

- (a) Prove that  $|x_1^n - x_1| \leq d_2(x^n, x)$  for any  $x = (x_1, x_2) \in \mathbb{R}^2$ .
- (b) Prove that the sequence  $\{x^n\}_{n \geq 1}$  converges in  $(\mathbb{R}^2, d_2)$  if and only if the sequences  $\{x_1^n\}_{n \geq 1}$  and  $\{x_2^n\}_{n \geq 1}$  of real numbers both converge.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why does the Euclidean distance  $d_2(x^n, x)$  control each coordinate difference?

---

2. In the forward direction, what is squeezed between 0 and  $d_2(x^n, x)$ ?

---

3. Why do the coordinate sequences converge to the coordinates of the same point  $x$ ?

---

4. In the reverse direction, how do the real limit laws imply  $d_2(x^n, x) \rightarrow 0$ ?

---

5. What would change if the problem were in  $\mathbb{R}^3$  or  $\mathbb{R}^m$ ?

---

**Part (a): coordinate estimate.**

**Write the formula for  $d_2(x^n, x)$ :**

---

**Use  $|x_2^n - x_2|^2 \geq 0$  to get the desired inequality:**

---

**Part (b), forward direction.**

**Assume  $x^n \rightarrow x = (x_1, x_2)$  in  $(\mathbb{R}^2, d_2)$ . Then  $d_2(x^n, x) \rightarrow$**

---

**Use part (a) and squeeze to prove  $x_1^n \rightarrow x_1$ :**

---

**Repeat the same argument for the second coordinate:**

---

Part (b), reverse direction.

Assume  $x_1^n \rightarrow x_1$  and  $x_2^n \rightarrow x_2$ . Set  $x =$

---

Use the formula for  $d_2(x^n, x)$  and the limit laws to conclude:

---

Use this page for part (a) and the forward direction of part (b).

Use this page for the reverse direction of part (b).

### Official Solution (Professor)

(a) Recall that

$$d_2(x^n, x) = \left(|x_1^n - x_1|^2 + |x_2^n - x_2|^2\right)^{1/2}.$$

Since  $|x_2^n - x_2|^2 \geq 0$ ,

$$d_2(x^n, x) \geq \left(|x_1^n - x_1|^2 + 0\right)^{1/2} \geq |x_1^n - x_1|.$$

(b)  $\Rightarrow$  Suppose  $\{x^n\}_{n \geq 1}$  converges to  $x$  in  $(\mathbb{R}^2, d_2)$ . Then

$$d_2(x^n, x) \rightarrow 0.$$

By part (a),

$$0 \leq |x_1^n - x_1| \leq d_2(x^n, x).$$

By the squeeze theorem,  $|x_1^n - x_1| \rightarrow 0$ , so  $\{x_1^n\}_{n \geq 1}$  converges to  $x_1$ . Similarly,

$$0 \leq |x_2^n - x_2| \leq d_2(x^n, x),$$

so  $\{x_2^n\}_{n \geq 1}$  converges to  $x_2$ .

$\Leftarrow$  Suppose  $\{x_1^n\}_{n \geq 1}$  converges to  $x_1$  and  $\{x_2^n\}_{n \geq 1}$  converges to  $x_2$ . Then

$$|x_1^n - x_1| \rightarrow 0 \quad \text{and} \quad |x_2^n - x_2| \rightarrow 0.$$

By the limit laws,

$$d_2(x^n, x) = \left(|x_1^n - x_1|^2 + |x_2^n - x_2|^2\right)^{1/2} \rightarrow 0.$$

Thus  $\{x^n\}_{n \geq 1}$  converges to  $x$  in  $(\mathbb{R}^2, d_2)$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---



## Problem 6

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### Problem Statement

### Homework 9, Problem 6

Let  $(X, d)$  be a metric space and  $\{x_n\}_{n \geq 1}$  a sequence in  $X$ . Prove that if  $\{x_n\}_{n \geq 1}$  converges to  $x \in X$ , then for any  $y \in X$ , the sequence  $\{d(x_n, y)\}_{n \geq 1}$  of real numbers converges to  $d(x, y)$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What real-valued sequence are we trying to prove converges?

---

2. Why should the triangle inequality compare  $d(x_n, y)$  and  $d(x, y)$ ?

---

3. How do two applications of the triangle inequality produce an absolute value estimate?

---

4. What quantity goes to 0, and why does it squeeze the distance error?

---

5. How is this problem a sequence version of continuity of the map  $z \mapsto d(z, y)$ ?

**Setup.**

**Fix  $y \in X$ . Since  $x_n \rightarrow x$ , we know  $d(x_n, x) \rightarrow$**

---

**Triangle inequality estimates.**

**First apply triangle inequality to bound  $d(x_n, y)$  in terms of  $d(x_n, x)$  and  $d(x, y)$ :**

---

**Rearrange to get  $d(x_n, y) - d(x, y) \leq$**

---

**Apply triangle inequality again to bound  $d(x, y)$  in terms of  $d(x, x_n)$  and  $d(x_n, y)$ :**

---

**Rearrange to get  $d(x, y) - d(x_n, y) \leq$**

---

Finish.

Combine the two inequalities into an absolute value estimate:

---

Use squeeze theorem to conclude:

---

Use this page for the complete proof.

### Official Solution (Professor)

Fix  $y \in X$ . By the triangle inequality,

$$\begin{aligned}d(x_n, y) &\leq d(x_n, x) + d(x, y) &\implies d(x_n, y) - d(x, y) &\leq d(x_n, x), \\d(x, y) &\leq d(x, x_n) + d(x_n, y) &\implies d(x, y) - d(x_n, y) &\leq d(x_n, x).\end{aligned}$$

Together,

$$|d(x_n, y) - d(x, y)| \leq d(x_n, x).$$

Since  $\{x_n\}_{n \geq 1}$  converges to  $x$  in  $(X, d)$ , the right-hand side converges to 0. Therefore, by the squeeze theorem,

$$|d(x_n, y) - d(x, y)| \rightarrow 0,$$

and thus  $d(x_n, y)$  converges to  $d(x, y)$ .



What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Homework 10

---

## Problem 1

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### Problem Statement

### Homework 10, Problem 1

Let  $(X, d)$  be a metric space and let  $Y \subseteq X$ . We call the set  $Y$  complete if the subspace  $Y$  with the induced metric is complete. Prove that if  $Y$  is complete, then  $Y$  is closed in  $X$ . (Hint: It suffices to show  $Y \subseteq \bar{Y}$ . Use sequences.)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does it mean for  $Y$  to be complete as a subspace of  $X$ ?

---

2. What sequential characterization of closure should be used after fixing  $x \in \overline{Y}$ ?

---

3. Why is a sequence converging in  $X$  automatically Cauchy in  $X$ ?

---

4. Why is the same sequence Cauchy in the induced metric on  $Y$ ?

---

5. Where is uniqueness of limits in a metric space used?

---

Goal: prove  $Y$  is closed.

It suffices to show  $\overline{Y} \subseteq Y$ . Fix:

---

Use the sequential characterization of closure to get a sequence  $\{y_n\} \subseteq Y$  such that:

---

Explain why  $\{y_n\}$  is Cauchy in  $Y$ :

---

Use completeness of  $Y$  to get  $y \in Y$  with:

---

Compare convergence in  $Y$  with convergence in  $X$ , then use uniqueness of limits:

---

Conclude  $\overline{Y} \subseteq Y$ , hence  $Y$  is closed:

---



Use this page for the complete proof.

### Student-Derived Solution (No official solution available)

We show that  $\overline{Y} \subseteq Y$ . Since always  $Y \subseteq \overline{Y}$ , this implies  $\overline{Y} = Y$ , and hence  $Y$  is closed in  $X$ .

Let  $x \in \overline{Y}$ . By the sequential characterization of closure, there exists a sequence  $\{y_n\}_{n \geq 1}$  in  $Y$  such that

$$y_n \rightarrow x \quad \text{in } (X, d).$$

Every convergent sequence in a metric space is Cauchy, so  $\{y_n\}$  is Cauchy in  $(X, d)$ . Since the induced metric on  $Y$  is the restriction of  $d$ , the same sequence is Cauchy in  $(Y, d_Y)$ .

Because  $Y$  is complete, there is some  $y \in Y$  such that

$$y_n \rightarrow y \quad \text{in } (Y, d_Y).$$

Convergence in the subspace metric is the same distance condition as convergence in  $X$ , so  $y_n \rightarrow y$  in  $(X, d)$ . Thus the sequence  $\{y_n\}$  converges to both  $x$  and  $y$  in  $X$ . By uniqueness of limits in metric spaces,  $x = y$ . Since  $y \in Y$ , we have  $x \in Y$ . Therefore  $\overline{Y} \subseteq Y$ , so  $Y$  is closed in  $X$ .

What was the key definition?:

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What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

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### Problem Statement

### Homework 10, Problem 2

Let  $(X, d)$  be a metric space. Prove that if  $A, B \subseteq X$  are both compact then  $A \cup B$  is compact.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What definition of compactness is most direct for a finite union?

---

2. Why does an open cover of  $A \cup B$  automatically cover  $A$  and  $B$  separately?

---

3. What finite subcover do you get from compactness of  $A$ ?

---

4. What finite subcover do you get from compactness of  $B$ ?

---

5. Why is the union of the two finite index sets still finite and enough to cover  $A \cup B$ ?

---

Open-cover setup.

Let  $\{U_i\}_{i \in I}$  be an arbitrary open cover of  $A \cup B$ :

---

Restrict the same cover to  $A$ , and use compactness of  $A$  to choose a finite  $J_A \subseteq I$ :

---

Restrict the same cover to  $B$ , and use compactness of  $B$  to choose a finite  $J_B \subseteq I$ :

---

Set  $J =$

---

Check  $J$  is finite and  $\{U_i\}_{i \in J}$  covers  $A \cup B$ :

---



Conclude compactness because the original open cover was arbitrary:

---

Use this page for the complete proof.

### Student-Derived Solution (No official solution available)

Let  $\{U_i\}_{i \in I}$  be an open cover of  $A \cup B$ . Thus each  $U_i$  is open in  $X$ , and

$$A \cup B \subseteq \bigcup_{i \in I} U_i.$$

Since  $A \subseteq A \cup B$ , the same family is an open cover of  $A$ . Because  $A$  is compact, there is a finite set  $J_A \subseteq I$  such that

$$A \subseteq \bigcup_{i \in J_A} U_i.$$

Similarly,  $\{U_i\}_{i \in I}$  is an open cover of  $B$ , so compactness of  $B$  gives a finite set  $J_B \subseteq I$  such that

$$B \subseteq \bigcup_{i \in J_B} U_i.$$

Set  $J = J_A \cup J_B$ . Then  $J$  is finite, and

$$A \cup B \subseteq \left( \bigcup_{i \in J_A} U_i \right) \cup \left( \bigcup_{i \in J_B} U_i \right) = \bigcup_{i \in J} U_i.$$

Thus every open cover of  $A \cup B$  has a finite subcover. Therefore  $A \cup B$  is compact.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

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#### Problem Statement

#### Homework 10, Problem 3

Let  $(X, d)$  be a metric space.

- (a) Prove that if  $A \subseteq B \subseteq X$ ,  $A$  is closed in  $X$ , and  $B$  is compact, then  $A$  is compact. (Hint: Given an open cover of  $A$ , we can add  $A^c$  to get an open cover of  $B$ .)
- (b) Prove that if  $C, D \subseteq X$  are both compact, then  $C \cap D$  is compact. (Hint: Use (a).)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why does adding  $A^c$  to an open cover of  $A$  help cover all of  $B$ ?

---

2. Where is the assumption that  $A$  is closed used?

---

3. After compactness of  $B$  gives a finite subcover, why can  $A^c$  be discarded when covering  $A$ ?

---

4. Which lecture result says compact subsets of a metric space are closed?

---

5. In part (b), which compact set plays the role of  $B$  in part (a)?

---



Part (a): closed subset of a compact set.

Let  $\{U_i\}_{i \in I}$  be an open cover of  $A$ . Since  $A$  is closed,  $A^c$  is:

---

Show  $\{A^c\} \cup \{U_i\}_{i \in I}$  covers  $B$ :

---

Use compactness of  $B$  to extract a finite subcover:

---

Restrict back to  $A$  and remove the  $A^c$  part:

---

Part (b): compact intersection.

Use compactness  $\Rightarrow$  closed to show  $C \cap D$  is closed in  $X$ :

---

Apply part (a) with  $A = C \cap D$  and  $B =$

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Let  $\{U_i\}_{i \in I}$  be an open cover of  $A$ . Since  $A$  is closed,  $A^c$  is open. Thus

$$\{A^c\} \cup \{U_i\}_{i \in I}$$

is an open cover of  $B$ , because

$$B \subseteq X = A^c \cup A \subseteq A^c \cup \bigcup_{i \in I} U_i.$$

Since  $B$  is compact, there is a finite subcover for  $B$ . We may write it as

$$\{A^c\} \cup \{U_i\}_{i \in J}$$

for some finite  $J \subseteq I$ . Then

$$B \subseteq A^c \cup \bigcup_{i \in J} U_i.$$

Since  $A \subseteq B$  and  $A \cap A^c = \emptyset$ , it follows that

$$A \subseteq \bigcup_{i \in J} U_i.$$

Hence  $\{U_i\}_{i \in J}$  is a finite subcover of  $A$ . Therefore  $A$  is compact.

(b) Since  $C$  and  $D$  are compact, they are closed by Lecture 27. Hence  $C \cap D$  is closed. Also  $C \cap D \subseteq C$ , and  $C$  is compact. By part (a),  $C \cap D$  is compact.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

---

### Problem Statement

### Homework 10, Problem 4

Let  $(X, d)$  be a metric space and let  $Y \subseteq X$ . Prove that if  $Y$  is compact, then  $Y$  is complete.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_



**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. To prove completeness, what kind of sequence in  $Y$  should you start with?

---

2. Which compactness theorem gives a convergent subsequence in  $Y$ ?

---

3. Why is a convergent subsequence not enough by itself to prove completeness?

---

4. How does the Cauchy property transfer convergence from a subsequence to the full sequence?

---

5. Why does the final limit lie in  $Y$ ?

---

**Setup:** prove every Cauchy sequence in  $Y$  converges in  $Y$ .

Let  $\{y_n\}$  be a Cauchy sequence in  $Y$ :

---

Use compactness and Lecture 26 to obtain a subsequence  $\{y_{k_n}\}$  and point  $y \in Y$  such that:

---

Fix  $r > 0$ . From subsequence convergence choose  $N_1$  such that:

---

From the Cauchy property choose  $N_2$  such that:

---

Use triangle inequality with  $y_n$ ,  $y_{k_n}$ , and  $y$ :

---

Conclude the whole sequence converges to  $y \in Y$ :

---

Use this page for the complete proof.

### Official Solution (Professor)

Let  $\{y_n\}_{n \geq 1}$  be a Cauchy sequence in  $Y$ . We show that it converges to some point of  $Y$ .

Since  $Y$  is compact, by Lecture 26 there exists a subsequence  $\{y_{k_n}\}_{n \geq 1}$  that converges to some point  $y \in Y$ .

We prove the whole sequence converges to  $y$ . Fix  $r > 0$ . Since  $y_{k_n} \rightarrow y$ , there exists  $N_1$  such that

$$d(y_{k_n}, y) < \frac{r}{2} \quad \text{for all } n \geq N_1.$$

Since  $\{y_n\}$  is Cauchy, there exists  $N_2$  such that

$$d(y_n, y_m) < \frac{r}{2} \quad \text{for all } n, m \geq N_2.$$

Set  $N = \max\{N_1, N_2\}$ . For  $n \geq N$ , we have  $k_n \geq n \geq N$ , and so

$$d(y_n, y) \leq d(y_n, y_{k_n}) + d(y_{k_n}, y) < \frac{r}{2} + \frac{r}{2} = r.$$

Since  $r > 0$  was arbitrary,  $y_n \rightarrow y$ . Because  $y \in Y$ , every Cauchy sequence in  $Y$  converges in  $Y$ . Therefore  $Y$  is complete.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Homework 11

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## Problem 1

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### Problem Statement

### Homework 11, Problem 1

Consider the metric space  $(\mathbb{R}, d)$  with the usual metric  $d(x, y) = |x - y|$ . Note that the set  $\mathbb{Q}$  is not an interval, and so from lecture we know that it must be disconnected.

- (a) Find two sets  $A, B \subseteq \mathbb{R}$  that are nonempty and separated so that  $\mathbb{Q} = A \cup B$ , and prove your answer.
- (b) Find a set  $\emptyset \neq C \subsetneq \mathbb{Q}$  that is both open and closed in  $\mathbb{Q}$ , and prove your answer.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why is cutting  $\mathbb{Q}$  at an irrational number better than cutting it at a rational number?

---

2. What does it mean for two subsets of  $\mathbb{R}$  to be separated?

---

3. What are the closures in  $\mathbb{R}$  of  $(-\infty, \sqrt{2}) \cap \mathbb{Q}$  and  $(\sqrt{2}, \infty) \cap \mathbb{Q}$ ?

---

4. Why does  $\sqrt{2} \notin \mathbb{Q}$  make the separation work?

---

5. How does the subspace characterization of open and closed sets in  $\mathbb{Q}$  prove part (b)?

Part (a): separated decomposition of  $\mathbb{Q}$ .  
Choose an irrational cut point and define  $A =$

---

Define  $B =$

---

Check  $A$  and  $B$  are nonempty and  $A \cup B = \mathbb{Q}$ :

---

Compute  $\overline{A}$  and  $\overline{B}$  in  $\mathbb{R}$ :

---

Use those closures to prove  $\overline{A} \cap B = \emptyset$  and  $A \cap \overline{B} = \emptyset$ :

---

Part (b): clopen subset of  $\mathbb{Q}$ .

Choose  $C =$

---

Prove  $C$  is nonempty and proper:

---

Show  $C$  is open in  $\mathbb{Q}$  and closed in  $\mathbb{Q}$  using subspace results:

---

Use this page for part (a).

Use this page for part (b).



**Student-Derived Solution (No official solution available)**

(a) Define

$$A = (-\infty, \sqrt{2}) \cap \mathbb{Q}, \quad B = (\sqrt{2}, \infty) \cap \mathbb{Q}.$$

Both sets are nonempty, for instance  $1 \in A$  and  $2 \in B$ . Also  $A \cup B = \mathbb{Q}$ , since every rational number is either less than  $\sqrt{2}$  or greater than  $\sqrt{2}$ , and  $\sqrt{2} \notin \mathbb{Q}$ . In  $\mathbb{R}$ ,

$$\overline{A} = (-\infty, \sqrt{2}], \quad \overline{B} = [\sqrt{2}, \infty).$$

Hence

$$\overline{A} \cap B = (-\infty, \sqrt{2}] \cap ((\sqrt{2}, \infty) \cap \mathbb{Q}) = \emptyset,$$

and similarly

$$A \cap \overline{B} = ((-\infty, \sqrt{2}) \cap \mathbb{Q}) \cap [\sqrt{2}, \infty) = \emptyset.$$

Therefore  $A$  and  $B$  are separated, nonempty, and satisfy  $\mathbb{Q} = A \cup B$ .

(b) Let

$$C = (-\infty, \sqrt{2}) \cap \mathbb{Q}.$$

Then  $C \neq \emptyset$ , since  $1 \in C$ , and  $C \subsetneq \mathbb{Q}$ , since  $2 \in \mathbb{Q} \setminus C$ .

The set  $(-\infty, \sqrt{2})$  is open in  $\mathbb{R}$ , so  $C = (-\infty, \sqrt{2}) \cap \mathbb{Q}$  is open in the subspace  $\mathbb{Q}$ . Also  $(-\infty, \sqrt{2}]$  is closed in  $\mathbb{R}$ , and since  $\sqrt{2} \notin \mathbb{Q}$ ,

$$(-\infty, \sqrt{2}] \cap \mathbb{Q} = (-\infty, \sqrt{2}) \cap \mathbb{Q} = C.$$

Thus  $C$  is closed in  $\mathbb{Q}$ . Therefore  $C$  is a nonempty proper subset of  $\mathbb{Q}$  that is both open and closed in  $\mathbb{Q}$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

---

### Problem Statement

### Homework 11, Problem 2

Let  $(X, d)$  be a metric space.

- (a) Prove that if  $F \subseteq X$  is disconnected and closed in  $X$ , then there exist  $A, B \subseteq X$  that are nonempty, separated, and closed in  $X$  so that  $F = A \cup B$ .
- (b) Prove that if  $U \subseteq X$  is disconnected and open in  $X$ , then there exist  $A, B \subseteq X$  that are nonempty, separated, and open in  $X$  so that  $U = A \cup B$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Which Lecture 29 characterization of disconnectedness gives a nontrivial clopen subset of  $F$  or  $U$ ?

---

2. In part (a), why does closed in  $F$  plus  $F$  closed in  $X$  imply closed in  $X$ ?

---

3. In part (b), why does open in  $U$  plus  $U$  open in  $X$  imply open in  $X$ ?

---

4. Why are the two pieces nonempty?

---

5. How do closedness or openness of the pieces imply they are separated?

---

Part (a): closed disconnected set.

Use disconnectedness of  $F$  to choose  $A \neq \emptyset, F$  that is both open and closed in  $F$ . Set  $B =$

---

Explain why  $A$  and  $B$  are nonempty and  $F = A \cup B$ :

---

Use  $F$  closed in  $X$  to prove  $A$  and  $B$  are closed in  $X$ :

---

Use closedness and disjointness to prove separatedness:

---

Part (b): open disconnected set.

Use disconnectedness of  $U$  to choose  $A \neq \emptyset, U$  clopen in  $U$ , and set  $B =$

---

Use  $U$  open in  $X$  to prove  $A$  and  $B$  are open in  $X$ :

---

Use complements of open sets to prove separatedness:

---



Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

- (a) Since  $F$  is disconnected, by Lecture 29 there exists a set  $A \neq \emptyset, F$  that is both open and closed in  $F$ . Set  $B = F \setminus A$ .

The sets  $A$  and  $B$  are nonempty because  $A \neq \emptyset$  and  $A \neq F$ . Since  $A$  is closed in  $F$  and  $F$  is closed in  $X$ ,  $A$  is closed in  $X$  by Lecture 24. Also  $B = F \setminus A$  is closed in  $F$ , because  $A$  is open in  $F$ ; hence  $B$  is closed in  $X$ .

Finally,  $A \cap B = \emptyset$ . Since  $A$  and  $B$  are closed in  $X$ ,  $\overline{A} = A$  and  $\overline{B} = B$ . Thus

$$\overline{A} \cap B = \emptyset, \quad A \cap \overline{B} = \emptyset.$$

Therefore  $A$  and  $B$  are nonempty, separated, closed in  $X$ , and  $F = A \cup B$ .

- (b) Since  $U$  is disconnected, by Lecture 29 there exists a set  $A \neq \emptyset, U$  that is both open and closed in  $U$ . Set  $B = U \setminus A$ .

The sets  $A$  and  $B$  are nonempty. Since  $A$  is open in  $U$  and  $U$  is open in  $X$ ,  $A$  is open in  $X$  by Lecture 24. Since  $A$  is closed in  $U$ ,  $B = U \setminus A$  is open in  $U$ ; because  $U$  is open in  $X$ ,  $B$  is open in  $X$ .

Since  $A \cap B = \emptyset$ , we have  $A \subseteq B^c$ . Because  $B$  is open,  $B^c$  is closed, so  $\overline{A} \subseteq B^c$ , and hence  $\overline{A} \cap B = \emptyset$ . Similarly,  $\overline{B} \cap A = \emptyset$ . Therefore  $A$  and  $B$  are separated, open in  $X$ , nonempty, and  $U = A \cup B$ .

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

---

#### Problem Statement

#### Homework 11, Problem 3

Suppose that  $(X, d)$  is a metric space such that for any pair of points  $a, b \in X$ , there exists a connected set  $C \subseteq X$  so that  $a, b \in C$ . Prove that  $X$  is connected.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why is contradiction the natural proof method here?

---

2. What does a separation of  $X$  give you?

---

3. Why can you choose one point  $a$  from one side and one point  $b$  from the other?

---

4. Which Lecture 29 result says a connected set inside a union of separated sets lies in one side?

---

5. Why does the connected set containing both  $a$  and  $b$  contradict the separation?

---



**Contradiction setup.**

Assume  $X$  is disconnected. Then choose nonempty separated sets  $A, B \subseteq X$  such that:

---

Choose  $a \in A$  and  $b \in B$ . By hypothesis, choose connected  $C \subseteq X$  such that:

---

Since  $C \subseteq X = A \cup B$ , apply the connected-set/separated-union result:

---

Use  $a \in A \cap C$  and  $b \in B \cap C$  to rule out both alternatives:

---

Conclude  $X$  is connected:

---

Use this page for the complete proof.

### Official Solution (Professor)

Suppose toward a contradiction that  $X$  is disconnected. Then there exist sets  $A, B \subseteq X$  such that

$$X = A \cup B, \quad A \neq \emptyset, \quad B \neq \emptyset,$$

and  $A$  and  $B$  are separated.

Choose  $a \in A$  and  $b \in B$ . By the hypothesis, there exists a connected set  $C \subseteq X$  such that  $a, b \in C$ . Since  $C \subseteq X = A \cup B$ , and  $A$  and  $B$  are separated, Lecture 29 implies that either  $C \subseteq A$  or  $C \subseteq B$ .

Because  $a \in A \cap C$ , the only possible case is  $C \subseteq A$ . But then  $B \cap C = \emptyset$ , contradicting  $b \in B \cap C$ . Therefore  $X$  cannot be disconnected, so  $X$  is connected.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

---

### Problem Statement

### Homework 11, Problem 4

Let  $(X, d)$  be a metric space and  $A \subseteq X$ . Prove that if  $A$  is connected, then  $\overline{A}$  is connected.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why should the proof begin by assuming  $\overline{A}$  is disconnected?

---

2. If  $\overline{A} = B \cup C$  with  $B, C$  separated, what does connectedness of  $A \subseteq B \cup C$  imply?

---

3. Why can one assume  $A \subseteq B$  after relabeling?

---

4. How does  $A \subseteq B$  imply  $\overline{A} \subseteq \overline{B}$ ?

---

5. Why does separatedness then force the other side to be empty?

---



**Contradiction setup.**

Assume  $\overline{A}$  is disconnected, so  $\overline{A} = B \cup C$  with  $B, C$  nonempty and separated:

---

Apply connectedness of  $A$  to the separated union  $B \cup C$ :

---

After relabeling, assume  $A \subseteq B$ . Then conclude  $\overline{A} \subseteq$

---

Use separatedness  $\overline{B} \cap C = \emptyset$  to show  $\overline{A} \cap C = \emptyset$ :

---

Explain why this contradicts  $C \neq \emptyset$  and  $C \subseteq \overline{A}$ :

---

Use this page for the complete proof.

### Official Solution (Professor)

Suppose toward a contradiction that  $\overline{A}$  is disconnected. Then there exist sets  $B, C \subseteq X$  such that

$$\overline{A} = B \cup C, \quad B \neq \emptyset, \quad C \neq \emptyset,$$

and  $B$  and  $C$  are separated.

Since  $A$  is connected and  $A \subseteq \overline{A} = B \cup C$ , Lecture 29 implies that either  $A \subseteq B$  or  $A \subseteq C$ . After swapping  $B$  and  $C$  if necessary, assume  $A \subseteq B$ . Then

$$\overline{A} \subseteq \overline{B}.$$

Because  $B$  and  $C$  are separated,  $\overline{B} \cap C = \emptyset$ . Hence

$$\overline{A} \cap C = \emptyset.$$

But  $C \subseteq \overline{A}$ , since  $\overline{A} = B \cup C$ . Therefore  $C = \emptyset$ , a contradiction. Thus  $\overline{A}$  is connected.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 5

---

### Problem Statement

### Homework 11, Problem 5

Let  $(X, d)$  be a metric space and  $A, B \subseteq X$ . Prove that if  $A$  and  $B$  are both connected and  $A \cap B \neq \emptyset$ , then  $A \cup B$  is connected.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why does the assumption  $A \cap B \neq \emptyset$  prevent  $A$  and  $B$  from lying in different sides of a separation?

---

2. If  $A \cup B = C \cup D$  is a separation, what does connectedness of  $A$  imply?

---

3. What does connectedness of  $B$  imply?

---

4. How does a shared point force  $A$  and  $B$  into the same separated side?

---

5. Why does this make the other side empty?

---



**Contradiction setup.**

Assume  $A \cup B$  is disconnected. Choose nonempty separated  $C, D \subseteq X$  such that:

---

Use connectedness of  $A \subseteq C \cup D$  to conclude:

---

After relabeling, assume  $A \subseteq C$ . Use  $A \cap B \neq \emptyset$  to show  $B$  has a point in  $C$ :

---

Use connectedness of  $B$  to conclude  $B \subseteq C$ :

---

Then  $A \cup B \subseteq C$ , contradicting:

---

Use this page for the complete proof.

### Official Solution (Professor)

Suppose toward a contradiction that  $A \cup B$  is disconnected. Then there exist sets  $C, D \subseteq X$  such that

$$A \cup B = C \cup D, \quad C, D \neq \emptyset,$$

and  $C$  and  $D$  are separated.

Since  $A \subseteq C \cup D$  and  $A$  is connected, Lecture 29 implies that either  $A \subseteq C$  or  $A \subseteq D$ . After swapping  $C$  and  $D$ , assume  $A \subseteq C$ .

Similarly, since  $B \subseteq C \cup D$  and  $B$  is connected, either  $B \subseteq C$  or  $B \subseteq D$ . But  $A \subseteq C$  and  $A \cap B \neq \emptyset$ , so at least one point of  $B$  lies in  $C$ . Therefore  $B$  cannot be contained in  $D$ , and we must have  $B \subseteq C$ .

Thus  $A \cup B \subseteq C$ . Since  $A \cup B = C \cup D$ , this forces  $D = \emptyset$ , contradicting the separation. Therefore  $A \cup B$  is connected.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

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## Problem 6

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### Problem Statement

### Homework 11, Problem 6

Let  $(X, d)$  be a metric space.

- (a) Let  $A, B, C \subseteq X$ . Prove that if the sets  $A$  and  $B$  are separated and the sets  $A$  and  $C$  are separated, then the sets  $A$  and  $B \cup C$  are separated.
- (b) Prove that if  $F, G \subseteq X$  are closed and  $F \cup G$  and  $F \cap G$  are connected, then  $F$  is connected.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What two closure-intersection conditions define separated sets?

---

2. In part (a), how do unions distribute over intersections with  $\overline{A}$ ?

---

3. For part (b), why is contradiction a natural strategy?

---

4. If  $F = A \cup B$  is a separation, how does connectedness of  $F \cap G$  force  $F \cap G$  into one side?

---

5. Why does proving  $B$  is separated from  $A \cup G$  contradict connectedness of  $F \cup G$ ?

---



Part (a): separated from a union.

Write what it means that  $A$  is separated from  $B$  and from  $C$ :

---

Compute  $\overline{A} \cap (B \cup C)$ :

---

Use  $\overline{B \cup C} = \overline{B} \cap \overline{C}$  to compute  $A \cap \overline{B \cup C}$ :

---

Conclude  $A$  and  $B \cup C$  are separated:

---

Part (b): contradiction.

Assume  $F$  is disconnected and choose  $F = A \cup B$  with  $A, B$  nonempty separated:

---

Use connectedness of  $F \cap G$  to assume, after relabeling,  $F \cap G \subseteq A$ :

---

Show  $B$  and  $G$  are separated, using closedness of  $G$ :

---

Apply part (a) to show  $B$  and  $A \cup G$  are separated, yielding a separation of  $F \cup G$ :

---

Use this page for part (a).

Use this page for part (b).

### Official Solution (Professor)

(a) Since  $A$  and  $B$  are separated,

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

Since  $A$  and  $C$  are separated,

$$\overline{A} \cap C = \emptyset \quad \text{and} \quad A \cap \overline{C} = \emptyset.$$

Therefore

$$\overline{A} \cap (B \cup C) = (\overline{A} \cap B) \cup (\overline{A} \cap C) = \emptyset.$$

Also, by Lecture 23,

$$\overline{B \cup C} = \overline{B} \cup \overline{C}.$$

Hence

$$A \cap \overline{B \cup C} = (A \cap \overline{B}) \cup (A \cap \overline{C}) = \emptyset.$$

Thus  $A$  and  $B \cup C$  are separated.

(b) Suppose toward a contradiction that  $F$  is disconnected. Then there exist  $A, B \subseteq X$  such that

$$F = A \cup B, \quad A, B \neq \emptyset,$$

and  $A$  and  $B$  are separated.

Since  $F \cap G$  is connected and  $F \cap G \subseteq A \cup B$ , Lecture 29 implies that either  $F \cap G \subseteq A$  or  $F \cap G \subseteq B$ . After swapping  $A$  and  $B$  if necessary, assume  $F \cap G \subseteq A$ . We claim that  $B$  and  $G$  are separated. First  $B \cap G = \emptyset$ , because any point in  $B \cap G$  would lie in  $F \cap G \subseteq A$ , contradicting  $A \cap B = \emptyset$ . Since  $G$  is closed,  $\overline{G} = G$ , and this gives  $B \cap \overline{G} = \emptyset$ . Also, if  $x \in G = \overline{G}$ , then either  $x \in G \setminus F$  or  $x \in F \cap G \subseteq A$ , so  $x \notin B$ . Hence  $\overline{B} \cap G = \emptyset$ . Thus  $B$  and  $G$  are separated.

Since  $A$  and  $B$  are separated, part (a) implies that  $B$  and  $A \cup G$  are separated. Both are nonempty, and

$$F \cup G = (A \cup B) \cup G = (A \cup G) \cup B.$$

Therefore  $F \cup G$  is disconnected, contradicting the hypothesis. Hence  $F$  is connected.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

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## Homework 12

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## Problem 1

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### Problem Statement

### Homework 12, Problem 1

Let  $(X, d_X)$  and  $(Y, d_Y)$  be metric spaces,  $f : X \rightarrow Y$ , and  $x_0 \in X$ . Prove that  $f$  is continuous at  $x_0$  if and only if for any open set  $V \subseteq Y$  containing  $f(x_0)$ , there is an open set  $U \subseteq X$  containing  $x_0$  so that  $x \in U$  implies  $f(x) \in V$ .



Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What is the  $\varepsilon$ - $\delta$  definition of continuity at  $x_0$ ?

---

2. In the forward direction, how does openness of  $V$  produce a ball around  $f(x_0)$ ?

---

3. Why is  $U = B_\delta(x_0)$  the natural choice after applying continuity?

---

4. In the reverse direction, why should  $V = B_\varepsilon(f(x_0))$  be used?

---

5. How does openness of  $U$  produce the  $\delta$  needed for continuity?

---

**Forward direction.**

**Assume  $f$  is continuous at  $x_0$ . Fix open  $V \subseteq Y$  with  $f(x_0) \in V$ . Choose  $\varepsilon > 0$  so that:**

---

**Use continuity at  $x_0$  to choose  $\delta > 0$  so that:**

---

**Set  $U =$**

---

**Check  $U$  is open, contains  $x_0$ , and  $x \in U \Rightarrow f(x) \in V$ :**

---

**Reverse direction.**

**Assume the open-neighborhood condition. Fix  $\varepsilon > 0$  and set  $V =$**

---

Use the hypothesis to get open  $U \ni x_0$  with  $f(U) \subseteq V$ :

---

Use openness of  $U$  to choose  $\delta > 0$  with  $B_\delta(x_0) \subseteq U$ :

---

Conclude the  $\varepsilon$ - $\delta$  condition:

---

Use this page for the forward direction.

Use this page for the reverse direction.

**Student-Derived Solution (No official solution available yet)**

$\Rightarrow$  Assume  $f$  is continuous at  $x_0$ . Let  $V \subseteq Y$  be open with  $f(x_0) \in V$ . Since  $V$  is open, there is  $\varepsilon > 0$  such that

$$B_\varepsilon(f(x_0)) \subseteq V.$$

By continuity at  $x_0$ , there is  $\delta > 0$  such that

$$d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

Let  $U = B_\delta(x_0)$ . Then  $U$  is open,  $x_0 \in U$ , and if  $x \in U$ , then  $f(x) \in B_\varepsilon(f(x_0)) \subseteq V$ .

$\Leftarrow$  Assume the open-set condition. Let  $\varepsilon > 0$ , and set  $V = B_\varepsilon(f(x_0))$ . Then  $V$  is open and contains  $f(x_0)$ , so there is an open set  $U \subseteq X$  with  $x_0 \in U$  and  $f(U) \subseteq V$ . Since  $U$  is open and  $x_0 \in U$ , there is  $\delta > 0$  such that  $B_\delta(x_0) \subseteq U$ . Thus if  $d_X(x, x_0) < \delta$ , then  $x \in U$ , so  $f(x) \in V$ , meaning

$$d_Y(f(x), f(x_0)) < \varepsilon.$$

Therefore  $f$  is continuous at  $x_0$ .



What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 2

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### Problem Statement

### Homework 12, Problem 2

Let  $(X, d_X)$  and  $(Y, d_Y)$  be metric spaces and  $f : X \rightarrow Y$  a function. Prove that  $f$  is continuous if and only if  $f(\overline{A}) \subseteq \overline{f(A)}$  for any  $A \subseteq X$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. What does  $x \in \overline{A}$  mean sequentially in a metric space?

---

2. In the forward direction, how does continuity move a convergent sequence from  $X$  to  $Y$ ?

---

3. Why does  $f(a_n) \rightarrow f(x)$  prove  $f(x) \in \overline{f(A)}$ ?

---

4. In the reverse direction, why choose  $A = X \setminus f^{-1}(V)$  for an open  $V \subseteq Y$ ?

---

5. Why does  $x \notin \overline{A}$  produce an open neighborhood  $U$  of  $x$  with  $f(U) \subseteq V$ ?

---

**Forward direction.**

Assume  $f$  is continuous. Fix  $A \subseteq X$  and  $x \in \overline{A}$ . Choose a sequence  $a_n \in A$  such that:

---

Use continuity to show  $f(a_n) \rightarrow$

---

Conclude  $f(x) \in \overline{f(A)}$ , hence  $f(\overline{A}) \subseteq \overline{f(A)}$ :

---

**Reverse direction.**

Assume  $f(\overline{A}) \subseteq \overline{f(A)}$  for every  $A \subseteq X$ . Fix open  $V \subseteq Y$  with  $f(x) \in V$ :

---

Define  $A = X \setminus f^{-1}(V)$ . Suppose for contradiction that  $x \in \overline{A}$ :

---

Use the closure hypothesis to force  $f(x) \in \overline{f(A)}$ :

---

Use  $f(A) \subseteq Y \setminus V$  and  $Y \setminus V$  closed to contradict  $f(x) \in V$ :

---

From  $x \notin \overline{A}$ , build an open  $U \ni x$  with  $U \subseteq f^{-1}(V)$ , then apply Problem 1:

---

Use this page for the forward direction.



Use this page for the reverse direction.

### Student-Derived Solution (No official solution available yet)

$\Rightarrow$  Assume  $f$  is continuous. Let  $A \subseteq X$ , and let  $x \in \overline{A}$ . By the sequential characterization of closure, there is a sequence  $\{a_n\} \subseteq A$  such that  $a_n \rightarrow x$ . Since  $f$  is continuous,

$$f(a_n) \rightarrow f(x).$$

Each  $f(a_n) \in f(A)$ , so  $f(x) \in \overline{f(A)}$ . Therefore

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

$\Leftarrow$  Assume that for every  $A \subseteq X$ ,

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

We prove continuity using Problem 1. Let  $V \subseteq Y$  be open and suppose  $f(x) \in V$ . Set

$$A = X \setminus f^{-1}(V).$$

We claim  $x \notin \overline{A}$ . If  $x \in \overline{A}$ , then by the hypothesis

$$f(x) \in f(\overline{A}) \subseteq \overline{f(A)}.$$

But if  $a \in A$ , then  $f(a) \notin V$ , so  $f(A) \subseteq Y \setminus V$ . Since  $Y \setminus V$  is closed,

$$\overline{f(A)} \subseteq Y \setminus V.$$

This would imply  $f(x) \notin V$ , contradicting  $f(x) \in V$ . Thus  $x \notin \overline{A}$ .

Since  $x \notin \overline{A}$ , there exists an open set  $U \subseteq X$  containing  $x$  such that  $U \cap A = \emptyset$ . Hence

$$U \subseteq X \setminus A = f^{-1}(V),$$

so  $f(U) \subseteq V$ . By Problem 1,  $f$  is continuous at  $x$ . Since  $x \in X$  was arbitrary,  $f$  is continuous.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

### Problem 3

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#### Problem Statement

#### Homework 12, Problem 3

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be the function

$$f(x) = \frac{1}{1+x^2}.$$

- (a) Prove that  $f$  is continuous.
- (b) Find an open set  $U \subseteq \mathbb{R}$  so that  $f(U)$  is not open.
- (c) Find a closed set  $F \subseteq \mathbb{R}$  so that  $f(F)$  is not closed.
- (d) Find a set  $A \subseteq \mathbb{R}$  so that  $f(\overline{A}) \neq \overline{f(A)}$ .

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Which standard limit laws prove continuity of  $f(x) = 1/(1 + x^2)$ ?

---

2. What is the range of  $f$  on all of  $\mathbb{R}$ ?

---

3. Why is  $(0, 1]$  not open in  $\mathbb{R}$ ?

---

4. Why is  $(0, 1]$  not closed in  $\mathbb{R}$ ?

---

5. For  $A = (0, \infty)$ , compare  $f(\overline{A})$  with  $\overline{f(A)}$ . Where does the extra endpoint come from?

---

Part (a): continuity.

Use sequences or limit laws: if  $x_n \rightarrow x$ , then  $x_n^2 \rightarrow$

---

Conclude  $1 + x_n^2 \rightarrow 1 + x^2$ , and since denominators stay nonzero:

---

Parts (b) and (c): image counterexamples.

For an open set, try  $U = \mathbb{R}$ . Compute  $f(U) =$

---

Explain why that image is not open:

---

For a closed set, try  $F = [0, \infty)$ . Compute  $f(F) =$

---

Explain why that image is not closed:

---



**Part (d): closure/image mismatch.**

**Try  $A = (0, \infty)$ . Then  $\overline{A} =$**

---

**Compute  $f(\overline{A})$ ,  $f(A)$ , and  $\overline{f(A)}$ :**

---

Use this page for parts (a) and (b).

Use this page for parts (c) and (d).

**Student-Derived Solution (No official solution available yet)**

(a) Let  $x_n \rightarrow x$  in  $\mathbb{R}$ . Then by the standard limit laws,

$$x_n^2 \rightarrow x^2, \quad 1 + x_n^2 \rightarrow 1 + x^2.$$

Since  $1 + x^2 \neq 0$ , the reciprocal limit law gives

$$\frac{1}{1 + x_n^2} \rightarrow \frac{1}{1 + x^2}.$$

Hence  $f(x_n) \rightarrow f(x)$ , so  $f$  is continuous.

(b) Let  $U = \mathbb{R}$ , which is open. Then

$$f(U) = (0, 1].$$

This is not open in  $\mathbb{R}$ , since  $1 \in (0, 1]$  but no open interval around 1 is contained in  $(0, 1]$ .

(c) Let  $F = [0, \infty)$ , which is closed in  $\mathbb{R}$ . Then

$$f(F) = (0, 1].$$

This set is not closed in  $\mathbb{R}$ , since 0 is a limit point of  $(0, 1]$  but  $0 \notin (0, 1]$ .

(d) Let  $A = (0, \infty)$ . Then  $\overline{A} = [0, \infty)$ , so

$$f(\overline{A}) = f([0, \infty)) = (0, 1].$$

On the other hand,

$$f(A) = f((0, \infty)) = (0, 1), \quad \overline{f(A)} = [0, 1].$$

Therefore

$$f(\overline{A}) = (0, 1] \neq [0, 1] = \overline{f(A)}.$$

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 4

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### Problem Statement

### Homework 12, Problem 4

Consider the metric spaces  $(\mathbb{R}, |\cdot|)$  and  $(\mathbb{R}^2, d_2)$ .

- (a) Prove that the functions  $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$  given by  $f(x, y) = x$  and  $g(x, y) = y$  are continuous. (Hint: Use sequences.)
- (b) Prove that if  $h : [0, 1] \rightarrow \mathbb{R}^2$  is a continuous function such that  $h(0) = (x_0, y_0)$ ,  $h(1) = (x_1, y_1)$ , and  $x_0 < 0 < x_1$ , then there exists a point  $t \in (0, 1)$  so that  $h(t)$  lies on the  $y$ -axis.

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**



1. How does convergence in  $(\mathbb{R}^2, d_2)$  imply coordinatewise convergence?

---

2. Why does  $|x_n - x| \leq d_2((x_n, y_n), (x, y))$  prove continuity of the first projection?

---

3. What scalar function should be formed from the path  $h$  in part (b)?

---

4. Why is  $\phi(t) = f(h(t))$  continuous?

---

5. How does the Intermediate Value Theorem produce a point on the  $y$ -axis?

**Part (a): projections are continuous.**

**Let  $(x_n, y_n) \rightarrow (x, y)$  in  $(\mathbb{R}^2, d_2)$ . Write the coordinate inequality for  $x_n$ :**

---

**Use squeeze to prove  $x_n \rightarrow x$ , hence  $f(x_n, y_n) \rightarrow f(x, y)$ :**

---

**Repeat the coordinate inequality for  $y_n$ :**

---

**Part (b): path crosses the  $y$ -axis.**

**Define  $\phi : [0, 1] \rightarrow \mathbb{R}$  by  $\phi(t) =$**

---

**Explain why  $\phi$  is continuous:**

---

Compute  $\phi(0)$  and  $\phi(1)$ , and record their signs:

---

Apply the Intermediate Value Theorem to get  $t \in (0, 1)$  with  $\phi(t) = 0$ :

---

Translate  $\phi(t) = 0$  into " $h(t)$  lies on the  $y$ -axis":

---

Use this page for part (a).

Use this page for part (b).

**Student-Derived Solution (No official solution available yet)**

(a) We prove continuity of  $f(x, y) = x$ . Let  $(x_n, y_n) \rightarrow (x, y)$  in  $(\mathbb{R}^2, d_2)$ . Then

$$d_2((x_n, y_n), (x, y)) = \sqrt{(x_n - x)^2 + (y_n - y)^2} \rightarrow 0.$$

Since

$$|x_n - x| \leq d_2((x_n, y_n), (x, y)),$$

the squeeze theorem gives  $x_n \rightarrow x$ . Thus

$$f(x_n, y_n) = x_n \rightarrow x = f(x, y),$$

so  $f$  is continuous. The proof for  $g(x, y) = y$  is the same: use

$$|y_n - y| \leq d_2((x_n, y_n), (x, y))$$

to get  $y_n \rightarrow y$ , hence  $g(x_n, y_n) \rightarrow g(x, y)$ .

(b) Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be the first-coordinate projection from part (a), and define

$$\phi : [0, 1] \rightarrow \mathbb{R}, \quad \phi(t) = f(h(t)).$$

Since  $h$  is continuous and  $f$  is continuous,  $\phi = f \circ h$  is continuous. Also

$$\phi(0) = f(h(0)) = f(x_0, y_0) = x_0 < 0,$$

and

$$\phi(1) = f(h(1)) = f(x_1, y_1) = x_1 > 0.$$

By the Intermediate Value Theorem, there exists  $t \in (0, 1)$  such that  $\phi(t) = 0$ . This means the first coordinate of  $h(t)$  is 0, so  $h(t) \in \{(0, y) : y \in \mathbb{R}\}$ . Therefore  $h(t)$  lies on the  $y$ -axis.

What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 5

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### Problem Statement

### Homework 12, Problem 5

Going forward, you may assume (no proof necessary) any fact about  $\sin x$  and  $\cos x$  from calculus, such as  $\sin, \cos : \mathbb{R} \rightarrow [-1, 1]$  are continuous functions.

Prove that the function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  given by

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum. (Hint: Notice that we cannot directly apply the extreme value theorem here. However, we can apply it to any compact subset of  $\mathbb{R}^2$ .)



Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

\_\_\_\_\_

**Hypotheses:**

---

**Conclusion:**

---

**Definitions to unpack:**

---

**Useful theorem/lemma:**

---

**Key inequality/identity:**

---

**One-sentence proof strategy:**

1. Why can the extreme value theorem not be applied directly on all of  $\mathbb{R}^2$ ?

---

2. Why is  $f$  continuous?

---

3. What lower bound follows from  $\sin(x - y) \geq -1$ ?

---

4. Why does the closed unit disk capture a global minimizer?

---

5. Why is it useful to find one point inside the disk where  $f < 0$ ?

---

Continuity and coercive estimate.

Show  $f$  is continuous as a sum/composition of continuous functions:

---

Use  $\sin(x - y) \geq -1$  to prove  $f(x, y) \geq$

---

Conclude that if  $x^2 + y^2 \geq 1$ , then  $f(x, y) \geq$

---

Find a compact region where the minimum occurs.

Choose  $p = (-1/2, 1/2)$ . Compute  $f(p) =$

---

Explain why  $f(p) < 0$ :

---

Let  $K = \{(x, y) : x^2 + y^2 \leq 1\}$ . Explain why  $K$  is compact:

Apply the extreme value theorem to  $f|_K$ :

---

Compare points inside  $K$  and outside  $K$  to prove the minimizer on  $K$  is global:

---

Use this page for the setup and compact restriction.

Use this page to prove the minimum is global.

### Student-Derived Solution (No official solution available yet)

First,  $f$  is continuous on  $\mathbb{R}^2$ . The map  $(x, y) \mapsto x^2 + y^2$  is continuous, the map  $(x, y) \mapsto x - y$  is continuous, and  $\sin$  is continuous by the allowed calculus facts. Hence  $(x, y) \mapsto \sin(x - y)$  is continuous, and so  $f$  is continuous. Since  $\sin(x - y) \geq -1$ , we have

$$f(x, y) = x^2 + y^2 + \sin(x - y) \geq x^2 + y^2 - 1.$$

In particular, if  $x^2 + y^2 \geq 1$ , then  $f(x, y) \geq 0$ .

Now consider

$$p = \left(-\frac{1}{2}, \frac{1}{2}\right).$$

Then

$$f(p) = \frac{1}{4} + \frac{1}{4} + \sin(-1) = \frac{1}{2} - \sin(1) < 0.$$

Also  $p$  lies in the closed unit disk

$$K = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

The set  $K$  is closed and bounded in  $\mathbb{R}^2$ , hence compact. By the extreme value theorem, since  $f$  is continuous, there exists  $q \in K$  such that

$$f(q) \leq f(z) \quad \text{for all } z \in K.$$

We claim  $q$  is a global minimizer on  $\mathbb{R}^2$ . If  $z \in K$ , this is true by the choice of  $q$ . If  $z \notin K$ , then  $z = (x, y)$  satisfies  $x^2 + y^2 > 1$ , so  $f(z) \geq 0$ . But

$$f(q) \leq f(p) < 0 \leq f(z).$$

Thus  $f(q) \leq f(z)$  for every  $z \in \mathbb{R}^2$ . Therefore  $f$  has a global minimum.



What was the key definition?:

---

What was the key proof move?:

---

What step would I forget under time pressure?:

---

What nearby exam variation could appear?:

---

## Problem 6

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### Problem Statement

### Homework 12, Problem 6

Let  $X$  and  $Y$  be metric spaces and  $f : X \rightarrow Y$  a function. Prove that if  $X$  is compact and  $f$  is continuous and bijective, then the inverse function  $f^{-1} : Y \rightarrow X$  is also continuous. (Hint: It suffices to show that the preimage of any closed subset of  $X$  under  $f^{-1}$  is closed in  $Y$ .)

Topic: \_\_\_\_\_

What is being asked?:

\_\_\_\_\_

Definitions likely relevant:

\_\_\_\_\_

Theorem/proof method likely relevant:

\_\_\_\_\_

What should the first line of the proof be?:

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**Hypotheses:**

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**Conclusion:**

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**Definitions to unpack:**

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**Useful theorem/lemma:**

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**Key inequality/identity:**

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**One-sentence proof strategy:**

1. Which closed-set criterion for continuity is suggested by the hint?

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2. If  $C \subseteq X$  is closed, why is  $C$  compact?

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3. Why is  $f(C)$  compact in  $Y$ ?

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4. Why is  $f(C)$  closed in  $Y$ ?

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5. Why is  $(f^{-1})^{-1}(C) = f(C)$  when  $f$  is bijective?

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Use the closed-set criterion for  $f^{-1}$ .

Let  $C \subseteq X$  be closed. Since  $X$  is compact, conclude  $C$  is:

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Since  $f$  is continuous, conclude  $f(C)$  is:

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Since  $Y$  is a metric space, compact subsets of  $Y$  are:

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Use bijectivity to compute  $(f^{-1})^{-1}(C) =$

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Conclude the preimage under  $f^{-1}$  of every closed set  $C \subseteq X$  is closed in  $Y$ :

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Apply the closed-set criterion to finish:

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Use this page for the complete proof.

### Student-Derived Solution (No official solution available yet)

We prove that  $f^{-1} : Y \rightarrow X$  is continuous using the closed-set criterion.

Let  $C \subseteq X$  be closed. Since  $X$  is compact and  $C$  is closed in  $X$ ,  $C$  is compact. Since  $f : X \rightarrow Y$  is continuous, the image of a compact set under  $f$  is compact. Therefore  $f(C)$  is compact in  $Y$ .

Because  $Y$  is a metric space, compact subsets of  $Y$  are closed. Hence  $f(C)$  is closed in  $Y$ .

Since  $f$  is bijective,

$$(f^{-1})^{-1}(C) = f(C).$$

Thus the preimage under  $f^{-1}$  of every closed subset  $C \subseteq X$  is closed in  $Y$ . By the closed-set criterion for continuity,  $f^{-1}$  is continuous.



What was the key definition?:

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What was the key proof move?:

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What step would I forget under time pressure?:

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What nearby exam variation could appear?:

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